3GPP TSG RAN WG1 #123 **R1-2508811**
Dallas, USA, Nov 17th – 21st, 2025

Agenda item: **11.6**
Source: **Samsung (Moderator)**
Title: **Moderator summary #4 on AI/ML for 6GR**
Document for: **Decision**

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1 Introduction

In RAN 1 #122, the following agreement was made for use case identification:

Agreement
For 6GR AI/ML use cases identification/categorization, for each (sub-)use case proposed, proponent companies are encouraged to study and report the following:

- Definition of each (sub-)use case, including at least AI/ML model input/output
- The evaluation assumption, methodology, KPIs, benchmark, and preliminary simulation results
- Assumption on training types, e.g.,
  - offline training, online training/finetuning
  - Label construction (if applicable), including whether/how to obtain label data for model training
- Assumption on model location for inference, e.g., UE-sided model, NW-sided model, and two-sided model
- Collaboration/interaction between UE and NW, e.g.,
  - no collaboration/interaction
  - UE/Network collaboration targeting at separate or joint ML operation
- High level potential specification impact

This paper summarized the proposed use cases for 6GR study on AI/ML.

2 Use cases

Proposal 1: potential agenda for further study
From RAN 1 perspective, for potential further discussion on the categorized 6GR AI/ML use cases, the following corresponding primary agenda/working group can be considered for handling the use cases

Alt 1: no note
majority
Alt 2: whether/when to study the use cases is a separate discussion
IDC, Ericsson

Use cases		Primary agendas /WG
Low overhead CSI-RS or CSI prediction with AI/ML		CSI-RS and CSI acquisition
Low overhead DMRS with AI/ML receiver		UL & DL DMRS associated with PUSCH/PDSCH Note: Sub-Case C: DMRS-free may be related to modulation
CSI compression and feedback		CSI acquisition Note: this may be related to uplink control
AI/ML for beam management and extension		Initial access for Sub-case D Beam management for other sub-cases Note: sub-case A/B/D maybe related to mobility
AI/ML for SRS	**Low overhead SRS with AI/ML**	SRS
	Low PAPR SRS sequence design	
	AI/ML based SRS power	RAN 4

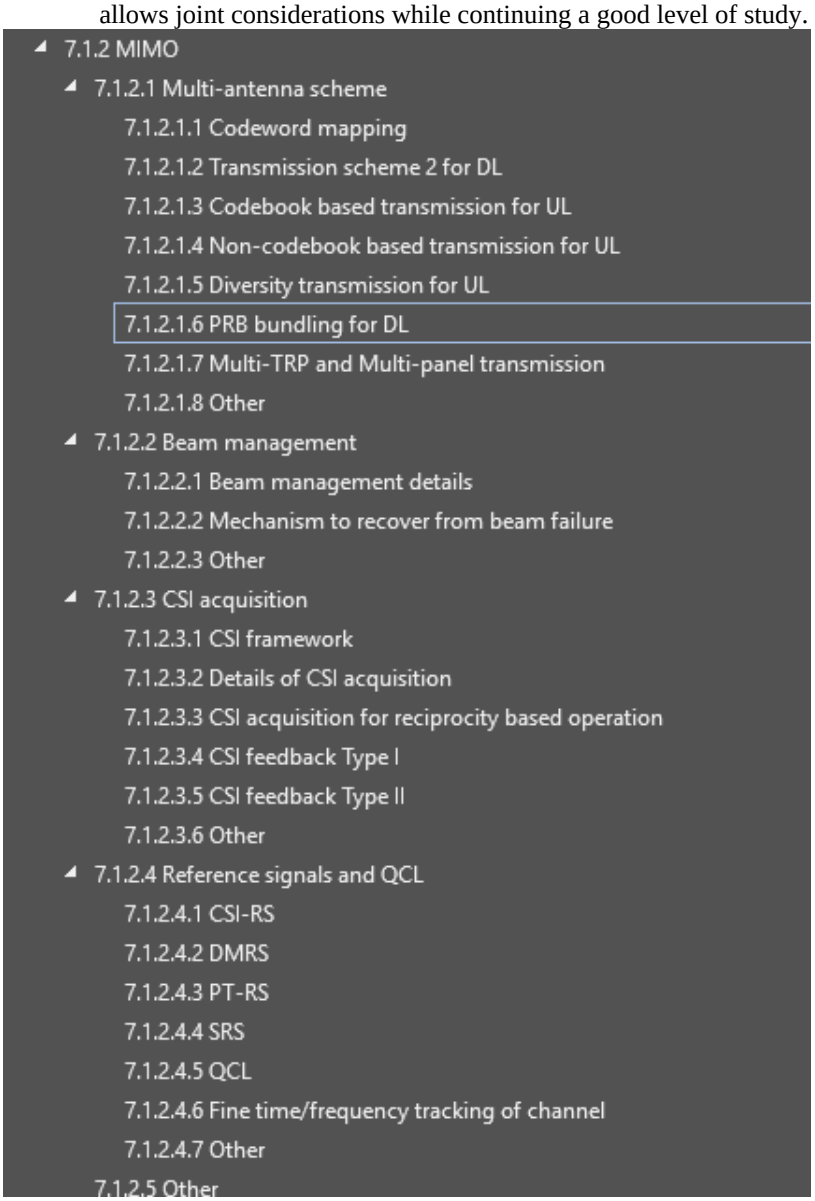
	imbalance compensation	
AI-enabled UL precoder indication		UL MIMO
AI-based non-linearity handling at transmitter or receiver		RAN 4 Note: this may be related to DMRS, SRS, Power control in RAN 1
AI/ML for (de)modulation		modulation Note: Sub-case B may be related to MIMO Note: assuming no change to DMRS
AI/ML based waveform for PAPR reduction		Waveform
AI/ML based HARQ-ACK feedback		Channel coding Note: this may be related to uplink control
PDCCH related	**Prior-Information-Aided DCI Decoding**	DL control
	Lossless DCI Compression	
Power control related	**UL closed-loop power control**	Power control
	Pathloss prediction	
RACH related design	**Early contention resolution in RACH**	Random access/PRACH
	Low PAPR PRACH sequence design	
RAN enhancement for AI/ML service	**Improved scheduling/HARQ for token traffic**	Alt 1: RAN 2 RAN 1 (scheduling/HARQ, others) SA 4 Alt 2: TBD Note: this may be related to scheduling/HARQ
	Transmission for audio token traffic	
Site Specific Learning for AI/ML using RAN Digital Twin		Depending on corresponding use case where site specific learning is applicable, e.g., DMRS
Digital twin construction related use cases	**AI/ML-enabled RAN digital twin with distributed model**	ISAC
	Sensing based RAN digital twin construction with NW-side AI/ML model	ISAC
AI for positioning		Positioning related agenda, if any

Company	Support or not	Comment
FL		For the TBD part, or even some use cases that has proposed agenda, they may be implementation. The spec impact is not very clear. Let's me collect input to see how to proceed the further discussion. 1. I don't think we need to spend time on this, if this cannot be agreed, we can drop this proposal. 2. if some of use cases can be agreed, for the use case that may not have clear understanding by the group on where to discuss, we can remove them from the table and add one sentence to address the concern, e.g., For other use cases, proponent can directly propose to corresponding agendas. Note: there is no intention to exclude/deprioritize any AI/ML use cases
Google		In our view, we think we should change the main-bullet as follows. The reason is that RAN1 has not identified any use cases. All the use cases in last meeting are based on companies' proposal. It is not practical for RAN1 to study all the cases in one release. We also need to notice

		that AI/ML is an optional feature according to the SID. Further, we think which agenda to discuss which topic is RAN1's chair's authority. Not sure whether we need any discussion. Further, the scheduling/HARQ should be RAN1's scope instead of RAN2. For the ~~identified /~~categorized 6GR AI/ML use cases, the following corresponding agendas can be considered for further discussion ~~potential further study~~
Apple		CSI compression has large impact on UL control channel design. Suggest adding it so it can be considered as a factor in corresponding agenda. Overall, guidance for next step is needed. When and how down-selection will happen. In each agenda, how the AI related topics and non-AI related topics will be organized, in separate sub-agenda or merged into one sub-agenda with related topics etc, or depending on the scope of each AI use cases?
MediaTek		In general, we believe it is necessary to have a preliminary down-selection of use cases by a mutually agreed timeline. Otherwise, it will be wasteful to discuss assumptions related to each use-case study in detail. For the DMRS-free sub-case in low overhead DMRS with AI/ML receiver, it would have impact to the modulation as well since constellation design is involved. Therefore, Modulation should also be added in the related agendas for this use case. For CSI compression and feedback, JSCCM-like techniques also affect the modulation and channel coding for uplink control information, therefore these agendas should also be added.
Huawei, HiSilicon		Firstly, we should clarify the rule of discussion – for any of the categorized use case, it should be allowed to be discussed at the corresponding agenda, and the discussion should not be challenged with the excuse of 'out of scope'. For the identified /categorized 6GR AI/ML use cases, the following corresponding agendas can be considered for potential further study Note: For any of the categorized use case, it is allowed to be discussed at the corresponding agenda; whether/when to study can be decided in the corresponding agenda. In addition, some views to specific use cases: Improved scheduling/HARQ for token traffic – It should be in "control, data scheduling and HARQ operation" agenda (11.9). Spec impact on differential scheduling and MCS are identified from our evaluations. Sensing based RAN digital twin construction with NW-side AI/ML model/distributed model – it is clear to be distributed to ISAC (11.14), the TBD should be removed. Low overhead CSI-RS or CSI prediction with AI/ML – it is not only involving CSI-RS, but related with CSI feedback. New CSI reporting type is identified, e.g., for NW-side model, channel matrix and/or multi-path power/angle/delay info is identified; for UE-side model, the predicted CSI is identified. Both need to discuss the CSI feedback format/content. AI/ML for beam management and extension – the 'Mobility' agenda should be clarified, since otherwise it sounds like RAN2 related. AI/ML based HARQ-ACK feedback – it is more appropriate to be assigned to "control, data scheduling and HARQ operation" agenda (11.9), since it is HARQ related. The small block coding is applicable to all UCI types. It is not clear which agenda should include "Power control" – 11.9 control channel?

		Regarding AI for positioning, it is not captured in the 6G SID.
CATT		For CSI compression and feedback, JSCC and JSCM have impact on UL control coding chain, suggest add control channel as the corresponding agenda.
Xiaomi		• For the AI-based non-linearity handling at transmitter or receiver, we think at current stage only keeping it in RAN4's study is sufficient. RAN4 should justify the performance and feasibility first. the potential impact on power control or DMRS depends on the detailed non-linearity compensation scheme which need RAN4 study. It is pre-mature to assign them to DMRS session and power control session. • For improved scheduling/HARQ for token traffic: - Firstly, we consider the name of this use case should be changed. The concept of token is not clearly defined. Here, we consider the token is minimum unit for the large generative model processing. But according to current discussion in RAN2 and SA4, some other AI services beyond token are also under study. Thus, we consider changing the "token" to "AI service" is safer. In addition, the potential enhancement for AI service is not limited to scheduling/HARQ, in RAN2 some other enhancement is also under study. These enhancements should be considered as a whole package. So, we consider the name of the use case can be changed as "RAN enhancement for AI service" - Secondly, for the potential enhancement related to RAN1 e.g., scheduling/HARQ, it highly depends on the AI service profile study in SA4 /RAN2 and higher layer design, it is premature to study it in RAN1 at current stage. So we consider we could wait for the study progress in SA4 and RAN2
CMCC		1. For Prior-Information-Aided DCI Decoding and Lossless DCI Compression, the leading group should be RAN 1 and the related agenda is DL control. RAN1 need to study the feasibility of the decoding based on AL/ML prior-information prediction. NW needs to know when UE will use AI based DCI decoder, so that NW can configure corresponding search space set and CORESET for PDCCH. The fallback mechanism to non-AI based decoder and configuration for AI based DCI should be designed together with non-AI DL control in RAN 1. Besides, like traditional PDCCH studies, the performance evaluation on PDCCH is first studied in RAN 1. This should be applied to both AI based and non-AI based methods. 2. For Low overhead CSI-RS or CSI prediction with AI/ML, the related agenda is not only reference signal of CSI-RS, but also CSI feedback/report design. The CSI report content will also be affected by CSI prediction. 3. For AI/ML based HARQ-ACK feedback, we think it could be also related to UL control, considering the signaling interaction/procedure between gNB and UE.
SPRD		At present, it seems that there is no specific agenda dedicated to discussing power control. Will a new agenda be introduced later, or will it be discussed within the MIMO agenda?
vivo		Support the FL proposal in general. The following details can be clarified: 1. For CSI-RS overhead reduction, the agenda is not only related to CSI-RS but also related to CSI feedback. 2. For PA nonlinearity, we are expecting SRS is related since it may require SRS transmission for data collection. Also PUSCH transmission scheduling is necessary. Thus SRS, PUSCH should both be added. "MIMO: reference signal -> DMRS, SRS, PUSCH"
OPPO		Thanks to FL for being inclusive for reported AI/ML use cases. We expect RAN1 to study as much as possible, but for other working groups (RAN2 and RAN4), can RAN1 make the decision on their consideration for study some specific AI/ML topic? If not, we would better note down the RAN1 only part. One typo might be **AI/ML based HARQ-ACK feedback** to channel coding->scheduling/HARQ, where the HARQ codebook should be designed in our understanding.

Lenovo		We doubt whether it would be feasible for RAN1 to study all the use cases listed in the proposal in Rel-20. Suggest changing the proposal statement to "For the listed/categorized 6GR AI/ML use cases, consider the corresponding agenda item as shown in the table for further discussion."
Panasonic		Thanks, FL, for the effort. We think it is necessary to have some levels of down-selection of use cases to be studied first, before discussing this topic. We can come back to discuss it later.
Fujitsu		For low overhead CSI-RS and CSI prediction with AI/ML, agree with other companies that it is also related with CSI feedback. For AI/ML based SRS power imbalance compensation, we think it may also have potential impact on RAN1. The following update is suggested. AI/ML based SRS power imbalance compensation RAN 4, RAN1
InterDigital		We share similar views as Google, and recommend to update the main bullet as follows: For the ~~identified /~~categorized 6GR AI/ML use cases, the following corresponding agendas can be considered for further discussion ~~potential further study~~ Secondly, the following note is not appropriate and should be removed/updated. *Note: whether/when to study can be decided in the corresponding agenda.* We cannot simply say that all of these will be now become added responsibilities for the identified agenda items to assess for study including the corresponding timing and work plan. Certainly, any use case that is to be studied in another agenda item would be eventually fit to the work plan considering the objectives of the agenda item. However, a first attempt at the question of "whether to study" should be addressed as part of the current discussions in 11.6. So, we propose to update the note as: Note: whether/when to study ~~can be decided in the corresponding agenda~~ each use case is a separate discussion. On "**AI-based non-linearity handling at transmitter or receiver**", we do not agree with the current mapping – the study should be primarily led by RAN4. Any impact to RAN1 would be geared primarily for later phase of the study, or even relevant only if normative work may be pursued. The assumptions on LCM aspects and impact to DMRS or power control can be adequately made by RAN4 during their study. With the main study, including evaluations and its analyses, being under RAN4's responsibility, the study outcome can clearly point to impact for RAN1 specifications. Thus, we do not see any reason to include any RAN1 study objectives for this use case. Lastly, in our view, it is not necessary to map each and every proposed use case. As can be seen from the above, companies have different views on target agenda items including whether multiple agenda items may be impacted. For the latter case, it is not even clear where the study may be conducted if impacting multiple agenda items. Moreover, most of these are for use cases that are not sufficiently motivated yet. We cannot simply commit from RAN1 perspective to study every use case that has been ever proposed.
Nokia		Few comments, • Some use-cases are related to each other.. E.g., o PA non-linearity and AI receiver for example are related to each other o Waveform for PAPR reduction -> this use-case may be related to PA non-linearity use-case as well ? • Some cases of use are related to multiple classical topics. How to do the proper study on them is not clear. E.g., o For low overhead DMRS → we think DMRS-free case have impact on modulation. So, it is related to modulation agenda items as well. o For JSCM → we think JSCM case have impact on modulation and channel

		coding. So, it is related to two other agenda items as well. o For Ul precoder indication → we think this case has impact on modulation. • The above list of agenda items is in very high-level. The following was an example MIMO agenda distribution in 5G times, and we may have to consider at least more detailed mapping. • One possibility is to **have a dedicated sub-agenda's under each classical topic to discuss and study AI/ML**. When the decisions are made in classical agenda items, that allows joint considerations while continuing a good level of study. 
NTT DOCOMO		We have a general issue with the agenda assignments. Some topics have an impact on multiple RAN1 sessions. We want to clarify where to discuss these topics and how to coordinate among sessions.
TCL		We need to prioritize these use cases or make down-selection to avoid duplicate discussion in the related agenda again.
Ericsson		For the main sentence, suggest to change to: "For the ~~identified /categorized~~ 6GR AI/ML use cases proposed by companies," About the note: • Suggest to identify a few use case for study, while the rest are decided in the corresponding agenda.

		• Since some use cases are marked as RAN2 or RAN4, suggest to update the note as follows: "Note: whether/when to study can be decided in the corresponding RAN1 agenda or other RAN group as marked."
LG Electronics		Thank FL for your great effort. We are generally fine with the proposal, but as other companies are mentioned, CSI compression may be related to MIMO, UL control channel, channel coding/modulation. Then, what is the guidance of next step discussion for use case related to multiple agendas?
ZTE		We are generally fine to list the related agenda for each AI/ML use case submitted by companies. From our perspective, we can firstly focus on this proposal and the update of the previous observations. The other proposals may not be urgent at this point.
Huawei, HiSilicon		Regarding "**RAN enhancement for AI traffic**", we do not have such use case in the previous observation. To Xiaomi's comments: The definition of token traffic has been clearly captured in RAN1 agreement, there is nothing unclear. Agreement For the study traffic model(s) for 6GR AI/ML services: • A representative AI/ML service is the generative AI, e.g., as defined in TR22.870. Send LS to SA4 (cc RAN2, SA1, SA2) requesting input if any on traffic characteristics for AI/ML services. Note: RAN1 is discussing the following options for the model: • Option-1a: The model is parameterized by Token, e.g., Token size, Token arrival rate, and Token delay budge o Token is the minimum unit of data generated in the application layer. From the RAN1 studied/proposed use case/sub-use cases, they are all related with HARQ/scheduling and serving token traffic, using "AI traffic" is too broad and too vague on its meaning. If RAN2/SA4 or other WGs have clear inputs, they can inform RAN1 via LS, then RAN1 can update the scope of study; otherwise, without the information of what is "other AI services beyond token" from other WG, or without additional inputs from RAN1 contributions, we do not know what to be additionally studied at RAN1. If Xiaomi has interests on this direction, it is welcome to bring evaluation results and solutions for extending the scope. Hence, we suggest to change back to the notation to "improved HARQ/scheduling to token traffic". In addition, it is not clear why the study on HARQ/scheduling is related with SA4 (which never defines RAN solution). It should be clear that RAN1 should at least start with evaluations to the token traffic model, similar to the procedure of R17 XR campaign. Considering there has not been clear initialization of this use case at other WGs (RAN2 or SA4), the appropriate agenda for study this use case at RAN1 is control/scheduling/HARQ agenda.
IMU		We do not support the inclusion of "LTM-related" under the line item AI/ML for beam management and extension. Beam prediction is not limited to LTM, nor is it uniquely triggered by an LTM-specific mobility procedure. In fact, in RAN2, discussions for 6GR mobility are moving toward a unified mobility framework, where handover procedures (e.g., baseline HO, CHO, LTM, future variants) are expected to share common measurement and prediction mechanisms. In such a unified structure, it is not appropriate to associate AI/ML-based beam prediction with only one specific HO procedure. Therefore, we propose to remove the "LTM related" reference from this row as: [Mobility, if triggered by RAN 2]

2.1 CSI prediction and CSI-RS overhead reduction

Proposal 2.1: study aspects

For the potential study on CSI prediction and CSI-RS overhead reduction with AI/ML, prioritize one-sided model use cases, at least the following aspects are to be considered:

- KPIs:
  - Performance related:
    - SGCS, NMSE, Throughput, CSI-RS overhead, NES gain (if applicable)
  - Model complexity
  - UE measurement complexity and feedback overhead (for NW-sided model)
  - Overhead for fine-tuning, when applicable.
- Other aspects for study:
  - Justification for UE-sided model and/or NW-sided model and/or two-sided model
  - Label quality impact on CSI prediction
  - Performance with practical impairments (e.g., phase drift and antenna mutual coupling among Tx ports)
  - Validation on the pre-requisite assumptions (e.g., stable correlation on Tx ports)
  - Generalization
    - Sub-Case A/B/D/E: deployment scenarios, antenna configurations, spatial domain factors (e.g., port number/patterns), frequency domain factors (e.g., bandwidth, subcarrier spacing, RS density), UE speed, beam domain factors (e.g., number of analog beams or antenna-to-port virtualization patterns), when appliable
    - Sub-Case C: frequency domain factors (e.g., bandwidth and frequency gap between the source and target carrier/frequency bands), when appliable
  - Feasibility/performance for finetuning, if applicable
  - Scalability, e.g., for frequency domain factor (e.g., bandwidth), spatial domain factor (e.g., number of antenna ports), for beam domain factors, when applicable (e.g., number of analog beams or antenna-to-port virtualization patterns)
  - Impact compression loss (for NW-sided model)
  - Coexistence with non-AI capable UE for UE-sided model
  - (Spec impact) CSI-RS pattern design and/or configuration
  - (Spec impact) CSI feedback format/content, if applicable
  - (Spec impact) signaling/procedure for
    - data collection
    - inference
    - fine-tuning, if applicable and feasible
    - performance monitoring
    - handling of NW-side additional conditions, if applicable
  - Note: the study and analysis consider the model location (i.e., UE-sided model, NW-sided model, two-sided model)

Company	Comment
FL	The proposal is to listing the potential AI/ML specific consideration. For spec impact part, based on our experience in NR, we need some LCM procedure.
Google	We do not think fine-tuning related should be studied. It is too premature to include this part.
Apple	For UE side model, SGCS need to consider codebook compression loss. For NW side model, methods to send raw channel matrix or PMI back to NW also need to be considered and evaluated. Suggest to add: on KPI: SGCS or SGCS ratio. On aspects to study: add of impact compression loss
Huawei, HiSilicon	Do not get the point of gain for CSI-RS overhead reduction and NES. Considering a typical network where cell-specific CSI-RS is configured, the BS may probably not reduce its CSI-RS port/density just due to the existence of some AI/ML enabled UEs in the cell, because other non-AI/ML UEs should still perform channel measurement based on the full port/density/band CSI-RS. On the other hand, if it is NW-side model, possible gain may be achieved by UE measurement complexity, overhead reduction. • KPIs:

	o Performance related: SGCS, NMSE, Throughput, ~~CSI-RS overhead, NES gain~~ o Model complexity o UE measurement complexity o CSI report overhead reduction In addition, for the study aspects, we need to justify the validity of UE-sided model – what is the benefit of it? We do not observe OH reduction gain or complexity reduction gain for UE-sided model. • Other aspects for study: o Justification of UE-sided model and NW-sided model o Label quality impact CSI prediction Regarding spec impact, we also need to consider CSI feedback format/content, which could be different from legacy. o (Spec impact) CSI-RS pattern design and/or configuration o (Spec impact) CSI feedback format/content It is not clear what does "Coexistence with non-AI capable UE" mean. If it means the design of CSI-RS pattern, as said, it is not likely to configure AI specific CSI-RS pattern. Regarding inter-vendor collaboration, we may focus on one-sided model for initial study.
Xiaomi	Regarding the online fine-tuning, we suggest to justify the feasibility and necessity first. And we also need to clarify what kind of the overhead is, e.g., RS overhead, reporting overhead for data collection? In addition, for other aspects, we suggest the following update o Label quality impact for CSI prediction o Generalization ▪ Sub-Case A/B/D/E: deployment scenarios, antenna configurations, spatial domain factors (e.g., port number/patterns), frequency domain factors (e.g., bandwidth, subcarrier spacing, RS density) ▪ Sub-Case C: frequency domain factors (e.g., bandwidth and frequency gap between the source and target frequency bands)
Qualcomm	We suggest to categorize the study aspects of CSI prediction and CSI-RS overhead reduction into three options 1) UE side model; 2) NW side model; 3) two sided model. Also, for "Performance with practical impairments", we suggest to be more specific to include: for example, phase drift and antenna mutual coupling among Tx ports.
CMCC	Support this use case, but we do not think NES gain should be included in the performance KPI. The power model haven't been decided in agenda 11.5. It is a little risky to take it as performance KPI if we do not have clear understanding of the meaning of NES gain.
SPRD	For the CSI prediction, it is necessary to take into account the changes of the channel over time. Therefore, the speed of the UE will also have an impact on performance. Hence, we suggest including the UE speed factor in the generalization part. o Generalization ▪ Sub-Case A/B/D/E: deployment scenarios, antenna configurations, spatial domain factors (e.g., port number/patterns), frequency domain factors (e.g., bandwidth, subcarrier spacing), UE speed ▪ Sub-Case C bandwidth and frequency gap between the source and target frequency bands
vivo	We would like to change the following highlighted parts. CSI reporting overhead is important aspects for NES on/off pattern prediction and report considerations for different model locations. For the potential study on CSI prediction and CSI-RS overhead reduction with AI/ML, at least the following aspects are to be considered:

	• KPIs: o Performance related: SGCS, NMSE, Throughput, CSI overhead reduction, CSI-RS overhead, NES gain o Model complexity o Overhead for fine-tuning, when applicable. o Inter-vendor collaboration overhead and complexity for two-sided model, when applicable. • Other aspects for study: o Label quality impact CSI prediction o Performance with practical impairments o Generalization ▪ Sub-Case A/B/D/E: deployment scenarios, antenna configurations, spatial domain factors (e.g., port number/patterns), frequency domain factors (e.g., bandwidth, subcarrier spacing), or beam domain factor (e.g., number of analog beams or antenna-to-port virtualization patterns) ▪ Sub-Case C bandwidth and frequency gap between the source and target frequency bands o Feasibility/performance for finetuning, if applicable o Scalability, e.g., for frequency domain factor (e.g., bandwidth), ~~and~~ spatial domain factor (e.g., number of antenna ports), or for beam domain factor (e.g., number of analog beams or antenna-to-port virtualization patterns) o Coexistence with non-AI capable UE o (Spec impact) CSI-RS pattern design and/or configuration o (Spec impact) signal/procedure for ▪ data collection ▪ inference ▪ fine-tuning, if applicable and feasible ▪ performance monitoring ▪ inter-vendor collaboration, if applicable and justified ▪ handling of NW-side additional conditions, if applicable Note: the study and analysis consider the model location (i.e., UE-sided model, NW-sided model, two sided model)
OPPO	For the KPI of overhead for fine-tuning, it seems not strong motivation to study it due to following reasons. 1. Most of companies assumed offline training for the model. 2. If the purpose of fine-tuning is to address the generalization issues, we already listed the generalization study and performance monitoring, it would be not very essential to tune the model via online solution (overhead introduced).
Lenovo	Considering that CSI prediction is also mentioned in the main bullet, suggest including CSI report configuration as part of the Spec impact bullets. Besides, suggest the minor changes to “Other aspects for study:” “Label quality impact on CSI prediction” “(Spec impact) signaling/procedure for”
Panasonic	We share similar views with other companies that NES gain is important KPI that can be evaluated. Regarding “overhead for fine-tuning”, we think it might not be early to be considered as companies almost assumed offline training and we do not have any observation about it.
Fujitsu	Agree with other companies that NES gain could be removed. Regarding Generalization, for Sub-Case C which is prediction across carrier/band/frequency band, the following update is suggested. The current wording seems only consider the prediction cross band.

	o Generalization ▪ Sub-Case A/B/D/E: deployment scenarios, antenna configurations, spatial domain factors (e.g., port number/patterns), frequency domain factors (e.g., bandwidth, subcarrier spacing) ▪ Sub-Case C bandwidth and frequency gap between the source and target carrier/frequency bands
InterDigital	We suggest removing references to "fine-tuning" and "inter-vendor collaboration" from the list. At this point, there is insufficient clarity on need/feasibility and exact nature of "fine-tuning" – companies seem to have very different ideas as to what "fine-tuning" may entail. If a clear need/benefit is identified during the course of the study, it can be considered then. The proposal here should not be seen as limiting the scope of the study in any way, especially when it comes to spec impact. This would be something that we would understand better as a group during the course of the study. Without any proper studies and evaluations conducted, the list of spec impact bullets has very limited relevance. We also suggest the following update to indicate that we are not somehow committing to these sub-use cases at this point – all the sub-use cases have not been analyzed and vetted, and thus, can only be considered as starting guidance at best, and thus, should not be limiting. At the same time, not all these sub-use cases may be studied in detail. o Generalization ▪ Sub-Case A/B/D/E: deployment scenarios, antenna configurations, spatial domain factors (e.g., port number/patterns), frequency domain factors (e.g., bandwidth, subcarrier spacing) ▪ Sub-Case C bandwidth and frequency gap between the source and target frequency bands ▪ Note that the sub-use cases referred above are intended to serve as examples and it does not imply that all sub-use cases will be studied or that the studied sub-use cases may only be limited to these. In general, these are just to serve as initial guidance and the study aspects should not be limiting in any way. Thus, the list of study aspects need not be perfectly comprehensive at this point.
TCL	We prefer to remove the NES gain as it is not clearly defined now.
Ericsson	**On KPI:** We don't think 'overhead for fine-tuning' should be included as a KPI since most of the companies assume offline training. In addition, it is not clear what overhead means here, and how to quantify it. Also, user perceived throughput should be the primary KPI for making conclusions/decisions. We don't think conclusions/decisions should be based on intermediate KPI such as SGCS/NMSE. Also, since CSI-RS overhead should impact the user perceived throughput, we don't think it needs to be listed as a separate KPI. It is too early to list 'NES gain' as a KPI as we don't have any agreements on network energy consumption model. Inter-vendor collaboration overhead is also very vague, it is unclear how to quantify it. Does model complexity include computational complexity or not? Seems the complexity related KPI is not consistent among different use cases. **On benchmark:** Lack of benchmarks **On Performance with practical impairments:** Regarding the suggestion to add 'phase drift and antenna mutual coupling among Tx ports', we think this is premature as it is possible for network to handle these via implementation (e.g., calibration at network side). **On other aspects for further study:**

	We don't think we need to create another category such as 'beam domain factor'. Spatial domain factors should capture this, and we don't think this new category should be added. What does "Coexistence with non-AI capable UE" mean?? **<u>On two-sided models:</u>** Sub use cases based on two-sided models shall be deprioritized in 6GR Rel-20 study, due to its high complexity in inter-vendor collaboration for training/testing, and to avoid overlapping discussion between NR and 6GR on two-sided AI/ML model. Suggest the following update: For the potential study on CSI prediction and CSI-RS overhead reduction with AI/ML, prioritize one-sided model use cases, at least the following aspects are to be considered:
LG Electronics	Since we believe this proposal is about further study aspects, we think any aspects should not be exclude in this step. And there is a clear benefit of this use case in terms of OH reduction or complexity reduction, so we suggest to study the both UE-sided model and NW-sided model in aspects of various KPIs. With the same perspective above, we agree with Xiaomi's update on sub case C.
IIT Madras	We support this use case. However, further clarification is required before NES gain can be agreed.
Tejas	We are in full agreement with the proposal. To better align we suggest the following categorization options: Model type: NW-sided and UE-sided models as the primary options. CSI overhead reduction domain: Frequency domain only, and Joint frequency and spatial domain as the available options.
ZTE	We tend to agree with other companies that it is difficult to achieve network power saving gain if only partial UEs support the UE-sided model.
Samsung	We agree with the proposal. Regrading NES gain, we believe it should be considered. The impact of non-AI/ML capable UEs including on NES gain is already captured with another bullet for further study.
Sony	We agree with Google's view. In this agenda item, we should try to shortlist use cases for AI/ML further study. We also agree that all proposed use cases cannot obviously be studied in one release but that prioritization is down to RAN1 recommendation to RAN.
OPPO2	By comparing DMRS OH reduction and CSI-RS OH reduction, we feel a little confused about the following aspects • Why is there no non-ideal labeling problem in measuring CSI-RS? But for DMRS, various assumptions on how to get the label. Shouldn't CSI-RS deserve such careful treatment? • Why is there only one option for channel estimation? Similarly as DMRS, there could be AI-based channel interpolation or channel prediction.
IMU	• We agree with prioritizing one-sided model use cases at this stage. The evaluation results of one-sided model use cases would serve as a baseline for assessing the performance of two-sided models at a later stage. • It is unclear what NES means in performance-related KPIs, and this can vary significantly depending on the AI/ML solution being designed. Therefore, we do not support including NES in the KPIs. • Considering overhead is a good perspective; however, as it stands, the item remains open-ended. If the details of the overhead are not going to be defined, we do not support having it listed here.

2.2 DMRS design with AI receiver

Companies have different receiver assumption, which may result in different input/output/label, and impact on generalization or study. Below is a brief summary on receiver assumptions:

Proposed observation 2.2-1: (Clarification on receiver assumptions for each sub-use cases)

For 6GR AI/ML use cases identification/categorization, for the use case of low overhead DMRS with AI/ML receiver, the following receiver assumptions are reported:

For Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain

- **Option 1: AI-based denoise on DMRS REs only**
  - Note: channel interpolation, equalization, demodulation rely on non-AI
  - **Input:** estimated channel on DMRS REs (e.g., LS)
  - **Output:** denoised channel on DMRS REs only
  - **Label:**
    - **Option M:** Received signal on adjacent RE as label (Self-supervise)
    - **Option N:** Known sequence/data (end to end training)
  - **Complexity:** 267K Para/4.35M Flops for 8PRB
  - **Reported by: Samsung**

- **Option 2: AI-based channel estimation for data REs (with or without denoise on DMRS REs)**
  - Note: equalization, demodulation rely on non-AI
  - **Input:**
    - **Option A**: Received signal on DMRS REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern]
    - **Option B**: Estimated channel on DMRS REs (e.g., LS or MMSE)
    - **Option C**: Received signal on DMRS REs and data REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern] (3)
  - **Output:**
    - **Option M:** Estimated channel on target REs (data REs with/without DMRS REs)
    - **Option N:** Estimated filtering coefficients for channel estimation.(2)
  - **Label:**
    - **Option X:** Ideal channel information on target REs (all)
    - **Option Y:** Estimated channel on target REs (in any SNR (2) or high SNR range (e.g. 30dB) (1))**Option Z:** Known sequence/data (end to end training)
  - **Complexity:**
    - CMCC: 1M Para/30M Flops for 10PRB
    - Qualcomm: 800K Para/230M Flops for 4PRB 4layer
    - ZTE: 7M Flops for 8PRB 4layer
    - Samsung: 150/190/340K Para, /2.4/5.9/3.7M Flops 8 PRB2layer
    - vivo: 11M Flops 8PRB 2layer (for direct AI/ML inferenc), 3.56M Flops for all RBs, 2 layers(for AIML generated filtering coefficients)
    -
  - **Reported by: CMCC, Apple, Huawei, Qualcomm, Spreadtrum, ZTE, Samsung(1), vivo(2), Fujitsu Ofinno, Futurewei(3)**

- **Option 3: AI based channel prediction for target REs**
  - Note: interpolation, equalization, demodulation rely on non-AI
  - **Input**: estimated channel on DMRS REs (e.g., MMSE)
  - **Output**: estimated channel on target REs (full or partial data REs)
  - **Label:**
    - Option X: estimated channel on target REs (e.g., MMSE)
    - Option Y: Known sequence/data (end to end training)
  - **Complexity:** 2~3M Para/up to 1M Flops for 10PRB 2layer 2T2R
  - **Reported by: xiaomi**

- **Option 4: AI-based joint {channel estimation + equalization + demodulation/LLR calculation}**
  - **Input**:
    - **Option A**: Received signal on DMRS REs and data REs (all)
      - Additionally: noise variance (1,4)
    - **Option B**: Estimated channel on DMRS REs (e.g., MMSE) (1)
  - **Output**:
    - **Option M:** LLRs (1,2,3,4,5, 6,7,8,9,11,13)
    - Note: decoding is based on non-AI
    - **Option N:** Estimated constellation points (4,12), equalizer coefficients (10)
    - Note: LLR calculation and decoding is based on non-AI
  - **Label:**
    - **Option X**: Known sequence/data (all other than 10)
    - **Option Y**: label free (10)
  - **Complexity:**
    - Samsung: 11M Para/84M Flops for 2PRB 2T2R
    - Lenovo: 6M Para/ for 14PRB 4 layer, 4T8R
    - MediaTek: 600k Para/128M Flops for 6PRB 1layer
    - InterDigital: 3177M/12700M/76248M Flops for 52PRB, SISO/8T2R (2layer)/32x8(3layer)
    - LGE: 4.88M Para/4.32MFLops 10PRB 16Rx
    - xiaomi (Option N): 0.3M Para/328.21M Flops for 5PRB 2layer 2T2R
    - Nokia: 630k Para for 16PRB 2T8R 2 layers
  - **Reported by:** Nokia (1), Ericsson (2), CATT (3), Samsung (4), Kyocera (5), Lenovo (6), MTK (7), Docomo (8), NVIDIA (9), Interdigital (10), LGE (11), xiaomi(12), Futurewei (13)

- **Option 5 : AI-based {channel estimation} + AI-based {equalization + demodulation/LLR calculation}**

For Model 1: Channel estimation

  - **Input:** Received signal on superimposed signal (all)
  - **Output:** Estimated channel on target REs (data REs)
  - **Label:** Ideal channel information on target REs (all)

For Model 2: equalization + demodulation/LLS calculation

  - **Input**: Received signal on DMRS REs and data REs (all)
  - **Output**: LLRs (all)
  - Note: decoding is based on non-AI
  - **Label:** Known sequence/data

For Sub-case B: Superimposed pilot

- **Option 2: AI-based channel estimation for data REs**
  - Note: equalization, demodulation rely on non-AI
  - **Input:**
    - **Option A**: Received signal on superimposed signal (all)
    - **Option B**: Estimated channel on superimposed signal (e.g., LS) (1)
  - **Output:** Estimated channel on target REs (data REs)
  - **Label:**
    - Option X: Ideal channel information on target REs (all) or power delay profile (1)
    - Option Y: Estimated channel on target REs (1)
  - **Complexity:**
    - Lenovo: 3M Para/ for [14]PRB 2 layer 32T4R
    - InterDigital: 3.177K/12.7K Flops for 52PRB, SISO/8T2R
    - Vivo: 744M Flops (time domain), 6.12M Flops (Doppler domain), 8PRB, 2layers, 16T4R
  - **Reported by:** vivo (1), Lenovo, Huawei, OPPO, InterDigital

- **Option 4: AI-based joint {channel estimation + equalization + demodulation/LLR calculation}**
  - **Input**: Received signal on DMRS REs and data REs (all)
  - **Output**:
    - **Option M:** LLRs (2, 3, 4, 5)

        - Note: decoding is based on non-AI
        - **Option N:** Estimated constellation points (1,3)
        - Note: LLR calculation and decoding is based on non-AI
    - **Label:** Known sequence/data
    - **Complexity:**
        - xiaomi (Option M): 0.4M Para/473.56M Flops for 5PRB 2layer 2T2R
        - Xiaomi(Option N): 0.4M Para/ 450.89M Flops for 5PRB 2layer 2T2R
    - **Reported by:** ZTE (1), Huawei (2), Xiaomi (3), Docomo (4), Kyocera (5)

- **Option 5 : AI-based {channel estimation} + AI-based {equalization + demodulation/LLS calculation}**

For Model 1: Channel estimation

- **Input:** Received signal on superimposed signal (all)
- **Output:** Estimated channel on target REs (data REs)
- **Label:** Ideal channel information on target REs (all)

For Model 2: equalization + demodulation/LLS calculation

- **Input**: Received signal on DMRS REs and data REs (all)
- **Output**: LLRs (all)
- Note: decoding is based on non-AI
- **Label:** Known sequence/data
- **Complexity:**
    - CMCC: 0.84M Para/ 674M Flops for 10PRB 2layer
    - OPPO: 0.08M/0.11M Para / 10.67M(8PRB 1layer) 90.44M Flops (8PRB 4layer)
- **Reported by:** CMCC, OPPO, NVIDIA, LGE

For Sub-case C: DMRS free

- **Option 2: AI-based channel estimation for data REs**
    - Note: equalization, demodulation rely on non-AI
    - **Input:** Received signal
    - **Output:** Estimated channel on target REs (data REs)
    - **Label:** Ideal channel information on target REs
    - **Reported by:** Huawei

- **Option 4: AI-based joint {channel estimation + equalization + demodulation/LLS calculation}**
    - **Input**: Received signal on DMRS REs and data REs (all)
    - **Output**: LLRs
    - **Label:** Known sequence/data
    - **Complexity:**
        - MediaTek: 3.2M Para/10,856M Flops for 2T4R 6PRB; 600k Para/128M Flops for 1T8R 6PRB
        - Lenovo: 6M Para (14PRB 4layer)
    - **Reported by:** NVIDIA, MediaTek, Lenovo

- **Option 5 : AI-based {channel estimation} + AI-based {equalization + demodulation/LLS calculation}**

For Model 1: Channel estimation

- **Input:** Received signal on superimposed signal (all)
- **Output:** Estimated channel on target REs (data REs)
- **Label:** Ideal channel information on target REs (all)

For Model 2: equalization + demodulation/LLS calculation

- **Input**: Received signal on DMRS REs and data REs (all)
- **Output**: LLRs (all)
- Note: decoding is based on non-AI
- **Label:** Known sequence/data

Proposal 2.2-1B: (receiver assumptions for each sub-use cases)

For 6GR AI/ML use cases identification/categorization, for the use case of low overhead DMRS with AI/ML receiver, the following receiver assumptions are reported by companies:

For Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain
Option 1, Option 2, Option 3, Option 4, Option 5
For Sub-case B: Superimposed pilot
Option 2, Option 4, Option 5
For Sub-case C: DMRS free
Option 2, Option 4, Option 5

Receiver options:

- **Option 1: AI-based denoise on DMRS REs only**
  - Note: channel interpolation, equalization, demodulation rely on non-AI
  - **Input:** estimated channel on DMRS REs (e.g., LS)
  - **Output:** denoised channel on DMRS REs only
  - **Label:**
    - **Option M:** Self-supervise (received signal on adjacent RE as label)
    - **Option N:** supervise (Known sequence/data (end to end training)

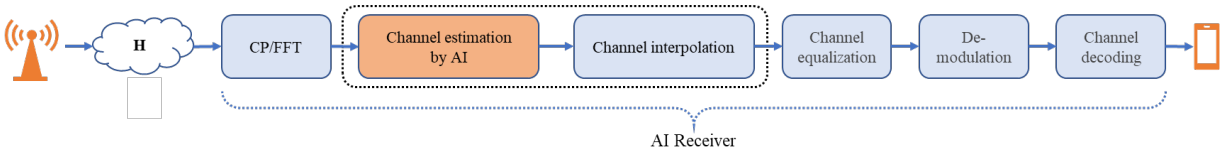
- **Option 2: AI-based channel estimation for data REs (with or without denoise on DMRS REs)**
  - Note: equalization, demodulation rely on non-AI
  - **Input:**
    - **Option A**: Received signal on DMRS REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern]
    - **Option B**: Estimated channel on DMRS REs (e.g., LS or MMSE)
    - **Option C**: Received signal on DMRS REs and data REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern] (3)
  - **Output:**
    - **Option M**: Estimated channel on target REs (data REs with/without DMRS REs)
    - **Option N:** Estimated filtering coefficients for channel estimation
  - **Label:**
    - **Option X:** Ideal channel information on target REs (all)
    - **Option Y:** Estimated channel on target REs (in any SNR or high SNR range)
    - **Option Z:** Known sequence/data (end to end training)

- **Option 3: AI based channel prediction for target REs**
  - Note: interpolation, equalization, demodulation rely on non-AI
  - **Input**: estimated channel on DMRS REs (e.g., MMSE)
  - **Output**: estimated channel on target REs (partial data REs)
  - **Label:**
    - **Option X**: estimated channel on target REs (e.g., MMSE)
    - **Option Y:** Known sequence/data (end to end training)

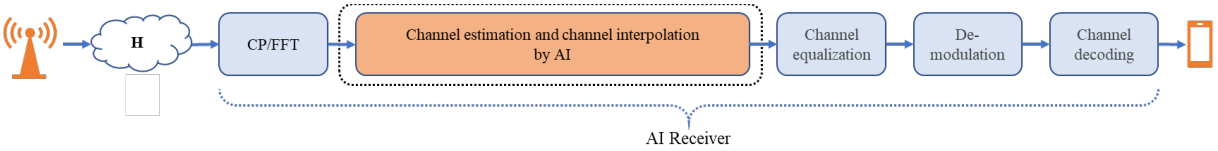
- **Option 4: AI-based joint {channel estimation + equalization + demodulation/LLR calculation}**
  - **Input**:
    - **Option A**: Received signal on DMRS REs and data REs (all)
      - Additionally: noise variance
    - **Option B**: Estimated channel on DMRS REs (e.g., MMSE)
  - **Output**:
    - **Option M:** LLRs
    - Note: decoding is based on non-AI
    - **Option N:** Estimated constellation points
    - Note: LLR calculation and decoding is based on non-AI
    - **Option O:** equalizer coefficients
  - **Label:**
    - **Option X**: Known sequence/data

- **Option Y**: label free

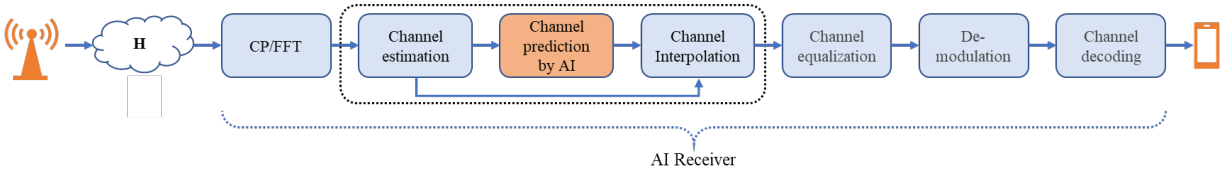
- **Option 5: AI-based {channel estimation} + AI-based {equalization + demodulation/LLR calculation}**
  - **For Model 1: Channel estimation**
    - **Input:** Received signal on superimposed signal
    - **Output:** Estimated channel on target REs (data REs)
    - **Label:**
      - **Option X:** Ideal channel information on target REs
      - **Option Y:** Estimated channel on target REs (in any SNR or high SNR range)
      - **Option Z:** Known sequence/data (end to end training)
  - **For Model 2: equalization + demodulation/LLS calculation**
    - **Input**: Received signal on DMRS REs and data REs
    - **Output**: LLRs
    - Note: decoding is based on non-AI
    - **Label:** Known sequence/data



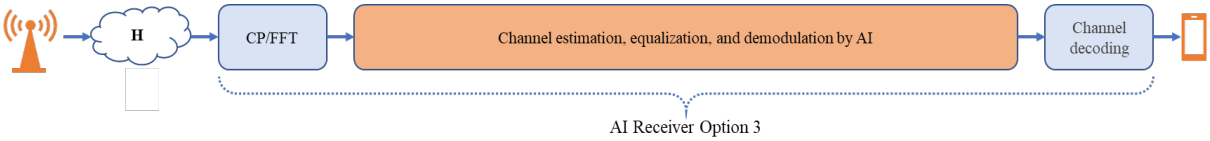
(a) AI Receiver Option 1



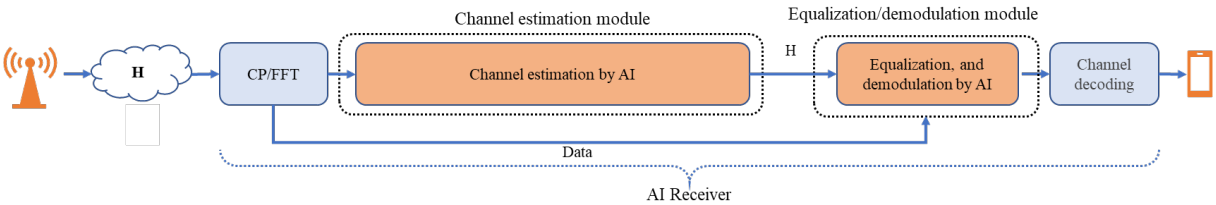
(b) AI Receiver Option 2



(d) AI Receiver Option 3



(e) AI Receiver Option 4



(f) AI Receiver Option 5

Company	Comment
FL	To facilitate potential further study, clarification on AI receiver is helpful. The intention of this proposal is to capture the above as observation, similar as the tables we agreed in previous meeting. For the highlighted part, please help to double check if my understanding is correct or not, or help to fill in the number of any (delete is also ok)

Google	For option 1 under sub-case A, if AI can only be used to cancel interference+noise, how can AI help to reduce the DMRS overhead? For option 4 under sub-case A/B/C, there seems to be a typo. “LLS” should be changed into “LLR”?
MediaTek	Added some corrections and information on number of parameters our reported complexity numbers highlighted. Corrected “Idea” to “Ideal” in a few places. It seems the figures (a)-(f) do not match to the options in the text for the AI-receivers.
Xiaomi	Thanks for FL’s great effort. We have identified a few issues regarding our results in this observation, as outlined below: ● For Option 3 of Sub-case A, we suggest to revise the complexity part as follows, where the proposed changes shown in red font: **For Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain** … • **Option 3: AI based channel prediction for target REs** o Note: (interpolation), equalization, demodulation rely on non-AI o **Input**: estimated channel on DMRS REs (e.g., MMSE) o **Output**: estimated channel on target REs ((full or) partial data REs) o **Label:** estimated channel on target REs (e.g., MMSE) o **Complexity:** 2~3M Para/up to 1M Flops for 10PRB 2layer 2T2R ▪ Note: For slot-level sparse DMRS, the AI model is invoked 0.5 times per slot; in other cases, it is invoked once per slot. o **Reported by: xiaomi** ● For Option 4 of Sub-case A, we provided the evaluation results using Option N in this meeting, but it is missed in the observation. So we revise this part as follows, where the proposed changes shown in red font: **For Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain** … • **Option 4: AI-based joint {channel estimation + equalization + demodulation/LLS calculation}** … o **Output**: ▪ **Option M:** LLRs (1,2,3,4,5, 6,7,8,9,10,11) ▪ Note: decoding is based on non-AI ▪ **Option N:** Estimated constellation points (4, 12) ▪ Note: LLS calculation and decoding is based on non-AI … o **Complexity:** ▪ Samsung: 11M Para/84M Flops for 2PRB 2T2R ▪ Lenovo: 6M Para/ for [xx]PRB 1 layer ▪ MediaTek: 128M Flops for 24PRB 1layer ▪ InterDigital (option Y): 0.01/0.129/1.05M Flops for 1/8/32 Rx ▪ LGE: 4.88M Para/4.32MFLops [xx]PRB 16Rx ▪ xiaomi (Option N): 0.3M Para/328.21M Flops for 5PRB 2layer 2T2R o **Reported by:** Nokia (1), Ericsson (2), CATT (3), Samsung (4), Kyocera (5), Lenovo (6), MTK (7), Docomo (8), NVIDIA (9), Interdigital (10), LGE (11), xiaomi (12) ● For Option 4 of Sub-case B, we provided the evaluation results using Option M in last meeting, but it is missed in the observation as well. So we revise this part as follows, where the proposed changes shown in red font: **For Sub-case B: Superimposed pilot** … • **Option 4: AI-based joint {channel estimation + equalization + demodulation/LLS calculation}** … o **Output**:

	▪ **Option M:** LLRs (2, 3, 4, 5) ▪ Note: decoding is based on non-AI ▪ **Option N:** Estimated constellation points (1,3) ▪ Note: LLS calculation and decoding is based on non-AI … o **Complexity:** ▪ xiaomi (Option M): 0.4M Para/473.56M Flops for 5PRB 2layer 2T2R ▪ xiaomi (Option N): 0.4M Para/450.89M Flops for 5PRB 2layer 2T2R … In addition, there is typo of "LLS". It should be "LLR".
Qualcomm	For orthogonal-DMRS, 1) Suggest combining Option 1,2,3, as they are pre-processing and post-processing variations of the "AI-based channel estimation based on sparse orthogonal DMRS". In general, non-AI and AI can be used in hybrid manner in many different ways, and we do not need to categorize them into many different options. 2) Suggest distinguishing between pilot-aided channel estimation and data-aided channel estimation.
vivo	We propose the following update based on our latest contribution. **For Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain** • **<u>Option 2: AI-based channel estimation for data REs (with or without denoise on DMRS REs)</u>** o Note: equalization, demodulation rely on non-AI o **Input:** ▪ **Option A**: Received signal on DMRS REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern] ▪ **Option B**: Estimated channel on DMRS REs (e.g., LS or MMSE) o **Output:** ▪ **Option M:** Estimated channel on target REs (data REs with/without DMRS REs) ▪ **Option N:** Estimated filtering coefficients for channel estimation. o **Label:** ▪ **Option X:** Ideal channel information on target REs (all) ▪ **Option Y:** Estimated channel in high SNR range (e.g. 30dB) (1) ▪ **Option Z: Estimated channel based correctly received data transmission.** o **Complexity:** ▪ CMCC: 1M Para/30M Flops for 10PRB ▪ Qualcomm: 800K Para/230M Flops for 4PRB 4layer ▪ ZTE: 7M Flops for 8PRB 4layer ▪ Samsung: 150/190/340K Para, /2.4/5.9/3.7M Flops 8 PRB2layer ▪ vivo: 11M Flops 8PRB 2layer (for direct AI/ML inferenc), 3.56M Flops for all RBs, 2 layers(for AIML generated filtering coefficients) o **Reported by: CMCC, Apple, Huawei, Qualcomm, Spreadtrum, ZTE, Samsung(1), vivo, Fujitsu** **For Sub-case B: Superimposed pilot** • **<u>Option 2: AI-based channel estimation for data REs</u>** o Note: equalization, demodulation rely on non-AI o **Input:** ▪ **Option A**: Received signal on superimposed signal (all) ▪ **Option B**: Estimated channel on superimposed signal (e.g., LS) (1) o **Output:** Estimated channel on target REs (data REs)

	○ **Label:** Ideal channel information on target REs (all) or **Estimated channel based correctly received data transmission** ○ **Complexity:** ▪ Lenovo: 3M Para/ for [xx]PRB 32T4R ▪ InterDigital: 3.177K/12.7K Flops for 52PRB, SISO/8T2R ▪ Vivo: 744M Flops (time domain), 6.12M Flops (Doppler domain), 8PRB, 2layers, 16T4R ○ **Reported by:** vivo (1), Lenovo, Huawei, OPPO, InterDigital ○
OPPO	It seems too many options for implementation-based AI/ML receivers are listed. To ease the follow-up study, we do suggest merging some receiver operations such as For AI-based channel estimation, Option 1, 2A and 2B can be categorized into one type, as they are all about channel estimation. How to interpolations (legacy or AI approach) seems implementation choice. Btw, as for channel estimation based on CSI-RS, should we also introduce such Options? If not, why only here? For AI-based joint receiver (from CE to demodulation), Option 4 and 5 can be considered as another type, since one model or two model (H transferred from one to another) also belong to implementation.
Ofinno	Added reported company in Option 2
Lenovo	The highlighted part is reported as below. **For Sub-case A:** Under Option-4 ▪ Lenovo: 6M Para/ for [14]PRB 4 layer, 4T8R **For Sub-case B:** Under Option-2 ▪ Lenovo: 3M Para/ for [14]PRB 2 layer, 32T4R **For Sub-case C** :Under Option-4 ▪ Lenovo: 6M ~~Flops~~ Para/ for 14 PRB ~~14~~layer, ~~32T4R~~ 4T8R
Futurewei	First, the studies/results we discussed in our contributions (in RAN1#122bis and RAN1#123) belong to **Option A and B under Option 2.** We experimented with 2 approaches in modeling, the first uses received signals on DMRS REs and the second one uses received signals on DMRS REs to perform rough channel estimation first, e.g., LS, then use LS outcome as input to AI/ML model. We made updates in the "Updates on observation for 6GR AIML use case identification" document. We suggest the following modifications to Option 2 (in red text below). • **Option 2: AI-based channel estimation for data REs (with or without denoise on DMRS REs)** ○ Note: equalization, demodulation rely on non-AI ○ **Input:** ▪ **Option A**: Received signal on DMRS REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern] ▪ **Option B**: Estimated channel on DMRS and/or data REs (e.g., LS or MMSE) ○ **Output:** Estimated channel on target REs (data REs with/without DMRS REs) ○ **Label:** ▪ **Option X:** Ideal channel information on target REs (all) ▪ **Option Y:** Estimated channel in high SNR range (e.g. 30dB) (1) ○ **Complexity:** ▪ CMCC: 1M Para/30M Flops for 10PRB ▪ Qualcomm: 800K Para/230M Flops for 4PRB 4layer ▪ ZTE: 7M Flops for 8PRB 4layer ▪ Samsung: 150/190/340K Para, /2.4/5.9/3.7M Flops 8 PRB2layer ▪ vivo: 4.84M Flops 8PRB 4layer ○ **Reported by: CMCC, Apple, Huawei, Qualcomm, Spreadtrum, ZTE, Samsung(1), vivo, Fujitsu, Futurewei** Another comment is that we share similar views as OPPO. There are too many options for AI/ML receivers and grouping/merging some of them together would be better.

InterDigital	On the categorization, we would like to second the suggestions from Qualcomm. The current version looks unnecessarily too fragmented and a bit convoluted to follow the distinctions. Also, some corrections to our use case and evaluation are provided below: model output for sub-case A, and complexity estimates for sub-case B. **For Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain** • **Option 4: AI-based joint {channel estimation + equalization + demodulation/LLS calculation}** o **Input**: ▪ **Option A**: Received signal on DMRS REs and data REs (all) • Additionally: noise variance (1,4) ▪ **Option B**: Estimated channel on DMRS REs (e.g., MMSE) (1) o **Output**: ▪ **Option M:** LLRs (1,2,3,4,5, 6,7,8,9,~~10~~,11) ▪ Note: decoding is based on non-AI ▪ **Option N:** Estimated constellation points (4), Equalizer coefficients (10) ▪ Note: LL**RS** calculation and decoding is based on non-AI o **Label:** ▪ **Option X**: Known sequence/data (all other than 10) ▪ **Option Y**: label free (10) o **Complexity:** ▪ Samsung: 11M Para/84M Flops for 2PRB 2T2R ▪ Lenovo: 6M Para/ for [xx]PRB 1 layer ▪ MediaTek: 600k Para/128M Flops for 6PRB 1layer ▪ InterDigital (option Y): 0.01/0.129/1.05M Flops for 1/8/32 Rx ▪ LGE: 4.88M Para/4.32MFLops [xx]PRB 16Rx o **Reported by:** Nokia (1), Ericsson (2), CATT (3), Samsung (4), Kyocera (5), Lenovo (6), MTK (7), Docomo (8), NVIDIA (9), Interdigital (10), LGE (11) **For Sub-case B: Superimposed pilot** • **Option 2: AI-based channel estimation for data REs** o Note: equalization, demodulation rely on non-AI o **Input:** ▪ **Option A**: Received signal on superimposed signal (all) ▪ **Option B**: Estimated channel on superimposed signal (e.g., LS) (1) o **Output:** Estimated channel on target REs (data REs) o **Label:** Ideal channel information on target REs (all) or power delay profile (1) o **Complexity:** ▪ Lenovo: 3M Para/ for [xx]PRB 32T4R ▪ ~~InterDigital: 3.177K/12.7K/ Flops for 52PRB, SISO/8T2R~~ ▪ **InterDigital: 3177M/12700M/76248M Flops for 52PRB, SISO/8T2R (2layer)/32x8(3layer)** ▪ Vivo: 744M Flops (time domain), 6.12M Flops (Doppler domain), 8PRB, 2layers, 16T4R o **Reported by:** vivo (1), Lenovo, Huawei, OPPO, InterDigital Question for those using “noise variance” as input to the channel. How is the noise variance calculated here? Is it based on sparse-DMRS and using non-AI/ML methods or something else (including ideal assumption), etc.?
Nokia	Changes in **Red:** **For Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain** • **Option 1: AI-based denoise on DMRS REs only** o Note: channel interpolation, equalization, demodulation rely on non-AI o **Input:** estimated channel on DMRS REs (e.g., LS) o **Output:** denoised channel on DMRS REs only

- **Label:** Self-supervise (received signal on adjacent RE as label) or supervise (Known sequence/data (end to end training)) % this is a possibility and does not have to rule out
- **Complexity:** 267K Para/4.35M Flops for 8PRB
- **Reported by: Samsung**

- **Option 2: AI-based channel estimation for data REs (with or without denoise on DMRS REs)**
  - Note: equalization, demodulation rely on non-AI
  - **Input:**
  - **Option A**: Received signal on DMRS REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern]
  - **Option B**: Estimated channel on DMRS REs (e.g., LS or MMSE)
  - Option C: Received signal on DMRS REs and data REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern] % this is a possibility and does not have to rule out
  - Option D : Option B + C % this is a possibility and does not have to rule out
  - **Output:** Estimated channel on target REs (data REs with/without DMRS REs)
  - **Label:**
    - **Option X:** Ideal channel information on target REs (all)
    - **Option Y:** Estimated channel in high SNR range (e.g. 30dB) (1)
    - Option Z: Known sequence/data (end to end training) % this is a possibility and does not have to rule out
  - **Complexity:**
    - CMCC: 1M Para/30M Flops for 10PRB
    - Qualcomm: 800K Para/230M Flops for 4PRB 4layer
    - ZTE: 7M Flops for 8PRB 4layer
    - Samsung: 150/190/340K Para, /2.4/5.9/3.7M Flops 8 PRB2layer
    - vivo: 4.84M Flops 8PRB 4layer
  - **Reported by: CMCC, Apple, Huawei, Qualcomm, Spreadtrum, ZTE, Samsung(1), vivo, Fujitsu**

- **Option 3: AI based channel prediction for target REs**
  - Note: (interpolation), equalization, demodulation rely on non-AI
  - **Input**: estimated channel on DMRS REs (e.g., MMSE)
  - **Output**: estimated channel on target REs ((full or) partial data REs)
  - **Label:**
    - Option X: estimated channel on target REs (e.g., MMSE)
    - Option Y: Known sequence/data (end to end training)
  - **Complexity:** 2~3M Para/up to 1M Flops for 10PRB 2T2R
  - Reported by: xiaomi

- **Option 4: AI-based ~~joint~~ {channel estimation + equalization + demodulation/LLR ~~LLS~~ calculation}**
  - **Input**:
    - **Option A**: Received signal on DMRS REs and data REs (all)
      - Additionally: noise variance (1,4)
    - **Option B**: Estimated channel on DMRS REs (e.g., MMSE) (1)
  - **Output**:
    - **Option M:** LLRs (1,2,3,4,5, 6,7,8,9,10,11)
    - Note: decoding is based on non-AI
    - **Option N:** Estimated constellation points (4)
    - Note: LLS calculation and decoding is based on non-AI
  - **Label:**
    - **Option X**: Known sequence/data (all other than 10)
    - **Option Y**: label free (10)
  - **Complexity:**

	▪ Samsung: 11M Para/84M Flops for 2PRB 2T2R ▪ Lenovo: 6M Para/ for [xx]PRB 1 layer ▪ MediaTek: 600k Para/128M Flops for 6PRB 1layer ▪ InterDigital (option Y): 0.01/0.129/1.05M Flops for 1/8/32 Rx ▪ LGE: 4.88M Para/4.32MFLops [xx]PRB 16Rx ▪ Nokia: 630k Para for 16PRB 2T8R 2 layers o **Reported by:** Nokia (1), Ericsson (2), CATT (3), Samsung (4), Kyocera (5), Lenovo (6), MTK (7), Docomo (8), NVIDIA (9), Interdigital (10), LGE (11) **For Sub-case C: DMRS free** • **Option 2: AI-based channel estimation for data REs** o Note: equalization, demodulation rely on non-AI o **Input:** Received signal ~~on superimposed signal~~ o **Output:** Estimated channel on target REs (data REs) o **Label:** Ideal channel information on target REs o **Reported by:** Huawei • **Option 4: AI-based ~~joint~~ {channel estimation + equalization + demodulation/LLS calculation}** o ~~—~~**Input**: Received signal ~~on DMRS REs and data REs (all)~~ o **Output**: LLRs o **Label:** Known sequence/data o **Complexity:** ▪ MediaTek: 3.2M Para/10,856M Flops for 2T4R 6PRB; 600k Para/128M Flops for 1T8R 6PRB ▪ Lenovo: 6M Flops (xx PRB 1layer, 32T4R) o **Reported by:** NVIDIA, MediaTek, Lenovo
LG Electronics	For option 4 under sub-case A, we assumed 10 PRB in our experiments. Please edit • LGE: 4.88M Para/4.32MFLops 10PRB 16Rx
ZTE	It is fine to list different receiver assumptions. But for this use case, it is premature to make observation on the model complexity due to the limited input at this stage.
Samsung	Suggest to not combine option 1/2/3 as one. For each assumption, different input/output/label are assumed. The procedure to obtain different types of labels, and the consideration aspects (e.g., generalization, scalability) are quite different for each receiver type. Therefore, some additional clarification on the receiver assumptions is needed to facilitate further study. To answer google's question. with denoise on DMRS, BLER gain is observed compared with legacy. It is possible to reduce the density of DMRS while keeping acceptable BLER performance.
FL	Two directions for consideration
Futurewei	We would like to clarify the evaluations we performed related to DRMS. We studied 3 variations: 1) First variation belongs to **Sub-case A**: Sparse orthogonal DMRS in frequency and/or time domain o Receiver assumption belongs to **Option 2:** AI-based channel estimation for data REs (with or without denoise on DMRS REs) ▪ Input: **Option A**: Received signal on DMRS REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern] ▪ Output: **Option M**: Estimated channel on target REs (data REs with/without DMRS REs) ▪ Label: **Option X:** Ideal channel information on target REs (all) 2) Second variation also belongs to **Sub-case A**: Sparse orthogonal DMRS in frequency and/or time domain o Receiver assumption also belongs to **Option 2** ▪ Input: **Option C**: Received signal on DMRS REs and data REs, [and DMRS sequence/pattern or for a given DMRS sequence/pattern] (3) ▪ Output: **Option M**: Estimated channel on target REs (data REs with/without

	DMRS REs) ▪ Label: **Option X:** Ideal channel information on target REs (all) 3) Third variation also belongs to **Sub-case A**: Sparse orthogonal DMRS in frequency and/or time domain o Receiver assumption belongs to **Option 4: AI-based joint {channel estimation + equalization + demodulation/LLR calculation}** ▪ Input: **Option A**: Received signal on DMRS REs and data REs (all) ▪ Output: **Option M:** LLRs ▪ Label: **Option X**: Known sequence/data In the current Proposed observation 2.2-1: (Clarification on receiver assumptions for each sub-use cases), our company's name is only captured in "Option 2: AI-based channel estimation for data REs (with or without denoise on DMRS REs)". We added our name to Option 4 as shown below. • **Option 4: AI-based joint {channel estimation + equalization + demodulation/LLR calculation}** o **Input**: ▪ **Option A**: Received signal on DMRS REs and data REs (all) • Additionally: noise variance (1,4) ▪ **Option B**: Estimated channel on DMRS REs (e.g., MMSE) (1) o **Output**: ▪ **Option M:** LLRs (1,2,3,4,5, 6,7,8,9,11,13) ▪ Note: decoding is based on non-AI ▪ **Option N:** Estimated constellation points (4,12), equalizer coefficients (10) ▪ Note: LLR calculation and decoding is based on non-AI o **Label:** ▪ **Option X**: Known sequence/data (all other than 10) ▪ **Option Y**: label free (10) o **Complexity:** ▪ Samsung: 11M Para/84M Flops for 2PRB 2T2R ▪ Lenovo: 6M Para/ for 14PRB 4 layer, 4T8R ▪ MediaTek: 600k Para/128M Flops for 6PRB 1layer ▪ InterDigital: 3177M/12700M/76248M Flops for 52PRB, SISO/8T2R (2layer)/32x8(3layer) ▪ LGE: 4.88M Para/4.32MFLops 10PRB 16Rx ▪ xiaomi (Option N): 0.3M Para/328.21M Flops for 5PRB 2layer 2T2R ▪ Nokia: 630k Para for 16PRB 2T8R 2 layers o **Reported by:** Nokia (1), Ericsson (2), CATT (3), Samsung (4), Kyocera (5), Lenovo (6), MTK (7), Docomo (8), NVIDIA (9), Interdigital (10), LGE (11), xiaomi(12), Futurewei (13)

Proposal 2.2-2: study aspects

For the potential study on low overhead DMRS with AI/ML receiver, at least the following aspects are to be considered:

- KPIs:
  - Performance related: Throughput, DMRS overhead, BLER, [MSE, when applicable], PAPR (when applicable)
  - Receiver complexity and latency at least for UE-sided model
  - Overhead for fine-tuning, when applicable.
  - Inter-vendor collaboration overhead and complexity for two-sided model, when applicable.
- Other aspects for study:
  - Label quality impact for DMRS design with AI receiver
  - Performance with SU-MIMO and MU-MIMO including imperfect precoder, open loop/close loop MIMO, and coexistence with non-AI capable UEs

- Scalability, e.g., frequency domain allocation, time domain allocation, number of layers, number of Tx/Rx, modulation, DMRS pattern/sequence
- Performance under non-ideal factor, at least including realistic Rnn estimation, TO, FO, phase noise, interference, other distortions (e.g., PA, IQ imbalance).
- Performance with practical inter-vendor collaboration solution for two-sided model, when appliable
- Generalization
  - Generalization performance of AI receiver over different channels, MCS levels
  - Generalization performance of AI receiver over different non-ideal factors, if the trained model is used to overcome non-ideal factors
- (Spec impact) DMRS related design and performance impact
  - Time-domain and frequency-domain sparse DMRS pattern for Sub-case A
  - DMRS sequence/pattern design, power allocation for Sub-case B
  - RE mapping, constellation design for Sub-case C
- (Spec impact) Whether/what signaling/procedure is needed for
  - data collection
  - inference
  - performance monitoring
  - inter-vendor collaboration, if applicable and justified
  - fine-tuning, if feasible and justified
- Note: the study and analysis consider the model location (i.e., UE-sided model, NW-sided model)

Company	Comment
FL	The proposal is to listing the potential AI/ML specific consideration for AI receiver. For spec impact part, it is not clear on whether LCM related procedure is needed or not.
Google	As we mentioned in last meeting, the potential spec impact should include the CSI calculation, unless we only study UL DMRS related aspects.
Apple	For performance evaluation, DL SU MIMO with open loop and close loop MIMO should be studied. For models option 1/2/3/5, channel estimation MSE can be added as KPI, on top of BLER. For DL, similar to CSI-RS, coexistence with non-AI capable UE needs to be studied for DL MU transmission as part of other aspects to study.
Huawei, HiSilicon	For KPI, 1) UL PAPR should be considered, since the new DMRS/data multiplexing would introduce PAPR increasing issue. The corresponding power backoff may impact the performance unless the PAPR issue is resolved. 2) The receiver complexity is mainly for UE; NW does not have such strong restriction on model complexity. 3) The performance for both UL and DL should consider SU-MIMO and MU-MIMO (both of which may occur in real network), to justify the applicability. • KPIs: o Performance related: Throughput, DMRS overhead, BLER, UL PAPR o Receiver complexity and latency (for UE side model) • Other aspects for study: o Label quality impact for DMRS design with AI receiver o Performance with uplink SU/MU-MIMO and downlink SU/MU-MIMO including imperfect precoder Benchmark should at least consider NR DMRS/data MUX pattern, since we do not have 6GR fundamental DMRS design yet. Why do we need to consider “Coexistence with the UE with non-AI based receiver”? AI and non-AI are anyway subject to separate processing.

	Regarding inter-vendor collaboration, we may focus on one-sided model for initial study.
CATT	For KPIs, raw BER can be added as an intermediate KPI. For generalization, the performance over different MCS level needs to be studied.
Xiaomi	Support in general. For receiver complexity of KPI, we think that how to compare it in a fair manner requires further study. As we known, the complexity of an AI model depends on factors such as the number of layers, the number of Rx, the number of RBs, and the number of operations or inference runs. The resulting complexity can vary significantly under different assumptions. Therefore, using an averaged complexity metric may be a reasonable method, e.g., Flops averaged per layer, per Rx, per RB and per slot. For the fine-tuning, what kind of fine-tuning is assumed here. If it is offline fine-tuning, we don't think we need to consider the overhead. If it is the online fine-tuning, the feasibility and benefits should be studied fisrt. In addition, the overhead mentioned here is also unclear. Regarding the "non-ideal factor" mentioned in the proposal, what does "inference" refer to? Do you actually mean "interference"? For the spec impacts of Sub-case A, in addition to DMRS pattern design, we suggest to further study the following two aspects: ● For frequency-domain sparse DMRS, study power boost enhancement when a low density DMRS in the frequency domain is used ● For time-domain sparse DMRS, study how to maintain phase continuity and power consistency of DMRS across slots.
SPRD	For KPIs, MSE can be added as an intermediate KPI.
OPPO	For the study, it seems we are introducing too many imperfect factors into the study. Of course, everyone wants the study to go as practically as possible. However, on the other hand, we need to consider the efforts to conduct the study. For instance, the following bullets can be removed, if not strongly justified. We understand the reason the imperfect precoder would impact data transmission, but it is up to MIMO (precoding), not the problem of effective channel estimation. ~~o Performance with uplink MIMO and downlink MU-MIMO including imperfect precoder~~ If we have put the following ones into consideration, then why not put it under CSI-RS overhead reduction, too? Both cases somehow face the similar imperfect situations. ~~Performance under non-ideal factor, at least including realistic Rnn estimation, TO, FO, phase noise, inference, other distortions (e.g., PA, IQ imbalance).~~
Lenovo	We suggest to add the following to the list of spec impact. We believe it should have the same weight as the other sub-cases and more accurate study can be conducted during the SI. ~~o~~ (Spec impact) DMRS related design and performance impact ▪ Time-domain and frequency-domain sparse DMRS pattern design for Sub-case A ▪ DMRS sequence/pattern design, power allocation for Sub-case B ▪ DMRS overhead reduction, RE mapping, constellation design for Sub-case C
Futurewei	At least for the option(s) that perform channel estimation, e.g., Option 2: AI-based channel estimation for data REs (with or without denoise on DMRS REs), MSE can be used as intermediate KPI as the labels of Option X and Option Y are ideal/estimated channels. • **Option X:** Ideal channel information on target REs (all) • **Option Y:** Estimated channel in high SNR range (e.g. 30dB) (1) We suggest the following changes: • KPIs:

	○ Performance related: Throughput, DMRS overhead, BLER, MSE ○ Receiver complexity and latency ○ Overhead for fine-tuning, when applicable. ○ Inter-vendor collaboration overhead and complexity for two-sided model, when applicable.
Panasonic	Regarding "overhead for fine-tuning", we think it might not be early to be considered as companies almost assumed offline training and we do not have any observation about it.
Fujitsu	For KPI, we suggest to include intermediate KPI, such as NMSE.
InterDigital	As for the CSI prediction and CSI-RS OH reduction case, for DM-RS OH reduction as well, we do not see a compelling reason to include consideration of two-sided models. Thus, the "inter-vendor collaboration" bullet should be removed from spec-impact bullets. Similarly, under KPIs, the bullet on "fine-tuning" is not acceptable at this stage and should be updated to refer to the OH associated with at least performance monitoring instead. • KPIs: ○ Performance related: Throughput, DMRS overhead, BLER ○ Receiver complexity and latency ○ Overhead for at least performance monitoring, when applicable. ○ ~~Inter-vendor collaboration overhead and complexity for two-sided model, when applicable.~~ Also, support CATT's suggestion to study generalization performance across different MCS levels.
Nokia	Some changes. • Benchmarks were listed last time. Does not have to list FFS. • Other aspects will create a debate. Let's keep the proposal simple by listing KPIs. For the potential study on low overhead DMRS with AI/ML receiver, at least the following aspects are to be considered: • KPIs: ○ Performance related: Throughput, DMRS overhead, BLER ○ Receiver complexity and latency ○ Overhead for fine-tuning, when applicable. ○ Inter-vendor collaboration overhead and complexity for two-sided model, when applicable. • ~~Benchmark~~ ○ ~~FFS on DMRS pattern assumption, receiver assumption, channel estimation assumption~~ • ~~Other aspects for study:~~ ○ ~~Label quality impact for DMRS design with AI receiver~~ ○ ~~Performance with uplink MIMO and downlink MU-MIMO including imperfect precoder~~ ○ ~~Scalability, e.g., frequency domain allocation, time domain allocation, number of layers, number of Tx/Rx, modulation, DMRS pattern/sequence~~ ○ ~~Performance under non-ideal factor, at least including realistic Rnn estimation, TO, FO, phase noise, inference, other distortions (e.g., PA, IQ imbalance).~~ ○ ~~Performance with practical inter-vendor collaboration solution for two-sided model, when appliable~~ ○ ~~Generalization~~ ▪ ~~Generalization performance of AI receiver over different channels~~ ▪ ~~Generalization performance of AI receiver over different non-ideal~~

	~~factors, if the trained model is used to overcome non-ideal factors~~ ~~o Coexistence with the UE with non-AI based receiver, if applicable~~ ~~o (Spec impact) DMRS design and performance impact~~ ~~▪ Time-domain and frequency-domain sparse DMRS pattern for Sub-case A~~ ~~▪ DMRS sequence/pattern design, power allocation for Sub-case B~~ ~~o (Spec impact) Whether/what signal/procedure is needed for~~ ~~▪ data collection~~ ~~▪ inference~~ ~~▪ performance monitoring~~ ~~▪ inter-vendor collaboration, if applicable and justified~~ ~~Note: the study and analysis consider the model location (i.e., UE-sided model, NW-sided model)~~
Ericsson	**<u>On KPI:</u>** We don't think 'overhead for fine-tuning' should be included as a KPI since most of the companies assume offline training. In addition, it is not clear what overhead means here, and how to quantify it. Also, user perceived throughput should be the primary KPI for making conclusions/decisions. We don't think conclusions/decisions should be based on intermediate KPI such as MSE and raw BER. **<u>On other aspects for further study:</u>** What does "Coexistence with non-AI capable UE" mean?? **<u>On two-sided models:</u>** Sub use cases based on two-sided models shall be deprioritized in 6GR Rel-20 study, due to its high complexity in inter-vendor collaboration for training/testing, and to avoid overlapping discussion between NR and 6GR on two-sided AI/ML model. Suggest the following update: For the potential study on low overhead DMRS with AI/ML receiver, prioritize one-sided model use cases, at least the following aspects are to be considered: **<u>On spec impacts:</u>** RAN 4 aspects are missing: e.g., demod requirement
LG Electronics	We also suggest to include intermediate KPI, e.g., NMSE. For initial study, one-side model can be studied first.
ZTE	Regarding the following bullet, we believe both DL and UL MIMO should consider MU-MIMO. o *Performance with uplink MIMO and downlink MU-MIMO including imperfect precoder*
Samsung	KPI: For receiver complexity and latency, even if NW side does not have such strong restriction on model complexity, it is still meaningful to study the impact of introducing AI receiver to PUSCH processing time. The complexity is related to varies aspects, e.g., number of RBs, number of layers. It is suggested to further study how to compare the complexity under different settings/assumptions. For Inter-vendor collaboration, suggest to study one-sided model first. Other aspects: Suggest to keep the sub-bullet: o Performance with uplink MIMO and downlink MU-MIMO including imperfect precoder Non-ideal precoder introduces cross-layer interference. Current research has not sufficiently addressed whether the model remains robust when the precoder changes

IMU	When using an AI/ML receiver to obtain low-overhead DMRS, it is necessary to consider the tradeoff if, on the other hand, there will be additional computational load or fine-tuning overhead. Moreover, the resulting solutions will be somewhat subjective unless we define the hardware-related and performance-related metrics for fine-tuning overhead. In its current form, we do not support including fine-tuning overhead. Moreover, we think that the impact of inter-vendor collaboration should be deprioritized at this stage.

Proposal 2.2-3: Labeling acquisition requirement

For the potential study on low overhead DMRS with AI/ML receiver, the following can be considered for label acquisition:

- For receiver option 1, option 2, option 3 and option 5 (model for channel estimation):
  - Option X: genie channel information on target REs
    - Label can be generated according to a certain channel model
      - FFS on the generalization performance to practical/unseen channel model to verify the feasibility
  - Option Y: Estimated channel on target REs
    - Label can be obtained by known DMRS/data received in high SNR range
  - [Option Z: based on the known sequence/data (end to end training)
- For receiver option 4 and option 5(model for equalization +demodulation/LLS calculation):
  - Option X: Known sequence/data
    - Label can be generated by the known TB/DMRS at receiver, e.g., TB passed CRC or predefined sequence
  - Option Y: label free
    - Data-aided joint channel estimation and equalization using unsupervised learning at the receiver that utilizes repetition of some data REs within the allocated resources

Company	Comment
FL	To facilitate potential further study, clarification on label acquisition is helpful. This is to provide some considerations on label acquisition, which is unique to AI receiver than other use cases.
Google	Can label be obtained by other signals, e.g., CSI-RS? Anyway, the label is to identify a channel. CSI-RS is one RS to identify the channel.
Apple	For option 2 where model used for channel estimation, data tone channel can be obtained via decision directed method, based on decoded PDSCH.
MediaTek	Question for clarification: for AI-based channel estimation based on DMRS in the DL, how do the reported companies intend to obtain the labeled data in practice, since the estimated channel may be precoded with an unknown precoder?
Huawei, HiSilicon	What is the difference between Option 1 and Option 2? Both seem need estimated channel under good channel status (e.g., high SNR, interference free, etc.).
CATT	We suggest the following update: • For receiver option 4 and option 5(model for equalization +demodulation/LLS calculation) o Option X: Known sequence/data ▪ Label can be generated by the known TB/DMRS at receiver, e.g., TB passed CRC, or predefined sequence
xiaomi	For receiver option 3, although the label used in our evaluation is the estimated channel, the genie channel information or estimated channel in high SNR can also be used to improve receiver performance. So we suggest to add “receiver option 3” in the second bullet for further study, i.e., For the potential study on low overhead DMRS with AI/ML receiver, the following can be considered for label acquisition: … • For receiver option 2, option 3 and option 5 (model for channel estimation): …
Qualcomm	Can we clarify that the label acquisition is not needed for training based on synthetic data? It is TBD whether models trained on synthetic data can generalize well in the field.

vivo	We would like to update as following ***Proposal 2.2-3: Labeling acquisition requirement*** For the potential study on low overhead DMRS with AI/ML receiver, the following can be considered for label acquisition: • For receiver option 1 and option 3: estimated channel o Label can be obtained by via conventional LS or MMSE • For receiver option 2 and option 5 (model for channel estimation): o Option X: genie channel information on target REs ▪ Label can be generated according to a certain channel model • FFS on the generalization performance to practical/unseen channel model to verify the feasibility o [Option Y: Estimated channel on target REs in high SNR ▪ Label can be obtained by known DMRS/data received in high SNR range • FFS on the performance, and how to verify the quality of the label] o Option Z: **Estimated channel based correctly received data transmission.** • For receiver option 4 and option 5(model for equalization +demodulation/LLS calculation) o Option X: Known sequence/data ▪ Label can be generated by the known TB/DMRS at receiver, e.g., TB passed CRC o Option Y: label free ▪ FFS on the feasibility and performance
OPPO	Too early to list all labeling methods, it can be up to companies' report. Normally, the label should experience better quality than all-whether measurements of UEs, otherwise how to train a good model? Above combinations of receiver options and labeling options may not fully cover the uses companies adopted in their evaluations. For channel estimation, the label can be nearly ideal under high SNR measurement.
Ofinno	Similar view with Huawei. Regarding acquisition of the label, the label is generated based on synthetic manner or measurement-based?
Fujitsu	Same question as Qualcomm.
InterDigital	The numberings for the "receiver options" are somewhat confusing and difficult to follow from Proposed Observation 2.2-1. Please correct the description for Option Y as below: • For receiver option 4 and option 5(model for equalization +demodulation/LLS calculation) o Option X: Known sequence/data ▪ Label can be generated by the known TB/DMRS at receiver, e.g., TB passed CRC o Option Y: label free ▪ ~~FFS on the feasibility and performance~~ Data-aided joint channel estimation and equalization using unsupervised learning at the receiver that utilizes repetition of some data REs within the allocated resources The feasibility and performance of the scheme have been demonstrated in our tdoc (R1-2508515), just like all other proposals being considered here.
Nokia	Some updates shall be made based on the 2.2.-1. So, better to wait on this proposal.
Ericsson	Support CATT's addition, "pre-defined sequence" as label.
Samsung	For the model using channel information as label, the quality of the label will have impact on the performance. Some investigation on the label quality impact is needed.

2.3 CSI compression and feedback

Proposal 2.3: study aspects

For the potential study on CSI compression and feedback, at least the following aspects are to be considered:

- KPIs
  - Intermediate KPIs: SGCS, NMSE
  - System KPIs: throughput, CSI feedback overhead (e.g., number of UL resource elements, number of bits)
  - Model/computational complexity at NW side
  - Model/computation complexity and computation latency at UE side
  - Inter-vendor collaboration overhead and complexity for two-sided model, if applicable.
  - Overhead of downloadable basis/codebook/compression matrix for NW-sided model, if applicable.
- Benchmark
  - NR SSCC approach with 2-sided model can be additionally considered in additional to other 6GR non-AI CSI feedback benchmark (if any)
- Study aspects:
  - Performance under fading channel with practical channel estimation
  - Generalization performance
    - Different NW-sided assumptions
    - Different UE side additional conditions, when applicable
    - Different UL fading channels (for Sub-case A and Sub-case B)
    - Note: for all Sub-Cases: generalization issue studied in NR are referred as baseline
  - Scalability: e.g., number of antenna ports, subband number, UL resources for UCI (when applicable), number of payloads(when applicable)
    - Note: for all Sub-Cases: Scalability studied in NR are referred as baseline
  - Performance impacted by non-ideal factors, such as channel estimation error, PA non-linearity, phase noise, IQ imbalance, interference measurement
  - For Sub-case A and Sub-case B: whether/how to introduce explicit reliability check mechanism
  - (Spec impact)
    - For Sub-case A: necessary signaling/ procedure to support JSCC transmission
    - For Sub-case B and Sub-case C: singling/procedure for downloadable compression matrix,
    - For Sub-case B: necessary signaling/ procedure to support JSCM transmission, including rate matching mechanism, multiplexing mechanism
  - (Spec impact) signaling/procedure is needed for
    - data collection
    - inference
    - performance monitoring
    - inter-vendor collaboration, if applicable and justified

Note: Depending on the 5GA progress, whether/how to support two-sided model can be decided in the later phase of Rel-20 6GR study.

Company	Comment
FL	The proposal is to listing the potential AI/ML specific consideration for AI receiver. For spec impact part, it is not clear on whether LCM related procedure is needed or not.
Google	In our view, we should firstly agree to study CSI compression. Currently we do not see a good reason to study it in the first release for 6G. Based on the lessons from 5G, AI/ML based CSI compression requires a lot of Tus. We need to notice that 6G has to study a lot of non-AI related aspects. It is not practical to study all of them.
Huawei,	CSI compression with two-sided model is being discussed at 5GA. For 6GR study, we may focus

HiSilicon	on NW-sided model only to avoid duplication.
CATT	For Sub-Case A and Sub-Case B, legacy rate-matching (including puncturing and shortening) are no longer applicable. Whether/how to multiplex with other UCI and/or with UL-SCH is also questionable. We suggest the following update: o For Sub-case A and Sub-case B: whether/how to introduce explicit reliability check mechanism, rate matching mechanism, multiplexing mechanism
Xiaomi	We should try to address the overlap between 5G-A and 6G for two-sided model. Due to it's complexity on inter-vendor collaborations, it's better to decide in the later phase of 6GR study on whether/how to support two-sided model depending on the 5GA progress. During the initial phase of 6GR study, we should focus on CSI compression use cases that don't not require inter-vendor collaborations on model training. JSCC and JSCM need a totally different PHY procedure from legacy SSCC, which may further complicate the evaluations and spec impact discussions. In come cases, SSCC performance may be improved via link adaption. Thus, SSCC approach should be considered as a baseline for further study. The following changes are proposed: For the potential study on CSI compression and feedback, at least the following aspects are to be considered: • KPIs o Intermediate KPIs: SGCS, NMSE o System KPIs: throughput, overhead o Model complexity at NW side o Model/computation complexity and computation latency at UE side ~~o Inter-vendor collaboration overhead and complexity for two-sided model, if applicable.~~ o Overhead of downloadable basis/codebook/compression matrix for NW-sided model, if applicable. • Study aspects: o Performance under fading channel with practical channel estimation (for Sub-case A and Sub-case B) o Generalization performance ▪ Different NW-sided assumptions ▪ Different UE side additional condition, when applicable ▪ Different UL channel (for Sub-case A and Sub-case B) ▪ Note: for all Sub-Cases: generalization issue studied in NR are referred as baseline o Scalability: e.g., number of antenna ports, subband number, UL resources for UCI (when applicable), number of payloads(when applicable) ▪ Note: for all Sub-Cases: Scalability studied in NR are referred as baseline o Performance impacted by non-ideal factors, such as channel estimation error, PA non-linearity, phase noise, IQ imbalance, interference measurement o For Sub-case A and Sub-case B: whether/how to introduce explicit reliability check mechanism o During the study of Sub-case A and Sub-case B in 6GR, the evaluation performances and specification impacts/complexity should be compared with SSCC approach. o (Spec impact) ▪ For Sub-case B and Sub-case C: singling/procedure for downloadable compression matrix, ▪ For Sub-case B: necessary signaling/ procedure to support JSCM transmission o (Spec impact) signal/procedure is needed for ▪ data collection

	▪ inference ▪ performance monitoring ▪ ~~inter-vendor collaboration, if applicable and justified~~ Note: Depending on the 5GA progress, whether/how to support two-sided model can be decided in the later phase of Rel-20 6GR study.
Qualcomm	We suggest to include the following in performance impacted by non-ideal factors. For Performance impacted by non-ideal factors, such as channel estimation error, PA non-linearity, phase noise, IQ imbalance, interference measurement, impact to waveform characteristics and potential Tx power back-off to meet RAN4 requirements.
CMCC	For the spec impact on CSI compression and feedback, we think the following should also be added: ● (Spec impact) For Sub-case A: necessary signaling/ procedure to support JSCC transmission
vivo	We would like to clarify that it is controversial do directly categorize the JSCM as NW sided model. We also view it as a NW-sided model. ***Proposal 2.3: study aspects*** For the potential study on CSI compression and feedback, at least the following aspects are to be considered: • KPIs o Intermediate KPIs: SGCS, NMSE o System KPIs: throughput, overhead o Model complexity at NW side o Model/computation complexity and computation latency at UE side o Inter-vendor collaboration overhead and complexity for two-sided model, if applicable. o Overhead of downloadable basis/codebook/compression matrix ~~for NW-sided model,~~ if applicable.
Panasonic	It seems there are many points that are overlapped with Rel. 19 CSI compression. We should avoid duplicated work in 5G-A and 6G. Can FL clarify on your intention for this proposal?
Fujitsu	Some clarification is needed for the generalization part. What does it mean by "Different UL channel" and why this should be considered for generalization? o Generalization performance ▪ Different NW-sided assumptions ▪ Different UE side additional condition, when applicable ▪ Different UL channel (for Sub-case A and Sub-case B) Note: for all Sub-Cases: generalization issue studied in NR are referred as baseline
InterDigital	We need to first agree whether we study CSI compression for 6GR in R20 and even so, whether to study two-sided models for this purpose. This is currently being worked on for 5GA with still open aspects (e.g., those requiring SA feedback related to solutions for inter-vendor collaboration, etc.), and the achievable performance for fully specified encoder-decoder reference model options are also pending outcome of RAN4 work on this. Accordingly, studies on two-sided models should be deferred to a later time once 5GA design is stable and available for 3GPP to learn from first. Otherwise, just mimicking what we did for 5GA may not be a good idea for 6GR. We could be more open to cases where AI/ML may be used for the CB/downloadable basis design that does not use online inferencing, and such cases can be directly considered within MIMO (with use of AI/ML as a tool) when CB design is considered in MIMO for non-AI/ML track, and not necessarily as an AI/ML use case.

Ericsson	Sub use cases based on two-sided models shall be deprioritized in 6GR Rel-20 study, due to its high complexity in inter-vendor collaboration for training/testing, and to avoid overlapping discussion between NR and 6GR on two-sided AI/ML model. Suggest the following updates: For the potential study on CSI compression and feedback, prioritize one-sided model use cases, at l east the following aspects are to be considered: • KPIs o Intermediate KPIs: SGCS, NMSE o System KPIs: throughput vs. CSI feedback overhead o Model/computational complexity at NW side o Model/computation complexity and computation latency at UE side ~~o Inter-vendor collaboration overhead and complexity for two-sided model, if applicable.~~ o Overhead of downloadable basis/codebook/compression matrix for NW-sided model, if applicable. • Study aspects: o Performance under fading channel with practical channel estimation ~~(for Sub-case A and Sub-case B)~~ o Generalization performance ▪ Different NW-sided assumptions ▪ Different UE side additional condition, when applicable ▪ Different UL channel (for Sub-case A and Sub-case B) ▪ Note: for all Sub-Cases: generalization issue studied in NR are referred as baseline o Scalability: e.g., number of antenna ports, subband number, UL resources for UCI (when applicable), number of payloads(when applicable) ▪ Note: for all Sub-Cases: Scalability studied in NR are referred as baseline o Performance impacted by non-ideal factors, such as channel estimation error, PA non-linearity, phase noise, IQ imbalance, interference measurement o For Sub-case A and Sub-case B: whether/how to introduce explicit reliability check mechanism o (Spec impact) ▪ For Sub-case B and Sub-case C: singling/procedure for downloadable compression matrix, ▪ For Sub-case B: necessary signaling/ procedure to support JSCM transmission o (Spec impact) signal/procedure is needed for ▪ data collection ▪ inference ▪ LCM aspects for one-sided model use cases, e.g., performance monitoring ▪ ~~inter-vendor collaboration, if applicable and justified~~
Samsung	We sympathize with other companies concern on workload of two-sided model. For efficiency, the NR study on two-sided model including issues related to inter-vendor collaboration can be leverages so that the study would focus on new proposals, e.g., NW-side model. Note: the NR's study outcomes on two-sided models including for inter-vendor collaboration can be considered to avoid repetition. [Gilwon] Also, since for the case of JSCM, number of UL resource elements (REs) is a key measure to indicate overhead, which is different from the conventional payload overhead,

	bits, we suggest to add the following: • KPIs o Intermediate KPIs: SGCS, NMSE o System KPIs: throughput, overhead (e.g., number of UL resource elements, number of bits)
OPPO	For the KPI part, the inter-vendor collaboration for two-sided model is being studied in 5G-A R20. The relationship between inter-vendor collaboration of 5G-A and 6GR should be clarified firstly. Whether to follow the same inter-vendor collaboration framework or design new inter-vendor collaboration framework for 6GR can be considered as different ways. In our opinion, in current study phase of 6GR, we prefer to reuse the intern-vendor collaboration framework studied/specified in 5G-A as a baseline. In later phase of 6GR, we can consider a new way from this aspect. For the study aspects, 1 the first bullet, the fairness between different sub-use cases should be ensured. The performance under fading channel with practical channel estimation should be considered for every sub-use case and possible baselines (e.g., eType II, SSCC in 5G-A), instead of sub-case A and usb-case B only. 2 for spec impact, for sub-case A, necessary signaling / procedure to support JSCC transmission should be studied, since it follows different framework to legacy CSI.

2.4 AI for beam management and extension

Proposal 2.4: study aspects

For the potential study on beam management and extension with AI/ML, at least the following aspects are to be considered:

- KPIs:
  - Performance related: RS overhead, prediction accuracy, throughput
  - Model complexity
- Other aspects for study:
  - Generalization
    - Sub-Case A: deployment scenarios, cell/TRP configurations
    - Sub-Case B: frequency domain factors (e.g., frequency domain frequency domain position_ frequency gap between the source and target frequency bands)
    - Sub-Case E: UE location (for UE-side model)
    - Note: for all Sub-Cases: generalization issue studied in NR are referred as baseline
  - Coexistence with non-AI capable UE for UE-sided model
  - (Spec impact) Initial access related to beam prediction (Sub-Case D)
  - (Spec impact) signaling/procedure for
    - data collection
    - inference
    - performance monitoring
    - handling of NW-side additional conditions, if applicable
  - Note: the study and analysis consider the model location (i.e., UE-sided model, NW-sided model)

Company	Comment
FL	The proposal is to listing the potential AI/ML specific consideration for AI receiver. For spec impact part, it is not clear on whether LCM related procedure is needed or not.
Google	We think the UE location related should be removed. We failed to see the necessity to include this part.
Apple	Agree with Google's comment. Remove UE location.
Huawei,	To clarify, Sub-Case E: UE location (for UE-side model)

HiSilicon	Why do we need to consider "Coexistence with non-AI capable UE"?
Xiaomi	For generalization of Sub-Case B, support the following update ▪ Sub-Case B: frequency domain factors (e.g., frequency domain position, or frequency gap between the source and target frequency bands)
Qualcomm	• Similar view as Google that UE location-related info should be removed for Sub-Case E. • For both sub cases A and B, since prediction of DL Tx beams are considered, generalization across different NW codebooks should also be considered for UE-side models, similar to Rel-18. • As with the other use cases, fine-tuning should also be added to the potential spec impact.
CMCC	For Inter-Cell beam prediction in Sub-case A, the model input is Set B measurement of Cell 1, and model output is Set A measurement of Cell 2. • So the generalization aspects for Sub-Case A are not mandatory "multi-cell configurations", we can simply say: "Sub-Case A: deployment scenarios, ~~multi-~~cell/TRP configurations".
Vivo	Could anyone clarify the motivation for adding UE location for generalization considerations?
Fujitsu	Same question as Huawei, "Coexistence with non-AI capable UE" is not needed for this case.
InterDigital	Agree with Google and others on removing "UE located-related info". We see a lot of overlap between the inter-cell/inter-TRP extension and the AI/ML-based mobility studies that recently concluded in RAN2 with normative work to start for R20 5GA soon. Thus, we do not think it would be appropriate to initiate parallel studies for 6GR. For initial access related use case, it expands the applicability to beyond RRC_CONNECTED to RRC_IDLE and INACTIVE states. There is strong dependency on the basic (non-AI/ML) initial access design (studies on which has not yet been started in RAN1). Further, the impact from complications regarding LCM procedures when extending to RRC_IDLE clearly affect WGs beyond RAN1 as well as raise feasibility questions that need to be addressed first in general. Thus, we do not think RAN1 is ready to initiate studies on this sub-use case at this point. Further, we cannot simply reuse results from 5GA as the design for BM is entirely open for 6GR and it is unclear if the 5GA results of comparison to benchmarks would be relevant. In summary, we think that the study of AI/ML-based BM and its extensions can be considered at a later stage in the Rel-20 6GR SI or at a subsequent release once designs for non-AI/ML-based BM and non-AI/ML-based initial access and details of UE behavior/expectation across RRC states are established for 6GR.
Ericsson	We agree with other companies' comment that UE location info should be removed from sub-case E.
LG Electronics	We also think that UE location should be deleted. Another two comments: - For generalization, Sub-Case D should be also included. - For coexistence with non-AI capable UE, Sub-Case D should be included since it is regarding initial access procedure when AI capable UE and non-AI capable UE are co-existed, but Sub-Case A/B seems not sure to be included.
Samsung	Ok to remove UE location of Sub-Case E in generalization aspect.

2.5 AI based (De-)Modulation

Proposed 2.5:

For the potential study on AI-based (de)modulation [and precoding],

- For constellation design, constellation designed with AI can be potentially study together with non-AI constellation.

- For AI receiver with constellation (and precoding) design, if justified, it can be studied together with DMRS-free use case.
-
- at least the following aspects are to be considered:
    - KPIs:
        - Performance related: Throughput, BLER
        - Receiver complexity and latency
        - Inter-vendor collaboration overhead and complexity for two-sided model, when applicable.
    - Other aspects for study:
        - Performance with (MU-)MIMO
        - Scalability, e.g., different frequency domain allocation, different time domain allocation, different number of layers, different modulation, different DMRS pattern/sequence
        - Performance gain under non-ideal factor, at least including realistic Rnn estimation, TO, FO, phase noise, inference, other distortions (e.g., PA, IQ imbalance).
        - Performance with practical inter-vendor collaboration solution for two-sided model, when appliable
        - Generalization
            - Generalization performance of AI receiver over different channels
            - Generalization performance of AI receiver over different non-ideal factors, if the trained model is used to overcome non-ideal factors
        - Coexistence with the UE with non-AI based receiver, if applicable
        - (Spec impact) modulation design
        - (Spec impact) Whether/what signal/procedure is needed for
            - data collection
            - inference
            - performance monitoring

Company	Comment
FL	In general, I think if proposal 1 can be agreed, no need to spend time on this. AI receiver needs to be justified.
Google	We do not think this should be studied in the first release of 6G
Apple	Agree with 1st bullet for constellation design. For AI receiver with constellation design, we think it is part of DMRS-less already. For joint constellation with precoding, it is more related to MIMO than modulation, as modulation agenda does not discuss precoder design. Suggest to remove the 2nd bullet.
Huawei, HiSilicon	We can focus on the 1st bullet for constellation design. This can be study together with non-AI constellation.
CMCC	If the constellation design is just produced with the help of AI, and then the constellation is specified or transferred to NW/UE side, inference may not be needed for gNB/UE. So, we suggest modifying: ▪ (Spec impact) Whether/what signal/procedure is needed for • data collection • inference, if applicable • performance monitoring
vivo	We would like to update as following: directly concluding non-AI specific aspects have been identified seems too rush, thus can be deleted for now. The second part is enough. ***Proposed 2.5:*** For the potential study on AI-based (de)modulation [and precoding], • For constellation design, ~~no AI specific aspects have been identified and~~ constellation designed with AI can be potentially study together with non-AI constellation. • For AI receiver with constellation (and precoding) design, if justified, at least the following aspects are to be considered:

	○ KPIs: ▪ Performance related: Throughput, BLER ▪ Receiver complexity and latency ▪ Inter-vendor collaboration overhead and complexity for two-sided model, when applicable. ○ Other aspects for study: ▪ Performance with (MU-)MIMO ▪ Scalability, e.g., different frequency domain allocation, different time domain allocation, different number of layers, different modulation, different DMRS pattern/sequence ▪ Performance gain under non-ideal factor, at least including realistic Rnn estimation, TO, FO, phase noise, inference, other distortions (e.g., PA, IQ imbalance). ▪ Performance with practical inter-vendor collaboration solution for two-sided model, when appliable ▪ Generalization • Generalization performance of AI receiver over different channels • Generalization performance of AI receiver over different non-ideal factors, if the trained model is used to overcome non-ideal factors ▪ Coexistence with the UE with non-AI based receiver, if applicable ▪ (Spec impact) modulation design ▪ (Spec impact) Whether/what signal/procedure is needed for • data collection • inference • performance monitoring
OPPO	For AI receiver with AI-designed constellation, its purpose is clearly not DMRS OH reduction. It targets better transmission, e.g. lower BLER or high throughput. For the 2nd bullet, the modulation and precoding can be jointly trained and operated to boost spectrum efficiency. We admit precoding in NR and LTE was considered as one of MIMO tech, but it would not be bother to study them jointly.
Lenovo	We are fine with the first main bullet. We are of the opinion that “AI based modulation plus precoding” should be considered under the MIMO agenda item, while AI-based modulation belongs to the “Modulation” agenda item.
Fujitsu	Generally fine with the proposal.
InterDigital	Same view as Google. Various aspects of constellation design, including UC, NUC (PCS/GCS) are under study already. It is not clear if parallel studies on AI/ML-based constellation design need to be started in parallel. Also, for the sub-case of using AI/ML as a tool to design constellation (no online inferencing), this can be directly considered in the modulation agenda, without defining as an AI/ML use case.
Ericsson	• Remove “when applicable” for **two** Inter-vendor collaboration bullets. It is known to be applicable if two-sided model. Also the upper level bullet has “if justified” already.

2.6 AI for DPoD/DPD/non-linearity compensation

Proposed 2.6:
For the potential study on AI for DPoD/DPD/non-linearity compensation, at least the following aspects are to be considered:

- Note: The study on performance/complexity are expected to be led by RAN 4.
- Other aspects for RAN 1 study:
  - For AI at receiver:
    - Feasibility/performance for finetune
    - (Spec impact) potential DMRS design/configuration impact with performance evaluation
    - (Spec impact) potential power control procedure impact
    - (Spec impact) Whether/what signal/procedure is needed for
      - data collection
      - inference
      - performance monitoring
      - online finetune, if applicable and feasible
  - For AI at transmitter:
    - Feasibility/performance for finetune
    - (Spec impact) potential power control procedure impact
    - (Spec impact) potential SRS/PUSCH design/configuration impact
    - (Spec impact) Whether/what signal/procedure is needed for
      - data collection
      - inference
      - performance monitoring
      - online finetune, if applicable and feasible

Company	Comment
FL	To facilitate potential further study, listing RAN 1 led part is helpful
Google	We do not think this should be studied in the first release of 6G. It is not a RAN1 issue as well.
Apple	RAN4 led, RAN1 should only study based on RAN4 request. No need for duplicated evaluation.
MediaTek	We have a similar view as Apple, i.e., the study should be led by RAN4 and RAN4 should be involved only based on RAN4 request.
Huawei, HiSilicon	RAN4 already had agreements on evaluation aspects. This use case can be led by RAN4. No need to discuss at RAN1.
CATT	RAN1's study is triggered only by RAN4's request.
vivo	We are supportive to study in RAN1. There are obvious RAN1 specification impact. For the AI at the transmitter side, the specific impact on the power control procedure such as power determination for data collection and PHR reporting for max power boosting also warrants further study. So, we propose to add one sub-bullet to address this. Moreover, for SRS/PUSCH design, there might be additional considerations for UE to collect data for finetuning. - For AI at transmitter: - Feasibility/performance for finetune - (Spec impact) potential power control procedure impact - (Spec impact) potential SRS/PUSCH design/configuration impact - (Spec impact) Whether/what signal/procedure is needed for - data collection - inference - performance monitoring - online finetune, if applicable and feasible
OPPO	We also suggest letting RAN4 handle the RF-related issue. RAN1 can response if RAN4 requests.
Lenovo	AI-based DPoD/non-linearity compensation can be made possible through an AI-module added to the receiver or achieving DPoD can also be made as an integral part of an AI-based receiver. In this context, we wonder can we conclude that AI-based DPoD is to be led by RAN4 without further discussion?

Panasonic	We suggest postponing this proposal because it would be handled by RAN4. If RAN4 needs any If RAN4 requests anything from RAN1, RAN1 would respond later.
Fujitsu	Agree that this case could be led by RAN4.
InterDigital	Same view as Google. We do not see any relevance for RAN1 to initiate studies on this topic. If any impact to RAN1 is identified by RAN4, that can be accommodated accordingly, e.g., during normative work if approved.
Ericsson	• Since this use case is expected to be RAN4 led, there is no need to discuss what RAN1 should study. RAN1 can wait for RAN4 to trigger the necessary aspects for study. • Remove the three bullets for finetune. We think online finetune is out of scope for the first release of 6G.
ZTE	Similar view as other companies that his use case can be led by RAN4. RAN1 can start further study once RAN4 LS comes.
Samsung	We see the comments like RAN4 should lead the discussion, which is already well captured in first note, "Note: The study on performance/complexity are expected to be led by RAN 4." The proposal means something since the RAN1 impacts are studied by us, the leading role of RAN4 in this use case did not prevent us to study the potential spec impact. Besides, for the power control part added by vivo, we think it's not only limited to AI at transmitter, it also applies to AI at receiver, since the different reception at network can also impact the transmission power at UE side. We suggest this one "(Spec impact) potential power control procedure impact" to be general for both sub-use cases.

2.7 AI for TMPI

Proposal 2.7:

For the potential study on AI-enabled UL precoder indication, at least the following aspects are to be considered:

- KPIs:
  - Performance related: SGCS, BLER, Throughput
  - Model complexity
  - Inter-vendor collaboration overhead and complexity for two-sided model, when applicable.
- Other aspects for study:
  - Generalization
    - deployment scenarios, UE antenna configuration, subband configuration (if appliable)
  - Coexistence with non-AI-enabled UL precoder
  - (Spec impact) The signaling/procedure related to the download/upload of UL codebooks/compressed UL precoder
  - (Spec impact) Whether/what signal/procedure is needed for
    - data collection
    - inference
    - performance monitoring
    - inter-vendor collaboration, if applicable and justified

Company	Comment
FL	I think this can be directly discussed in UL MIMO. If this is not related to two-sided model. this use case can be treated as AI-enabled use case. Proposal 1 may be enough.
Google	We do not think this should be studied in the first release of 6G. We need to support UL subband precoding first.
CMCC	For the generalization part, we think the subband configuration of UL MIMO is also needed if the AI/ML based TPMI is SB TPMI. So, we suggest modifying: o Generalization ▪ deployment scenarios, UE antenna configuration, subband configuration
vivo	AIML based UL TPMI indication does not necessarily rely on sub-band precoding. Even for wide-band precoding, we see potentials for improvement, for e.g., 3 to 4 times throughput increase for cell edge UEs.

Fujitsu	As mentioned by vivo, this case could be applicable to wideband precoding. In 6G, the channel bandwidth may be even larger. In this case, sub-band precoding is beneficial and the performance can be even further improved with the help of AI/ML.
InterDigital	Same view as Google – the relevance depends on design of basic UL MIMO.
Ericsson	We agree with Google's view.
LG Electronics	Similar view with Fujitsu. Also, in 6G, SB UL precoding is promising, and we see clear benefit from AI-based TPMI in the sub-band precoding case. Typo: SCGS->SGCS
Samsung	Similar view as vivo. The AI-enabled UL precoder is beneficial for wideband case. There is no need to mix up the discussion.

2.8 AI for SRS

Proposal 2.8-A: study aspects

For the potential study on low overhead SRS with AI/ML, at least the following aspects are to be considered:

- KPIs:
  - Performance related: SGCS, throughput, [PAPR, Cross-correlation between SRS sequences]
  - Model complexity
- Other aspects for study:
  - Performance (including generalization performance) under fading channel with practical channel estimation (for Sub-case A and Sub-case B)
  - Coexistence with non-AI designed sequence/pattern, if applicable
  - (Spec impact) SRS design and performance
    - Time-domain and frequency-domain and spatial-domain sparse SRS pattern for Sub-case A
    - SRS sequence/pattern design for Sub-case B
  - (Spec impact) The signaling/procedure related to the download/upload of SRS sequence/pattern
  - (Spec impact) Whether/what signal/procedure is needed for
    - data collection
    - inference
    - performance monitoring

Proposal 2.8-B:

For SRS design, AI/ML receiver can be considered as an implementation choice.

- FFS on whether additional procedure for data collection/performance monitoring is needed for not.

Company	Comment
FL	I think this can be directly discussed in SRS design If this is not related to two-sided model. this use case can be treated as AI-enabled use case. Proposal 1 may be enough.
Google	We do not think this should be studied in the first release of 6G
CMCC	Generally support this use case. Just one clarification on the fist Note "this use case focus on SRS based channel acquisition only.", we think it includes DL and UL channel acquisition, i.e. base on SRS for CB, NCB and AS.
vivo	The cross correlation between SRS sequences should not be put int brackets. It is one of the important aspects to be considered for SRS design.
Fujitsu	Agree with CMCC. The note "this use case focus on SRS based channel acquisition only" can be removed.
InterDigital	Same view as Google. If AI/ML is used as a tool to design SRS sequences, it can be directly considered as part of basic SRS design and not necessarily as an AI/ML use case.
LG	For sub-case A, spatial-domain sparse SRS pattern can be considered. So, we suggest following,

Electronics	(Spec impact) SRS design and performance Time-domain, frequency-domain and spatial-domain sparse SRS pattern for Sub-case A
IMU	Agree with LG Electronics. Spatial-domain sparse SRS pattern can be considered for sub-case A.

2.9 AI for PAPR reduction for waveform

Proposal 2.9 A:

For the potential study on AI-based waveform for PAPR reduction, at least the following aspects are to be considered:

- KPIs:
  - Performance related: BLER, CCDF of PAPR (UL), throughput (DL),
  - Model complexity
  - Inter-vendor collaboration overhead and complexity for two-sided model, when applicable
- Benchmark
  - FFS: Non-AI or implementation-based method
- Other aspects for study:
  - Generalization
    - The number of subcarriers
    - Modulation order
  - Coexistence with DFT-s-OFDM and CP-OFDM, e.g., UL MU-MIMO scenario
  - (Spec impact) Whether/what signal/procedure is needed for
    - data collection
    - inference
    - performance monitoring
    - inter-vendor collaboration, if applicable and justified

Proposed conclusion 2.9B:

AI based PAPR reduction for waveform can be study together with non-AI low PAPR waveform design.

Company	Comment
FL	Unless two-sided model can be justified. I don't see any special LCM related procedure is needed. conclusion should be enough. Anyway, I think if proposal 1 can be agreed, no need to spend time on this.
Google	We do not think this should be studied in the first release of 6G
vivo	Seems google is against most use cases , even for study purpose…
Lenovo	We think this use case should have a low priority for the first release of 6G. Further, we think the study should be focusing on only one-sided AI models.
InterDigital	Same view as Google, and in response to the comment from FL, we don't see a clear justification to study two-sided models for this use case at present. Even the study would require serious time and efforts and need proper dimensioning within the WG for any meaningful outcome. Thus, we need to be very careful on what we commit to study and there should be sufficient clarity and motivation behind that.
Samsung	As mentioned by FL, LCM-related operations are not required. AI-based waveforms for PAPR reduction can be discussed on the waveform agenda.
Sony	We do not support this proposal. This AI is meant for AI/ML use cases. We should first decide whether or not PAPR reduction is a plausible AI/ML use case. We think RAN1 should make this assessment and provide a recommendation at the check point on which use cases to progress the study on.
IMU	It is clear that, we further investigate the applicability of AI/ML to obtain PAPR reduced waveform belirlenen bir zaman kadar. Günün sonunda performans değerlendirmelerimizi sub use case'lere bağlı başarım sonuçları, generalization capability ve LCM açısından non-AI/ML ile karşılaştırmamız gerekiyor. Böylelikle 6G için applicable bir sonuç elde edeibliriz.

2.10 AI for Sequence based HARQ-ACK

Proposal 2.10: study aspects

For the potential study on AI/ML based HARQ-ACK feedback (up to11bit), at least the following aspects are to be considered:

- KPIs
  - BLER
  - Receiver complexity
- Study aspects:
  - Performance of joint source channel coding.
  - Performance of sequence-based vs. DMRS-based approaches.
  - Performance under fading channel with practical channel estimation
  - Generalization performance
    - Different UL channel
    - Different ACK/NACK probability, different correlation among ACK/NACK bits
  - Scalability: e.g., number HARQ-ACK payload, UL resources for UCI
  - Performance impacted by non-ideal factors, such as channel estimation error, PA non-linearity, phase noise, IQ imbalance, interference measurement
  - (Spec impact)
    - Learned sequences/modulated symbols design for up to 11 bits HARQ-ACK
    - Whether/how to introduce explicit reliability check mechanism
    - Necessity and singling/procedure for downloadable learned sequences/modulated symbols
    - Necessity and signaling/procedure to support learned sequences/modulated symbols transmission, if needed
  - (Spec impact) whether/what signal/procedure is needed for
    - data collection
    - inference
    - performance monitoring

Company	Comment
FL	The proposal is to listing the potential AI/ML specific consideration for AI receiver. For spec impact part, it is not clear on whether LCM related procedure is needed or not.
Google	We do not think this should be studied in the first release of 6G
Huawei, HiSilicon	This use case should be similar to AI for TMPI (for generating TPMI codebook), Demodulation (for generating constellation pattern), and SRS (for generating sequence), that AI is used to generate HARQ sequence/LUT. It can be studied together with non-AI HARQ sequence design.
Qualcomm	We think this is the lowest hanging fruit in 6G AI. A use case with obvious gain and only offline AI training (without necessity of inference), which add no complexity/latency to 6G DL/UL data/control path. We also made some suggested update in the “study aspects” to capture more comprehensive aspects of the study. Also, can FL clarify what is “Whether/how to introduce explicit reliability check mechanism”? Finally, a minor comment: It is better to change the section title from “AI for Sequence based HARQ-ACK” to “AI/ML based HARQ-ACK feedback”, as the latter is more generic and matches with the scope of the proposal better.
vivo	Also commenting on “Whether/how to introduce explicit reliability check mechanism. I think this is due to sequence-based transmission does not have explicit CRC check. But we don’t think this problematic since already in NR the RM codes less than 11 bits is not relying on CRC and as long as the target BLER is achievable, there should not be any problem for non-CRC checked transmission.

Lenovo	It is not clear about (up to11bit) in the proposal.
InterDigital	We tend to agree with Huawei's comment that this can be considered as use of AI/ML as a tool for specifying the HARQ-ACK sequence/mapping and thus, need not be listed here as an AI/ML use case. This can be studied directly as part of HARQ-ACK studies and HARQ-ACK CB design.

2.11 AI for PDCCH

Proposal 2.11: study aspects

For the potential study on AI/ML for PDCCH related use case, at least the following aspects are to be considered:

- Validation on the pre-requisite assumption before detailed study.
- Note: the study is not expected to be start before finishing PDCCH design.
- KPIs
  - BLER, DCI overhead
  - Decoding complexity and latency
  - Inter-vendor collaboration overhead and complexity for UE-sided and NW-sided model, if applicable.
- Study aspects:
  - Performance under fading channel with practical channel estimation
  - Generalization performance over different DL channel, missing of historical PDCCH
  - Scalability: DCI payload, aggregation level, DCI format
  - Performance impacted by non-ideal factors, e.g., interference measurement
  - (Spec impact) Whether/what signal/procedure is needed for
    - data collection
    - inference
    - performance monitoring
    - Inter-vendor collaboration overhead and complexity for UE-sided and NW-sided model, if applicable.

Proposed conclusion 2.12:

AI/ML for PDCCH related use case can be discussed in RAN 4 and/or RAN 1when PDCCH design has been finished.

Company	Comment
FL	If this is UE-sided model, I feel like this is implementation issue… which can be ran 4 for requirement. Pre-requisite assumption may need to be validated in Ran 1. UE-sided/NW-sided model should be justified first. Anyway, I think if proposal 1 can be agreed, no need to spend time on this.
Google	We do not think this should be studied in the first release of 6G
CMCC	Support Proposal 2.11 with modification. 1. Validation on the pre-requisite assumption: temporal correlation and stability between historical DCI field patterns. Lower information entropy in historical DCI field brings higher DCI content prediction accuracy. 2. Besides, we do not agree "the study is not expected to be start before finishing PDCCH design. The Prior-Information-Aided DCI Decoding is not related with PDCCH resource mapping but only related with the channel coding of PDCCH. We can start the study just after deciding PDCCH coding schemes. So, the Note in Proposal 2.11 should be modified: Note: the study is not expected to start before ~~finishing PDCCH design~~ PDCCH coding schemes are decided. 3. And we do not think it is only implementation issue for UE-sided model and only left to RAN4 for requirement.

	NW needs to know when UE will use AI based DCI decoder, so that NW can configure corresponding search space set and CORESET for PDCCH. The fallback mechanism to non-AI based decoder and configuration for AI based DCI should be designed together with non-AI DL control in RAN 1.
Ericsson	As the note stated, this use case should not start before finishing PDCCH design. Therefore, there is no need to consider this use case in Rel-20 SI. No need to agree on potential aspects to study at this stage.

2.12 (P)RACH related design

Proposal 2.12A: study aspects

For the potential study on early contention resolution in RACH, at least the following aspects are to be considered:

- KPI
  - Prediction accuracy of UE multiplicity, first-attempt success probability, and false alarm detection rate
  - System gain, e.g., RACH access delay
  - Receiver complexity
  - Addition signaling overhead
  - Note: The collision probability of more than one UE to select the same PRACH needs to be validated.
- Study aspects:
  - Performance under fading channel
  - Performance impacted by non-ideal factors, such as timing advance, TO/FO, PA non-linearity, phase noise, IQ imbalance, interference measurement
  - Generalization performance
    - Different UL channel
    - Different received power from different UE
  - (Spec impact) necessary signaling/procedure related to random access, if any
  - (Spec impact) Whether/what signal/procedure is needed for
    - data collection
    - inference
    - performance monitoring

Company	Comment
FL	I feel this can be led by RAN 4. Not sure what is the clear RAN 1 impact. I think if proposal 1 can be agreed, no need to spend time on this.
Google	We do not think this should be studied in the first release of 6G

Proposed conclusion 2.12:

AI based lower PAPR design for PRACH can be study together with non-AI PRACH design.

Company	Comment
FL	I don't see any special LCM related procedure is needed. conclusion should be enough. I think if proposal 1 can be agreed, no need to spend time on this.
Google	We do not think this should be studied in the first release of 6G
vivo	We are fine with this one.

2.13 AI for power control

Proposed conclusion 2.13:

For AI/ML for power control related use case, the study outcome of L1-RSRP prediction can be considered.
Proponent can propose necessary enhancement to corresponding agenda, e.g., power control.

Company	Comment
FL	One subcase is NW-sided model which can be handled by implementation, I don't feel any special study is needed. another use case is related to L1-RSRP, which is more like extension of beam management. Therefore, I don't see the necessary of special handle. Any other view? In addition, I think if proposal 1 can be agreed, no need to spend time on this.
Google	We think the "L1-RSRP" should be changed into "RSRP". Pathloss is usually based on RSRP.
IMU	It is clear that we need to conduct further study for RSRP prediction. However, it would not be appropriate to restrict this to only L1-RSRP. In RAN2, measurement prediction topics include L1–L3 direct and indirect measurement prediction. A solution obtained here could also contribute to those efforts.

2.14 AI/ML based SRS power imbalance compensation

Proposal 2.14A: study aspects

For the potential study on AI/ML based SRS power imbalance compensation, at least the following aspects are to be considered:

- Note: The study on performance/complexity are expected to be led by RAN 4.
- Study aspects:
  - (Spec impact) Whether/what signal/procedure is needed for
    - data collection
    - inference
    - performance monitoring

Proposed conclusion 2.14-B:

For AI/ML based SRS power imbalance compensation, AI/ML receiver can be considered as an implementation choice, or up to RAN 4.

- FFS on whether additional procedure for data collection/inference/performance monitoring is needed for not.

Company	Comment
FL	I feel this can be led by RAN 4. Not sure what is the clear RAN 1 impact. Maybe the proposed conclusion B is sufficient? Or proposal 1 is enough
Google	Support conclusion 2.14B
CMCC	This use case is also proposed in last RAN4 meeting, we support Conclusion 2.14-B, it is up to RAN4 study.
Fujitsu	We think the signal/procedure for data collection/inference/performance monitoring may have impact to RAN1. The following update is suggested. ***Proposal 2.14A: study aspects*** For the potential study on AI/ML based SRS power imbalance compensation, at least the following aspects are to be considered: • Note: The study on performance/complexity are expected to be led by RAN 4. • Study aspects in RAN1: o (Spec impact) Whether/what signal/procedure is needed for ▪ data collection ▪ inference ▪ performance monitoring ***Proposed conclusion 2.14-B:*** For AI/ML based SRS power imbalance compensation, AI/ML receiver can be considered as an implementation choice, or up to RAN 4. • FFS in RAN1 on whether additional procedure for data collection/inference/performance monitoring is needed for not.

2.15 AI for positioning

Proposed conclusion 2.15:

AI for positioning related use case can be considered in position related design.

Company	Comment
FL	Ericsson proposed to consider AI for Pos. Maybe proposal 1 is enough
Google	We think AI for positioning for NTN can be considered, since GNSS resilient is already agreed for NTN. For TN, currently we failed to see the urgency.
Huawei, HiSilicon	We did not find positioning in 6G SID scope
Ericsson	Since positioning (including indoor positioning) is a KPI/TPR for 6G, it is necessary to include AI/ML positioning in study for 6G first release.

2.16 RAN for token traffic

Proposal 2.16: study aspects

For the potential study on improved scheduling/HARQ for token traffic, at least the following aspects are to be considered:

- Evaluation methodology:
  - Traffic model (depending on the outcome of traffic model discussion in agenda 11.2)
  - Note: AI/ML models are located in the application layer. RAN1 don't need to discuss the model related.
  - Note: The coordination with other WGs on the performance evaluations is needed
- KPI: supported number of UEs, achievable throughput
- Benchmark: NR scheduling/HARQ mechanism without knowledge of token traffic
- Study aspects: Improved RAN solutions to facilitate Token communication, e.g., awareness of Token traffic unit and its characteristic at RAN, scheduling and HARQ mechanism

Proposed conclusion 2.16-B:

For proposed token related use case, AI/ML models are located in the application layer, which is out of the discussion scope of RAN 1.
Token traffic related system design/optimization can be further study after defining token related traffic model.
Proponent can propose the token related design/optimization to corresponding agenda.

Company	Comment
FL	Some new RAN for AI use cases have been submitted. We can fix the name later. Two potential directions. Personally, I am not so confident on direction A, maybe we can conclude for option B? Or proposal 1 is enough.
Google	The scheduling and HARQ is in RAN1's scope. The CQI accuracy is also important for AI traffic, since AI traffic usually requires higher reliability due to higher economic cost than other traffic. We also see the evaluation agenda has some agreement related to AI traffic. There is no reason to preclude the study to facilitate AI traffic at current stage.
Huawei, HiSilicon	Support direction A. There is no need to mention "depending on the outcome of traffic model discussion in agenda 11.2" at direction A, or "after defining token related traffic model" at direction B. All the AI related use cases are dependent on some fundamental design, e.g., AI MIMO also needs to wait for fundamental MIMO outcome (Tx/Rx port number, TxRU mapping, benchmark non-AI solution, etc.) for evaluations – that should not be the reason of suspending the discussion of any use case. Assuming we determine to discuss this feature at any certain RAN1 agenda 11.x (regardless it is the control/scheduling/HARQ agenda, evaluation agenda, or AI/ML agenda), this 11.x should be responsible to study the traffic model on behalf of RAN1.

	• Evaluation methodology: o Traffic model ~~(depending on the outcome of traffic model discussion in agenda 11.2)~~ o Note: AI/ML models are located in the application layer. RAN1 don't need to discuss the model related. o Note: The coordination with other WGs on the performance evaluations is needed • KPI: supported number of UEs, achievable throughput • Benchmark: NR scheduling/HARQ mechanism without knowledge of token traffic • Study aspects: Improved RAN solutions to facilitate Token communication, e.g., awareness of Token traffic unit and leveraging its characteristic at RAN, scheduling and HARQ mechanism
Qualcomm	We agree with the FL that GenAI traffic characteristics (including token traffic as a special case) should be studied first in RAN1.
CMCC	In the LS to other WGs for study traffic model(s) for 6GR AI/ML services in last meeting agenda 11.2, it says "RAN1 can continue the study before SA4 potential response.", so we think the Note in Evaluation methodology is not needed: • Evaluation methodology: o ~~Note: The coordination with other WGs on the performance evaluations is needed~~
vivo	We would like to clarify our scenario: for the audio token case, RAN1 should also consider the specific requirements of narrowband audio transmission over GEO satellite systems, where the RTT is long and link budget is constrained. For this type of service, it is beneficial to explore joint source– channel coding and cross-layer design. In such an approach, part of the traditional physical-layer channel-coding functionality can be shifted to, or jointly optimized with, the application-layer decoder. This allows the physical layer to adopt simplified or error-tolerant transmission, while relying on the application-layer decoding capability to achieve high-quality reconstruction of the audio signal.Under this framework, the relevant KPI for RAN1 evaluation should be the required SNR to obtain an equivalent reconstructed-audio quality compared with the existing narrowband-audio PHY processing chain. Accordingly, the benchmark is the current NB-audio physical-layer procedures in NR NTN. ***Proposal 2.16: study aspects*** For the potential study on improved scheduling/HARQ for token traffic, at least the following aspects are to be considered: • Evaluation methodology: o Traffic model (depending on the outcome of traffic model discussion in agenda 11.2) o Note: AI/ML models are located in the application layer. RAN1 don't need to discuss the model related. o Note: The coordination with other WGs on the performance evaluations is needed • KPI: supported number of UEs, achievable throughput, coverage improvement • Benchmark: NR scheduling/HARQ mechanism without knowledge of token traffic, physical-layer procedures/design in IoT-NTN • Study aspects: Improved RAN solutions to facilitate Token communication, e.g., awareness of Token traffic unit and its characteristic at RAN, scheduling and HARQ mechanism

OPPO	Proposal 2.16 (study aspects) is the right way to go. We would like to emphasize that in this agenda (11.6), the main purpose of the study is to a) clarify the token comm. (a brand-new traffic type in 3GPP); b) conduct evaluations for token comm.; and c) unveil the performance benefits assuming 6GR optimized for token comm. With above being said, if we go with 2.16-B, then we may heavily lag behind the progress of traffic model in agenda 11.2.
Panasonic	As token traffic related system design/traffic model is not studied well in RAN1, hence it can be deprioritized as for now.
Fujitsu	Some clarification is needed regarding the following bullet. For “awareness of Token traffic unit”, should it be done by RAN2? • Study aspects: Improved RAN solutions to facilitate Token communication, e.g., awareness of Token traffic unit and its characteristic at RAN, scheduling and HARQ mechanism
InterDigital	The characterization of “token traffic” is being considered in agenda 11.2, and we do not think initiating studies as an AI/ML use case is sufficiently justified at this point given we do not really know how they may be visible to RAN1 via traffic modelling.
TCL	Agree with Huawei, the token traffic model may be separately discussed on its related agenda.
Ericsson	Agree with the proposed conclusion. With this understanding, there is no need to discuss Proposal 2.16 under 11.6.
LG Electronics	Agree with Ericsson and InterDigital.
ZTE	We are ok to go with the conclusion. The potential research points for this use case are traffic model, scheduling/HARQ/Control enhancements. Since the AI model is at application layer, we think companies can directly propose their proposal regarding the traffic model, scheduling/HARQ/Control enhancements in the corresponding agenda.

2.17 Digital twin related use cases

Proposal 2.17-A: study aspects

For the potential study on digital twin related use cases, at least the following aspects need to be clarified:

- Note: digital twin related use cases at least including:
  - Sensing based RAN digital twin construction with NW-side AI/ML model
  - AI/ML-enabled RAN digital twin with distributed model
  - Site Specific Learning for AI/ML and RAN Digital Twin
- Label:
  - how to obtain/verify the ground-truth label for digital twin related information, when applicable.
- KPI:
  - BLER, throughput
  - FFS on other KPIs
- Benchmark
  - The performance without the help of digital twin
  - FFS on other benchmark
- Other study aspects
  - How to verify the performance of digital twin

- Generalization performance over different scenarios, FFS on how to define different scenarios
- Note: whether/how to support the generation of digital twin is up to ISAC

Proposal 2.17-B:

Whether/how to support the generation of digital twin is up to ISAC or other agenda.

Digital twin related use case can be discussed later.

Company	Comment
FL	It is hard to proceed the study on digital twin related use case study without knowing defining the function/performance of digital twin. Therefore, I suggest to go for proposal 2.11-B. we can come back when digital twin itself has been clarified. Maybe proposal 1 is enough.
Google	In our view, digital twin related is premature to be included in the first release of 6G.
Huawei, HiSilicon	There is no need for Proposal 2.17-B. Proposal 1 already distributes it to the ISAC agenda. Whether/how to discuss the AI/ML enabled digital twin generation can be discussed at ISAC, and for what we further study, Proposal 2.17-A can provide good directions.
Ericsson	We think digital twin is transparent to RAN1 spec, and there is no need to study digital twin. For sensing related aspects, at least a note should be added: "Note: AI/ML for sensing related use cases is not expected to be discussed until after non-AI based sensing is completed."

3 Framework and evaluation

3.1 Evaluation and KPIs

Several companies discussed aspect on EVM and KPIs. Several companies proposed for comprehensive evaluation of AI/ML use cases by considering KPIs including system performance, system and model complexity, inter-vendor collaboration complexity, power consumption. In addition to intermediate and system KPIs that were adopted in 5G NR, companies proposed new KPIs such power consumption and inference latency to be considered in 6GR.

Proposal 3.1 (RAN 1 #122b) :

For evaluation of AI/ML use cases in 6GR, consider

- Performance related metrics, including intermediate (model) performance KPIs, link level KPIs (e.g., BLER) and system level KPIs (e.g., throughput, overhead)
  - Where the overhead including the overhead associated with inferencing, performance monitoring and (re)training
- AI/ML Model related metrics, including model/computational complexity, inference latency, training latency (when applicable),
  - FFS on whether/how to measure power consumption
- Inter-vendor collaboration when applicable
- Generalization performance under a wide range of conditions
  - FFS on whether and how to consider realistic deployment scenarios

Note: Detailed metrics to be discussed per use case.

Company	Comment
FL	Leftover from last meeting. The red part may need to explained by proponent companies
Lenovo	Model storage requirement should also be considered in the second bullet. And just a wording suggestion: ▪ Inter-vendor collaboration complexity analysis when applicable
NTT DOCOMO	We suggest a further clarification for the Note in the last line, Note: Which types of KPI (intermediate, link-level, system-level) will be adopted and detailed

	metrics to be discussed per use case.
Futurewei	In the case that the model requires online fine-tuning, the overhead should also include the fine-tuning aspects. We suggest the following modification in the sub-bullet of the first main bullet: o Where the overhead including the overhead associated with inferencing, performance monitoring ~~and~~ (re)training and fine-tuning (if any).
Huawei, HiSilicon	1, the system level KPI should also consider the metric to evaluate the efficiency of serving AI/ML traffic type (e.g., tokens) For evaluation of AI/ML use cases in 6GR, consider ▪ Performance related metrics, including intermediate (model) performance KPIs, link level KPIs (e.g., BLER) and system level KPIs (e.g., throughput, overhead) o Where the overhead including the overhead associated with inferencing, performance monitoring and (re)training o FFS performance metric for AI/ML services such as token communication
CATT, CICTCI	(1) Training latency is difficult to evaluate as it is highly implementation or even hardware dependent. And it does not matter for most use cases unless they rely on fast online training. Suggest to remove it. (2) For two-sided model use cases (e.g. CSI compression related), we think 'scalability' should also be looked into, just the same as CSI compression study in NR.
SPRD	For the last bullet, what is the meaning of "a wide range of conditions"?
Xiaomi	1. For the AI/ML related metrics, it is unclear how to measure the inter-vendor collaboration 2. For the power consumption, with the increase of AI use cases, the power consumption will be increased accordingly. From that sense, guranteeing the power efficiency is critical to the user throughput. Thus, power consumption should be one criteria for the use case evaluation. As for the how to measure the power consumption, we don't think it could be simply reflected by the model complexity, it also affected by the inference frequency. Even when there is no task, certain base electronic current is also needed to maintain the AI engine on, that would also cause some energy consumption 3. As for the training latency, we share similar view with CATT
Nokia	Direction is fine. Better to discuss in the offline session for wording changes.
SK Telecom	For the third bullet, the wording needs to be improved like: "Complexity for inter-vendor collaboration when applicable". For the last bullet, it is a bit unclear what it implies. What should be considered for generalized performance?
CMCC	If model (re)training or model fine-tuning is adopted, then model can be changed when UE moved into different areas. So, generalization performance is not needed under a wide range of conditions, maybe generalization under one or several cells is enough. ▪ Generalization performance ~~under a wide range of conditions~~
Tejas	Fine with the direction
IIT Kanpur	We agree with Futurewei's comment regarding the inclusion of overhead aspects during fine-tuning, should the model require fine-tuning.
OPPO	Fine with this direction. But for some details, such as inference latency or training latency, we don't really know how to evaluate it. Btw, training latency in Rel-18 study, it can be considered as not time critical, so why would we study it again for 6GR.
Ericsson	Regarding performance related metrics bullet, suggest adding a sub-bullet: o Conclusions shall be drawn based on final performance metric (e.g., throughput) instead of intermediate KPIs In addition, please add a bullet • RAN4 interoperability testing feasibility and complexity
InterDigital	We support the sub-bullet in red. The OH considering inferencing, training/retraining, and performance monitoring are critical aspects that determine the overall benefits of specifying an AI/ML-based solution for a functionality. In addition, we also would like to second the suggestion from Ericsson on consideration of final

	performance metrics.

Proposal 3.1(new) :

For study/evaluation of AI/ML use cases in 6GR, consider

- Performance related metrics, including, intermediate (model) performance KPIs, link level KPIs (e.g., BLER) and system level KPIs (e.g., throughput, overhead), etc
  - including the overhead associated with inferencing, performance monitoring, online fine-tuning (if any)
  - Note: conclusions shall be drawn based on final performance metric (e.g., throughput) instead of intermediate KPIs
- AI/ML Model related metrics, including model/computation complexity/power consumption, inference latency, training latency (for online training/fine-tuning, ~~when~~ if applicable), scalability, etc
  - FFS on whether/how to measure power consumption
- Inter-vendor collaboration complexity and potential performance impact, when applicable
- Generalization performance ~~under a wide range of conditions~~
  - FFS on whether and how to consider realistic deployment scenarios

Note: Which types of KPI (intermediate, link-level, system-level) will be adopted and detailed metrics to be discussed per use case.

Note: RAN4 interoperability testing feasibility and complexity is up to RAN 4

Company	Comment
FL	Leftover from last meeting. The red part may need to explained by proponent companies
Google	Probably current agreement is sufficient. If we want to make some progress, we should try to resolve the FFS in previous agreement
Qualcomm	Suggest the following change, because Intermediate metrics also provide valuable insights. Final performance metrics may not be as widely available as intermediate metrics. o Note: conclusions shall be drawn based on final performance metric (e.g., throughput) in addition to ~~instead of~~ intermediate KPIs
CMCC	If we have the above proposals on the study aspects in each use case, maybe this proposal is not needed. Besides, we have concern on "Note: conclusions shall be drawn based on final performance metric (e.g., throughput) instead of intermediate KPIs", it is possible there is no so-called final performance metric for some use cases.
vivo	We would like to update as following: For study/evaluation of AI/ML use cases in 6GR, consider ▪ Performance related metrics, including, intermediate (model) performance KPIs, link level KPIs (e.g., BLER) and system level KPIs (e.g., throughput, overhead), etc o including the overhead associated with inferencing, performance monitoring, online/site specific fine-tuning (if any) o Note: conclusions shall be drawn based on final performance metric (e.g., throughput) instead of intermediate KPIs ▪ AI/ML Model related metrics, including model/computation complexity/power consumption, inference latency, training latency (for online training/fine-tuning, ~~when~~ if applicable), scalability, etc o FFS on whether/how to measure power consumption ▪ Inter-vendor collaboration complexity and potential performance impact, when applicable ▪ Generalization performance ~~under a wide range of conditions~~ o FFS on whether and how to consider realistic deployment scenarios Note: Which types of KPI (intermediate, link-level, system-level) will be adopted and detailed metrics to be discussed per use case. Note: RAN4 interoperability testing feasibility and complexity is up to RAN 4

Lenovo	It is also necessary to consider the AI-model size, especially with respect to the storage requirements. Consider • AI/ML Model related metrics, including model/computation complexity/power consumption, required storage, inference latency, training latency (for online training/fine-tuning, ~~when~~ if applicable), scalability, etc
InterDigital	Under performance KPIs, we support the inclusion of OH associated with inferencing, performance monitoring as these are real considerations when deciding on practical deployment of AI/ML-based solutions. Without consideration of this, the performance gains may be significantly overestimated. Also, we would like to second Qualcomm's suggestion regarding intermediate metrics.

3.2 LCM framework

Many companies proposed enhancement on NR's LCM, encompassing aspects such as data and model management, including model transfer, applicability of the associated ID, support for localized models, advanced training methods, e.g., online and federated learning, meta-learning for handling network-side additional conditions. Moreover, a number of companies proposed to 5G NR's LCM framework including functionality-based LCM as a starting point. Enhancement on AI/ML processing unit framework was proposed by a few companies, e.g., 1 company (Samsung) proposed to introduce AI/ML memory unit (MU) on the concurrently activated AI/ML feature/models

Proposal 3.2(RAN 1 #122b):

For 6G LCM framework for AI/ML for air interface, consider the 5G NR AI/ML LCM framework as adopted in 5G Rel-19 specifications (e.g., TS38.331) as a starting point.

- Study the necessity of potential enhancements for LCM, and if justified, the enhancement details. The examples to study include:
  - Data and model management
  - Handling of additional conditions
  - Enablers for continuous (online) on-device model training/finetuning
  - framework for AI/ML processing and memory

[Note: the study strived to provide a unified LCM across use cases]

Company	Comment
FL	Some modification according to Ericsson's proposal
Lenovo	Generally fine with the proposal. One suggestion is to list some aspects, e.g., data collection for online training, model validation, under the third subbullet.
NTT DOCOMO	We are generally fine with directions in the main bullet, but feel the examples are not clearly defined currently. We also suggest keeping the last note, which should be an essential enhancement of 6GR AI/ML compared with 5GA.
Futurewei	We are ok with the proposal in general with a small comment regarding online model fine-tuning. We believe this should be applicable on either UE (side) or NW, so we suggest removing the wording of "on-device": o Enablers for continuous (online) ~~on-device~~ model training/finetuning
Fujitsu	If the examples to study are listed in the proposal, then performance monitoring should be included which is important part of LCM. The following change is suggested. ▪ Study the necessity of potential enhancements for LCM, and if justified, the enhancement details. The examples to study include: o Data and model management o Performance monitoring o Handling of additional conditions o Enablers for continuous (online) on-device model training/finetuning o framework for AI/ML processing and memory
CATT, CICTCI	We think 'continuity of UE-side AI/ML features across cells' should be studied. This is to guarantee the UE-side AI/ML model/functionality can be smoothly functioning/switching among different cells around handover time.

SPRD	For the last example, it is necessary to clarify what components are included in the framework for AI/ML processing.
Huawei, HiSilicon	We do not think RAN1 can determine the 6GR AI/ML LCM framework on behalf of RAN2 – notice that TS38.331 spec is RAN2 spec, they are more appropriated to determine whether to reuse 5G or make a new architecture. For 6G LCM framework for AI/ML for air interface, ~~consider the 5G NR AI/ML LCM framework as adopted in 5G Rel-19 specifications (e.g., TS38.331) as a starting point.~~
Ofinno	Generally fine. On model management, is it correct understanding to include performance monitoring aspects also?
Xiaomi	1. For the LCM Framework, we share similar view with Huawei. The determination of LCM framework is RAN2 work. 2. For the third sub bullet, we share similar view with Futurewei, the “on-device” should be removed 3. For the Note part, we suggest to remove the bracket, it is quite important principle for the LCM framework
Nokia	Ok with the proposal with Futurewei update. Mentioning of e.g., TS38.331 is not needed.
SK Telecom	We think defining the unified AI/ML framework across use cases is very important in order to avoid any excessive framework, and thus prefer to keep the last note. Regarding the examples, it seems still necessary to be further discussed, and maybe the wording would be better to be open a bit more.
CMCC	We are generally supportive of this proposal. But if we consider two-sided model, 5G Rel-20 specifications are also needed as a starting point, and the 5G NR AI/ML LCM framework is not only described in TS 38.331, but also TS 38.214, TS 38.212, etc. For 6G LCM framework for AI/ML for air interface, consider the 5G NR AI/ML LCM framework as adopted in 5G Rel-19/20 specifications ~~(e.g., TS38.331)~~ as a starting point.
LG Electronics.	We are generally fine, but we may need to further study on efficient and faster model switching by considering fall back operation to non-AI and allowing performance monitoring for inactive model. So, we suggest following: ▪ Study the necessity of potential enhancements for LCM, and if justified, the enhancement details. The examples to study include: o Data and model management o Handling of additional conditions o Enablers for continuous (online) on-device model training/finetuning o Faster and more efficient model/functionality switching (e.g. via fall back operation to non-AI, performance monitoring for inactive model) o framework for AI/ML processing and memory
TCL	Agree
Tejas	We are supportive to this proposal
IIT Kanpur	We are fine with the proposal.
NEC	“5G NR AI/ML LCM framework as adopted in 5G Rel-19 specifications” is use case dependent and is very CSI framework dependent. It may not fit 6G use cases especially those non-MIMO use cases. We think it should be removed for this early disucssion.
vivo	@Futurewei, online model training/finetuning is different from on device. The major difference is the latency required. For online model training, it may require stringent timeline but on device training/finetuning may not necessarily have stringent timeline. If you want to add NW side online model training/finetuning, it can be explicitly added. But on device finetuning should not be removed.
OPPO	The LCM of 5G-A supports CSI use case and positioning. If we take the CSI report configured in TS 38.331 as a starting point, it may not be applicable to other non-CSI use cases, such as DMRS OH reduction. At the beginning of 6G, it would be a good chance to strive for a unified LCM framework.
InterDigital	We support the suggested change in the first bullet from Huawei. In fact, we suggest to simplify the first and second bullets as: • For 6G LCM framework for AI/ML for air interface, study potential enhancements relative to 5G

	NR AI/ML LCM framework. We are fine to keep the examples (including performance monitoring bullet suggested by Fujitsu) but do not think they are critical since they encompass all of LCM aspects.
Samsung	Support
MediaTek	We wish to add a bullet point: "leveraging cross-use case similarities" e.g., CSI-related use cases could have same model input data (e.g., CSI-RS measurements) and some parts of the model architecture could also be reused across CSI-related use cases.
Futurewei	@vivo, we understand your view related to on-device fine-tuning. We can clarify the second bullet as follows (in blue text): ▪ AI/ML Model related metrics, including model/computation complexity/power consumption, inference latency, training latency (for online training/fine-tuning, at device and/or NW ~~when~~ if applicable), scalability, etc o FFS on whether/how to measure power consumption
Panasonic	For main bullet, we support considering the 5G NR AI/ML LCM framework as a starting point for 6G LCM framework for AI/ML for air interface. 4) For red part of "as adopted in 5G Rel-19 specifications (e.g., TS38.331)", it is not needed. For a part of example, it might not be necessary as well, but we can live with it if majority companies want to keep it.

Proposal 3.2:

For 6G LCM framework for AI/ML for air interface, consider the 5G NR AI/ML LCM framework ~~as adopted in 5G Rel-19 specifications (e.g., TS38.331)~~ as a starting point.

- Study the necessity of potential enhancements for LCM, and if justified, the enhancement details. The examples to study include:
  - Data and model management (led by RAN 2)
  - Performance monitoring
  - Handling of additional conditions, e.g., continuity of UE-side AI/ML features across cells
  - Enablers for continuous (online) ~~on-device~~ model training/finetuning
  - Framework for AI/ML processing and memory
  - Leveraging cross-use case similarities for a unified LCM

~~[Note: the study strived to provide a unified LCM across use cases]~~

Companies	Comment
FL	*FL 1) delete adopted… for more open study 2) mentioned LCM, 3) UE-sided model 4) delete on device (maybe on NW), add last bullet and delete the notes.* *Also, in my understanding, other than data collection, all other aspects maybe discussed per use case.* *I understand companies want to strive to a common "framework". However, how can the of AI receiver be the same as CSI feedback or BM? But I support MTK's comment on similarities across use cases. Not sure whether this proposal is needed or workable.*
Google	We failed to see the necessity for "Enablers for continuous (online) on-device model training/finetuning"
Qualcomm	Suggest revert back to "on-device", as we're discussing specification support for UE-side on-device training. Our second comment is that the training doesn't need to be continuous.
CMCC	We are generally OK with this proposal. But if we have the above proposals on the study aspects in each use case, maybe this proposal is not needed.
vivo	We would like to emphasize site specific finetuning. For 6G LCM framework for AI/ML for air interface, consider the 5G NR AI/ML LCM framework ~~as adopted in 5G Rel-19 specifications (e.g., TS38.331)~~ as a starting point. ▪ Study the necessity of potential enhancements for LCM, and if justified, the enhancement details. The examples to study include: o Data and model management (led by RAN 2)

	o Performance monitoring o Handling of additional conditions, e.g., continuity of UE-side AI/ML features across cells o Enablers for continuous (online/site specific) ~~on-device~~ model training/finetuning o Framework for AI/ML processing and memory o Leveraging cross-use case similarities for a unified LCM
Panasonic	We are fine with the proposal.
NTT DOCOMO	We would like to clarify that the first 3 sub-bullets are use case-specific, which can be discussed along with each use cases. But the last 3 sub-bullets are common issues across use cases, which should be treated in a common session for all use cases.
Ericsson	Suggest to split the bullets into two groups. The first group (first two bullets) for issues that need to be studied. The second group (other bullets) for issues that the necessity should be studied. Suggest to update: • "Data collection and model management (led by RAN 2)"
LG	As highlighted by DCM, we also think at least last three bullet is use case common. So, what is your plan to studying those three bullets?

3.3 Data collection framework

A number of companies discussed data collection framework in their contribution. The following summarizes the discussion points

1. Enhancement in the data collection framework for future-proof and unified (across working groups) design.
2. Scope and restrictions, e.g., whether to restrict data collection to use cases or to support generic purpose data collection.
3. Whether to introduces a new AI/ML data management plane

Some of the proposals may not be under the realm of RAN1. However, RAN1 may identify requirements which may consequently suggest enhancement in the relevant working group. With this in mind, the RAN1 study may focus in identifying requirements that may lead to data collection framework enhancement.

Conclusion 3.3(RAN 1 #122bis):

For AI/ML study in 6GR, RAN1 to study on the content and format for data collection for each use case.

Company	Comment
Lenovo	It should be clarified that the data collection in this conclusion at least includes the date collection for model training (at least for offline training). Both NW-side and UE-side data collection should be considered.
NTT DOCOMO	Support.
Fujitsu	Is this proposal applicable to offline training, online training and finetuning?
CATT, CICTCI	Support. But another question is that, does RAN1 need to analyze data collection framework in physical layer in high-level? For example, should RAN1 analyze the pros and cons if CSI related AI/ML data collection is imbedded in CSI reporting framework in physical layer?
SPRD	OK
Nokia	Let RAN2 discuss this.
SK Telecom	Support in principle for studying this but not sure if this would be the scope of RAN1.
CMCC	Support. When we discuss the requirements of data collection, both offline training, online training can be considered subject to each use case.
LG Electronics	Agree in principle, but what is the motivation of this conclusion. Anyway, when we discuss each use case, this part can be also studied.
TCL	Agree
vivo	Support
OPPO	Open to discuss. At least the content of data collection of RAN1 studied use cases should fall into RAN1 territory.

Ericsson	Support in general. Suggesting updating “each use case” to “each use case led by RAN1”
Samsung	Support
Panasonic	Support

Conclusion 3.3(new):
For AI/ML study in 6GR, RAN1 to study on the content and format for data collection at least for offline training for each use case.

Company	Comment
Google	OK, we think another aspect is the UE complexity related aspect, e.g., CPU.
CMCC	When we discuss the requirements of data collection, we think both offline training and online training can be considered subject to each use case.
Lenovo	We are OK. Does this statement imply data collection at the UE side and the NW side? Consider adding that clarity.

4 2nd round

Proposal 3.1B:
For study/evaluation of AI/ML use cases in 6GR, consider

- Use case Pperformance related metrics (as for non-AI), including, intermediate (model) performance KPIs, link level KPIs (e.g., BLER) and system level KPIs (e.g., throughput, overhead), etc
  - For AI/ML, considering including the additional configuration/report overhead associated with inferencing, performance monitoring, online/site specific fine-tuning (if applicable), inter-vendor collaboration (if applicable)
  - Note: conclusions for use cases recommendation shall be drawn based on final performance metric (e.g., throughput) instead of intermediate KPIs
- AI/ML Model related metrics, including model/computation complexity/power consumption, required storage (at least for UE-sided model), inference latency, training latency (for online training/fine-tuning, if applicable), scalability, Inter-vendor collaboration complexity, when applicable, etc
  - FFS on whether/how to measure power consumption per use case
- Inter-vendor collaboration complexity and potential performance impact, when applicable
- Generalization performance
  - FFS on whether and how to consider realistic deployment scenarios

Note: Which types of KPI (intermediate, link-level, system-level) will be adopted and detailed metrics to be discussed per use case.
Note: RAN4 interoperability testing feasibility and complexity is up to RAN 4

Proposal 3.1C:
For study/evaluation of AI/ML use cases in 6GR, at least considering may be considered

- Common metrics as for non-AI, for example,
  - Intermediate performance KPIs (e.g., SGCS), link level KPIs (e.g., BLER) and system level KPIs (e.g., throughput vs, overhead), etc
  - Computation complexity/latency (inference/monitoring)
  - **Power consumption, if appliable**
  - Model required storage (at least for UE-sided model)
  - latencyModel required storage (at least for UE-sided model)
  - Data collection impact
  -
- Additional metrics for AI/ML, for example,
  - Scalability (refer to TR 38.843)
  - Generalization performance
    - FFS on whether and how to consider realistic deployment scenarios
  - Additional cConfiguration/report overhead associated with data collection, inferencing, performance monitoring, online/site specific fine-tuning (if applicable), inter-vendor collaboration (if applicable)
  - Model required storage (at least for UE-sided model)

- Online training/fine-tuning training latency, if applicable~~,~~
- Inter-vendor collaboration_ ~~–~~impact~~complexity~~, if applicable
- ~~—~~

Note: Details to be discussed per use case.
Note: the metrics may be applicable for both AI/ML and non-AI counter part
~~Note: RAN4 interoperability testing feasibility and complexity is up to RAN 4~~

Company	Comment
FL	It is not easy to discuss per use case, let's try to provide some guidance for further discussion for each use case distributed to each agenda
Kyocera	We are fine with the proposal in general. However, better clarification on the "including the overhead associated with inferencing" point may be needed. We understand that there is an overhead related to performance monitoring to generate ground truth reference for the model's performance evaluation. Moreover, there is also an overhead associated with fine tuning, e.g., online fine tuning where data samples are acquired by the AI/ML model residing entity (NW or UE) to fine tune the model to the new environment conditions. However, it is unclear for us the definition of the overhead associated with inference. Moreover, we think that the model complexity, power consumption and storage should be focused on UE sided AI/ML model. This should not be an issue for NW-sided models. Hence, we recommend the following update to the second bullet: .. • AI/ML Model-related metrics, including model/computation complexity, power consumption, and required storage (at least for UE-sided AI/ML models), inference latency, training latency (for online training/fine-tuning, if applicable), scalability, etc. ..
MediaTek	Since AI/ML-based techniques are intended to be discussed within each agenda along with non-AI/ML techniques, it would be better to classify the metrics in terms of those that are common for both AI/ML and non-AI/ML techniques and those that are specific to the use of AI/ML. A better wording would be as follows: For study/evaluation of AI/ML techniques for use cases in 6GR, consider ▪ Use case performance related metrics, including, link level KPIs (e.g., BLER) and system level KPIs (e.g., throughput, overhead). o ▪ AI/ML related metrics, including intermediate (model) performance KPIs (e.g., SGCS), computation complexity/power consumption, required storage, inference latency, training latency (for online training/fine-tuning, if applicable), scalability o including the overhead associated with inferencing, performance monitoring, online fine-tuning/site specific learning (if applicable) o FFS on whether/how to measure power consumption ▪ Inter-vendor collaboration complexity and potential performance impact, when applicable ▪ Generalization performance o FFS on whether and how to consider realistic deployment scenarios Note: Which types of KPI (intermediate, link-level, system-level) will be adopted and detailed metrics to be discussed per use case. Note: RAN4 interoperability testing feasibility and complexity is up to RAN 4
LG Electronics	Thanks FL for your effort. For the first note, we think the note is not needed. This is because, for the use case recommendation should be the based on the outcome by comparing AI/ML based solution and non-AI/ML based solution which is the guidance from chairman. So, we suggest the updated proposal as following: ***Proposal 3.1B:*** For study/evaluation of AI/ML use cases in 6GR, consider at least ▪ Performance related metrics, including, intermediate (model) performance KPIs, link level KPIs (e.g., BLER) and system level KPIs (e.g., throughput, overhead), etc o including the overhead associated with inferencing, performance monitoring, online

	fine-tuning/site specific (if applicable) o ~~Note: conclusions for use cases recommendation shall be drawn based on final performance metric (e.g., throughput) instead of intermediate KPIs~~ ▪ AI/ML Model related metrics, including model/computation complexity/power consumption, required storage, inference latency, training latency (for online training/fine-tuning, if applicable), scalability, etc o FFS on whether/how to measure power consumption ▪ Inter-vendor collaboration complexity and potential performance impact, when applicable ▪ Generalization performance o FFS on whether and how to consider realistic deployment scenarios Note: Which types of KPI (intermediate, link-level, system-level) will be adopted and detailed metrics to be discussed per use case. Note: RAN4 interoperability testing feasibility and complexity is up to RAN 4

Proposal x:
Endorse observation 2.1 ~observation 2.17 in R1-250xxxx.

Company	Comment
FL	I will separately submit a Tdoc for observation updates. If you have any comments. Please comment here. The version with some change mark can be found here. I will accept all changes for formal submission. http://10.10.10.10/ftp/RAN/RAN1/Inbox/drafts/11.6%20(AI_ML)/2nd%20round/R1-250xxxx%20Updates%20on%20observation%20for%206GR%20AIML%20use%20case%20identification_clean_v000.docx
Kyocera	We are ok with the cleaned version. Thanks for the effort.
MediaTek	2.1 Table B: Low overhead CSI-RS or CSI prediction with AI/ML Sub-case B: The specific model input attributed to MTK last time "Channel matrix over K CSI-RS occasions with >20 ms periodicity" can be removed, since our model input is not necessarily restricted to >20 ms. The model input in our case can be categorized in "1. Channel matrix over K CSI-RS occasions". This was updated in the earlier document, but seems it was lost in the latest version. FL: done 2.3 CSI compression and feedback with AI FL: I relied to you that if this output is bit, then it is Sub-Case A. I already marked MTK as it. Had replied to FL's comment in the 1st round document as follows. We have three separate evaluations for CSI compression in our Tdoc R1-2509148. Our JSCC and JSCCM evaluations are with Eigenvectors as model output of decoder (Section 3.1.1 of R1-2509148). These are listed in Sub-case A and B. However, we also have an evaluation of an extension of NR AI/ML CSI compression with explicit H as model output of decoder (Section 3.1.2 of R1-2509148). This does not fit into sub-case A as it is not trained with the channel model. Hence, we would like to keep it separate as e.g., Sub-case E with details as below:

Sub-use case	Sub-case E: Extension of NR AI/ML CSI compression with explicit channel feedback
Model input of decoder or model output of encoder, when applicable	Compressed CSI bits
Model output of decoder or model input of encoder, when applicable	Explicit H
Label	Explicit H
Training types	Offline training
KPI	NMSE

		Benchmark	TransNet (two-sided model from the literature)
		Model location for inference	Two-sided model
		Collaboration/interaction between UE and NW	Similar to two-sided model in NR
		Potential specification impact	1. Necessary signaling/procedure to support explicit channel feedback 2. Signalling/ procedure related to LCM for two-sided model including inter-vendor collaboration

SK Telecom

2.2 Low overhead DMRS with AI/ML receiver

FL: done

In Table C, there is still missing information as highlighted in cyan below. (And duplicated referencing for company '1')

Table C For low overhead DMRS with AI/ML receiver

Sub-use case	Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain	Sub-case B: Superimposed pilot	Sub-case C: DMRS free
Reported companies	(25) Nokia[1], Futurewei[2], Ericsson[3], ZTE[4], {Spreadtrum, UNISOC}[5], Interdigital[6], vivo[7], xiaomi[8], CMCC[9], {CATT, CICTCI}[10], Fujitsu[11], Apple[12], Samsung[13], Kyocera[14], Lenovo[15], Huawei[16], Qualcomm [17], Ofinno[18], NVIDIA[19], MediaTek[20], Lekha[21], LGE[22], DocoMo[23], BJTU[24], IIT Kanpur[25]	(12) vivo[1], CMCC[2], ZTE[3], Lenovo[4], Huawei[5], OPPO[6], NVIDIA[7], LGE[8] Xiaomi[9], InterDigital[10], DocoMo[11] Kyocera[12]	(9) BJTU[1], Huawei[2], NVIDIA[3], MediaTek[4], Lenovo[5], SKT[6], Nokia[7], DocoMo[8], NTT[9]
Model input	1. Received signal/estimated channel at DMRS and/or data REs and received signal on data [1,2,6,13, 22,15,3,17,10,4, 20,6,18,23,19,8, 24] 1a. additionally noise variance [1,13,24] 2. Received signal/estimated channel at DMRS[2,7, 8,11,12,13,16,5,23,25]	1. Received signal and DMRS sequence (superimposed signal) (Majority) 2. Estimated channel (in delay and/or doppler domain) from the received signal (superimposed signal) [1] For Tx side of two-sided model: modulated symbols and DMRS symbol[5]	Received signal[3, 4,6-9] For Tx side of two-sided model: coded bit[2,5,1]
Model output	1. Estimated channel at target data and/or DMRS REs[2,4,5,7,8,9,11, 12, 13,16,17,18,19,21,22,23,25] 1a. Estimated noise variance [12] 2. LLRs[1,2,3, 4, 6,10, 13,14,15,19,20, 22,24] 3. Filtering coefficients for channel estimation [7] 4. Estimated modulation	1. Estimated channel at target data REs[1,3,4,5,6,8, 10] 2.LLR[2,3, 5,6,7,8,9,11,12] 3.Estimated modulation symbols[9] For Tx side of two-sided model: superimposed signal[5]	1.LLR (majority) 2.Estimated channel[2] For Tx side of two-sided model: modulated data symbols [5,2,1]

		symbols[8] 5. Equalizer coefficients[6]		
	Label	1. Ideal channel information [2, 4, 5,7,8,9,11,12,13,15,16,17,18,22,23,25] 2. Known sequence/data[1,2,3, 4,8, 10, 13,14,15,16,20,22,19, 24] 3. Label free (unsupervised)[6, 21] 4. Estimated channel using legacy/new DMRS pattern with legacy receiver[4,8,13] 5. Estimated channel of adjacent RE (self-supervised)[13]	1. Known sequence/data [2,3,11,12,7,9] 2. Ideal channel information[1,8, 10] 3.Transmitted modulation symbols[9] 4. Estimated channel information using orthogonal DMRS[4]	1. Known sequence/data[1,3,1, 6-9] 2 Ideal channel information[2]
	Training types assumption	offline training	offline training	offline training
	KPI	MSE, BLER, throughput, raw BER	MSE, BLER, throughput	MSE, BLER, throughput
	Benchmark	With ideal channel information With conventional receiver with sparse or legacy DMRS	With ideal channel informal With conventional receiver with legacy DMRS overhead	With ideal channel information With conventional receiver with legacy DMRS overhead
	Model location for inference	UE-sided model for DL or NW-sided model for UL	UE-sided model for DL NW-sided model for UL Two-sided model[5]	NW-sided model for UL[3,4,6-9] Two-sided model[2,5,1]
	Collaboration /interaction between UE and NW	Similar to UE-sided or NW-sided model as NR	Similar to UE-sided or NW-sided model as NR Similar to two-sided model as NR	Similar to UE-sided model as NR Similar to NW-sided model as NR Similar to two-sided model as NR
	Potential spec impact	1. DMRS design 2. RAN 4: Demod requirement 3. Signalling/ procedure related to LCM for UE and/or NW sided model Etc.	1. DMRS design (Including power allocation between DMRS and data) 2. RAN 4: Demod requirement 3. Signalling/ procedure related to LCM for UE and/or NW sided model or two-sided model (including inter-vendor calibration), when applicable Etc.	1. RAN 4: Demod requirement 2. Signalling/ procedure related to LCM for UE and/or NW sided model or two-sided model (including inter-vendor calibration), when applicable 3. Constellation design and related signalling/procedure Etc.
FL	Updated in v001			

5 Proposals for online

Proposal:
Endorse observation 2.1 ~observation 2.17 in R1-2508811 section 6.

Note: this is to replace the corresponding observation in RAN 1 #122bis

Proposal 3.1C:
For study/evaluation of AI/ML use cases in 6GR, at least considering may be considered

- Intermediate performance KPIs (e.g., SGCS), link level KPIs (e.g., BLER) and system level KPIs (e.g., throughput vs overhead), etc
- Computation complexity/latency (inference/monitoring)
- Power consumption, if appliable
- Model required storage (at least for UE-sided model)
- Data collection impact
- Scalability (refer to TR 38.843)
- Generalization performance
  - FFS on whether and how to consider realistic deployment scenarios
- Configuration/report overhead associated with data collection, inferencing, performance monitoring, online/site specific fine-tuning (if applicable), inter-vendor collaboration (if applicable)
- Online training/fine-tuning training latency, if applicable
- Inter-vendor collaboration impact, if applicable

Note: Details to be discussed per use case.
Note: the metrics may be applicable for both AI/ML and non-AI counter part

6 Updated observations

6.1 Low overhead CSI-RS or CSI prediction with AI/ML

Observation 2.1
For 6GR AI/M use cases identification/categorization, [24 sources] provided preliminary simulation results and analysis on low overhead CSI-RS or CSI prediction with AI/ML.

- [24 sources] provided preliminary simulation results and analysis on frequency and/or spatial domain CSI prediction with sparse/low overhead CSI-RS with AI/ML. Detailed evaluation assumptions (model input/output/label/benchmark/KPI/ training type) and initial analysis can be found in Table A.
- [7 sources] provided preliminary simulation results (or by citing to NR study for CSI time domain prediction) and analysis on CSI time domain prediction with AI/ML wherein [3 sources] assumed Rel-19 CSI prediction while [3 sources] assumed differently. Detailed evaluation assumptions (model input/output/label/benchmark/KPI training type) and initial analysis can be found in Table B.
- [4 sources] provided preliminary simulation results and analysis on CSI prediction cross carrier/band/frequency block with AI/ML. Detailed evaluation assumptions (model input/output/label/benchmark/KPI/training type) and initial analysis can be found in Table B.
- [3 sources] provided preliminary simulation results and analysis on CSI prediction across analog beams with AI/ML. Detailed evaluation assumptions (model input/output/label/benchmark/KPI training type) and initial analysis can be found in Table B.
- [1 source] provided preliminary simulation results and analysis on CSI prediction with linear projection as pre-processing. Detailed evaluation assumptions (model input/output/label/benchmark/KPI training type) and initial analysis can be found in Table B.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table A low overhead CSI-RS or CSI prediction with AI/ML

Sub-use case	Sub-Case A: Frequency and/or spatial domain CSI prediction with sparse/low overhead CSI-RS with AI/ML
Reported companies	(24) Ericsson[1], ZTE[2], vivo[3], OPPO, Xiaomi, CMCC, Huawei[4], Samsung, Fujitsu, Apple, Qualcomm[5], Kyocera[6], Nokia[7], {Spreadtrum, UNISOC}[8], Interdigital[9], Lenovo, E[10], DoCoMo[11], CEWiT, IITM, IIT Kanpur, Tejas, {CATT, CICTCI}[12], MediaTek

Model input (for decoder of 2-sided model, when applicable)	1. Measurement of channel with sparse/low overhead CSI-RS (majority) 1a. Additional long-term multi-path power/angle/delay info information as assistance information[4] 1b. Received SRS info as assistance information for NW-sided model[6] 2. Reported CSI for NW-sided model[3,4,5] 2a. For spatial domain CSI prediction cross on-off pattern, the reported CSI can be associated with full port channel[3]
Model output (for decoder of 2-sided model, when applicable)	1. Full channel matrix (majority) 2. Eigenvector [3] for NW-sided model 3. Channel matrix/eigenvector with different/targeted antenna on/off patterns[3, 12]
Label	1. Estimated/ideal channel matrix based on full CSI-RS density(majority) 2. Ideal precoding matrix with full dimension[3] 3. Estimated/ideal channel matrix/eigenvector with different/targeted antenna on/off patterns[3, 12]
Training types	Offline training(majority) Online finetuning for UE-sided model (for NW-sided model + UE sided model without training collaboration)[4]
KPI	NMSE, SGCS, throughput, ratio of CSI-RS overhead, NES gain, CSI feedback overhead reduction[3]
Benchmark	1. non-AI based on full CSI-RS 2. non-AI based on sparse CSI-RS
Model location for inference	UE-sided model NW-sided model[2,3, 4,5,6,10] Two-sided model[2, 3] NW-sided model + UE-sided model without training collaboration[4]
Collaboration/ interaction between UE and NW	As UE-sided model in NR As NW-sided model in NR As two-sided model for CSI compression[4] in NR
Potential spec impact	1.Sparse CSI-RS design/configuration and corresponding feedback (especially for NW-sided model) 2. Signaling/ procedure related to LCM 3. Inter-vendor collaboration for two-sided model, when applicable

Table B low overhead CSI-RS or CSI prediction with AI/ML (cont.)

Sub-use case	Sub-Case B: CSI time domain prediction (as Rel-19 CSI prediction or extension)	Sub-case C: CSI prediction cross carrier/band/frequency band	Sub-Case D: CSI prediction across analog beams, or across different antenna to port virtualization	Sub-Case E: prediction with linear projection as pre-processing
Reported Companies	(7) Ericsson[2], BJTU, Samsung, MediaTek[3], LGE, vivo[1], Lenovo	(4) Samsung, Apple, LGE, DoCoMo[1]	(3) Samsung, vivo[1], Apple[2]	(1) Huawei
Model input	1. Channel matrix over K CSI-RS occasions [all] 2. Measurements of interference over K CSI-RS occasions[1] 3. Channel matrix with one P CSI-RS with 20ms periodicity and K-1 AP CSI-RS[2]	Channel matrix of carrier/band/frequency block A	1.Channel matrix/precoding matrix of Set B of beams, 2. Channel matrix of antenna to port virtualization pattern 1[2]	K past CSI information after linear projecting
Model output	1. Channel matrix of future instances 2. Interference in future instances[1]	Channel matrix of carrier/band/frequency block B	1.Channel matrix/precoding matrix of Set A of beams 2. Channel matrix of	Predicted CSI information after linear projecting at a future time instance

			antenna to port virtualization pattern 2[2]	
Label	Measurement(s) of future time occasion(s).	Channel matrix of carrier/band/frequency block B	1.Channel matrix of Set A of beams, 2. Channel matrix of antenna to port virtualization pattern 2 [2]	Ground-truth CSI information after linear projecting, based on the measurement at the future time instance
Training types assumption	offline training	offline training	offline training	Online finetuning
KPI	NMSE, SGCS, throughput, ratio of CSI-RS overhead	SGCS, NMSE, throughput, ratio of CSI-RS overhead, SGCS ratio when e-type II codebook is used for PMI feedback, NES gain.	SGCS, NMSE, throughput, ratio of CSI-RS overhead, NES gain.	SGCS
Benchmark	1. Sample and hold 2. Ground truth of future time occasions	1.Ground truth of target frequency block 2. Sample and hold	Ground truth of Set A of beams	1.Non-AI based CSI prediction 2.AI-based CSI prediction based on CSI information without linear projection
Model location for inference	UE-sided model NW-sided model[1]	UE-sided model NW-sided model[1]	UE-sided model NW-sided model[1] Two-sided model[1]	UE-sided model
Collaboration/ interaction between UE and NW	As UE-sided model in NR As NW-sided model in NR[1]	As UE-sided model in NR	As UE-sided model in NR As two-sided model in NR[1]	Similar to UE-sided model in NR
Potential spec impact	1. As AI based CSI prediction in NR 2. Reporting content, signalling and procedure for LCM for extension cases[1]	1. Cross carrier/frequency switching procedure enhancement based on predicted CSI 2. signalling/ procedure related to LCM	1.CSI-RS configuration for predicted beams or different antenna to port virtualization 2. signalling/ procedure related to LCM	Signaling/ procedure related to LCM considering online finetuning

6.2 Low overhead DMRS with AI/ML receiver

Observation 2.2
For 6GR AI/ML use cases identification/categorization, [23 sources] provided preliminary simulation results and analysis on low overhead DMRS with AI/ML receiver.

- [25 sources] provided preliminary simulation results and analysis on sparse orthogonal DMRS in frequency and/or time domain with AI/ML receiver.
- [12 sources] provided preliminary simulation results and analysis on superimposed pilot with AI/ML receiver.
- [9 sources] provided preliminary simulation results and analysis on DMRS free with AI/ML receiver.
- Detailed evaluation assumptions (model input/output/label/benchmark/KPI/ training type) and initial analysis can be found in Table C.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table C For low overhead DMRS with AI/ML receiver

Sub-use case	Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain	Sub-case B: Superimposed pilot	Sub-case C: DMRS free

Reported companies	(25) Nokia[1], Futurewei[2], Ericsson[3], ZTE[4], {Spreadtrum, UNISOC}[5], Interdigital[6], vivo[7], xiaomi[8], CMCC[9], {CATT, CICTCI}[10], Fujitsu[11], Apple[12], Samsung[13], Kyocera[14], Lenovo[15], Huawei[16], Qualcomm [17], Ofinno[18], NVIDIA[19], MediaTek[20], Lekha[21], LGE[22], DocoMo[23], BJTU[24], IIT Kanpur[25]	(12) vivo[1], CMCC[2], ZTE[3], Lenovo[4], Huawei[5], OPPO[6], NVIDIA[7], LGE[8] Xiaomi[9] , InterDigital[10], DocoMo[11], Kyocera[12]	(9) BJTU[1], Huawei[2], NVIDIA[3], MediaTek[4], Lenovo[5], SKT[6] , Nokia[7], DocoMo[8], NTT[9]
Model input	1. Received signal/estimated channel at DMRS and/or data REs and received signal on data [1,2,6,13, 22,15,3,17,10,4, 20,6,18,23,19,8, 24] 1a. additionally noise variance [1,13,24] 2. Received signal/estimated channel at DMRS[2,7, 8,11,12,13,16,5,23,25]	1. Received signal and DMRS sequence (superimposed signal) (Majority) 2. Estimated channel (in delay and/or doppler domain) from the received signal (superimposed signal) [1] For Tx side of two-sided model: modulated symbols and DMRS symbol[5]	Received signal[3, 4,6-9] For Tx side of two-sided model: coded bit[2,5,1]
Model output	1. Estimated channel at target data and/or DMRS REs[2,4,5,7,8,9,11, 12, 13,16,17,18,19,21,22,23,25] 1a. Estimated noise variance [12] 2. LLRs[1,2,3, 4, 6,10, 13,14,15,19,20, 22,24] 3. Filtering coefficients for channel estimation [7] 4. Estimated modulation symbols[8] 5. Equalizer coefficients[6]	1. Estimated channel at target data REs[1,3,4,5,6,8, 10] 2.LLR[2,3, 5,6,7,8,9,11,12] 3.Estimated modulation symbols[9] For Tx side of two-sided model: superimposed signal[5]	1.LLR (majority) 2.Estimated channel[2] For Tx side of two-sided model: modulated data symbols [5,2,1]
Label	1. Ideal channel information [2, 4, 5,7,8,9,11,12,13,15,16,17,18,22,23,25] 2. Known sequence/data[1,2,3, 4,8, 10, 13,14,15,16,20,22,19, 24] 3. Label free (unsupervised)[6, 21] 4. Estimated channel using legacy/new DMRS pattern with legacy receiver[4,8,13] 5. Estimated channel of adjacent RE (self-supervised)[13]	1. Known sequence/data [2,3,11,12,7,9] 2. Ideal channel information[1,8, 10] 3.Transmitted modulation symbols[9] 4. Estimated channel information using orthogonal DMRS[4]	1. Known sequence/data[1,3-9] 2 Ideal channel information[2]
Training types assumption	offline training	offline training	offline training
KPI	MSE, BLER, throughput, BER	MSE, BLER, throughput	MSE, BLER, throughput
Benchmark	With ideal channel information With conventional receiver with sparse or legacy DMRS	With ideal channel informal With conventional receiver with legacy DMRS overhead	With ideal channel information With conventional receiver with legacy DMRS overhead
Model location for inference	UE-sided model for DL or NW-sided model for UL	UE-sided model for DL NW-sided model for UL Two-sided model[5]	NW-sided model for UL[3,4,6-9] Two-sided model[2,5,1]
Collaboration/ interaction between UE and NW	Similar to UE-sided or NW-sided model as NR	Similar to UE-sided or NW-sided model as NR Similar to two-sided model as NR	Similar to UE-sided model as NR Similar to NW-sided model as NR Similar to two-sided model as NR
Potential spec	1. DMRS design	1. DMRS design	1. RAN 4: Demod

impact	2. RAN 4: Demod requirement 3. Signalling/ procedure related to LCM for UE and/or NW sided model Etc.	(Including power allocation between DMRS and data) 2. RAN 4: Demod requirement 3. Signalling/ procedure related to LCM for UE and/or NW sided model or two-sided model (including inter-vendor calibration), when applicable Etc.	requirement 2. Signalling/ procedure related to LCM for UE and/or NW sided model or two-sided model (including inter-vendor calibration), when applicable 3. Constellation design and related signalling/procedure Etc.

6.3 CSI compression and feedback with AI

Observation 2.3

For 6GR AI/ML use cases identification/categorization, [13 sources] provided preliminary simulation results and analysis on CSI compression and feedback with AI.

- [10 sources] provided preliminary simulation results and analysis on CSI compression with joint source and channel coding (JSCC)
- [12 sources] provided preliminary simulation results and analysis on CSI compression with joint source, channel coding and modulation (JSCM)
- [2 sources] provided preliminary simulation results and analysis on CSI feedback with downloadable basis/codebook.
- [4 sources] provided preliminary simulation results (or cite to NR AI/ML for CSI compression simulation results) and analysis on CSI reconstruction with CSI feedback with SRS (assuming separate source and channel coding).
    - Besides, [1 source] citing to NR study
- Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table D.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table D CSI compression and feedback with AI

Sub-use case	Sub-case A: CSI compression with JSCC	Sub-case B: CSI compression with JSCM	Sub-case C: DLable basis/codebook	Sub-case D: CSI reconstruction with CSI feedback with SRS (assuming SSCC)
Reported companies	(10), Samsung[2], vivo[3], {Pengcheng, ZGC}, Lenovo, OPPO, MediaTek[1], Fujitsu, BJTU, {BUPT, ZGC}, CMCC	(12) BJTU[1], Samsung[2], OPPO[3], {Pengcheng, ZGC}[4], vivo[5], CMCC, ZTE, {BUPT, ZGC}, Fujitsu, Apple[6], Lenovo, MediaTek[7]	(2) ZTE[1], Samsung	(4) Qualcomm[1], vivo[2], Samsung[3], ZTE[4] (1)LGE[5] (citing to NR study)
Model input of decoder (or model output of encoder, when applicable)	UE-part output/NW-part input: 1. (Received) compressed CSI bits[all] 1a. additionally estimated channel based on SRS[2,3]	UE-part output/NW-part/NW-sided input: 1. (Received) compressed CSI complex values for UE-part model[all] 2. (Received) compressed CSI complex values via a projection matrix[1,2,3,5,7]	1.Amplitudes and phases obtained by a look up table based on feedback CSI bits 1a. Selected basis[1]	UE-part output/NW-part/NW-sided input: 1. Compressed CSI bits 1a CSI measurement and channel matrix feedback for subset of CSI-RS ports[3] 2. Estimated channel based on SRS
Model output of decoder (or model input of	NW-part output or UE-part input:	NW-part/sided output or UE-part input:	Reconstructed Eigenvectors	NW-part/sided output or UE-part input:

encoder, when applicable)	1. (Reconstructed) Eigenvectors [all] 2. (Reconstructed) Explicit $H^{2,3}$	1. (Reconstructed) Eigenvectors[all] 1a: SNR as additional input to encoder [6] 2. (Reconstructed) Explicit H^{2} UE-part input: 3. Received signal at sparse CSI-RS and CSI-RS sequence [1,4]		(Reconstructed) Eigenvectors Explicit $H^{3,4,5}$
Label	1.Eigenvectors[all] 2.Explicit $H^{2,3}$	1.Eigenvectors[all] 2.Explicit H^{2}	Eigenvectors	1. Eigenvectors[1] 2. Explicit $H^{2,3,4,5}$
Training types	Offline training	Offline training	Offline training	Offline training
KPI	SGCS, NMSE, SE, UE complexity	SGCS, NMSE, SE, UE complexity, UPT vs overhead	SGCS, UPT vs overhead	SGCS, NMSE, UPT
Benchmark	eType II NR separate source and channel coding NR AI/ML CSI compression with separate source and channel coding and link adaptation[1]	eType II NR separate source and channel coding JSCM with two-sided model[1,2,3] NR AI/ML CSI compression with separate source and channel coding and link adaptation[7]	eType II	eType II NR AI/ML CSI compression without SRS SRS without CSI feedback
Model location for inference	Two-sided model	Two-sided model NW-sided model[1,2,3,7]	NW-sided model	Two-sided model NW-sided model[3]
Collaboration/ interaction between UE and NW	Similar to two-sided model in NR	Similar to two-sided model in NR For NW-sided model: no collaboration or Similar to NW-sided model in NR	No collaboration or Similar to NW-sided model in NR	Similar to two-sided model in NR For NW-sided model: no collaboration or Similar to NW-sided model in NR
Potential specification impact	1. Necessary signalling/ procedure to support JSCC 2. Signalling/ procedure related to LCM for two-sided model including inter-vendor collaboration	1. Necessary signalling/ procedure to support JSCM 2. Projection matrix design for NW-sided model, when applicable 3. Signalling/ procedure related to LCM with NW-sided model or two-sided model including inter-vendor collaboration, when applicable 4. RAN4 requirements, e.g., EVM	1. Downloadable basis/codebook related signalling/ procedure 2. Signalling/ procedure related to LCM with NW-sided model	1. Necessary signaling/procedure to support lower overhead and/or simpler CSI feedback 2. Signalling/ procedure related to LCM for two-sided model including inter-vendor collaboration

6.4 AI/ML for beam management and extension

Observation 2.4

For 6GR AI/ML use cases identification/categorization, [13 sources] provided preliminary simulation results and analysis on AI/ML for beam management and extension.

- [7 sources] provided preliminary simulation for DL Tx beam management and analysis on inter-cell/inter-TRP/M-TRP DL Tx beam prediction and management.
    - Besides, [6 sources] citing to NR study for DL Tx beam management and analysis on inter-cell/inter-TRP/M-TRP DL Tx beam prediction and management.
- [6 sources] provided preliminary simulation results and analysis on cross frequency DL Tx beam/beam pair prediction.
- [2 sources] provided preliminary simulation results and analysis on Tx-Rx beam pair prediction.
- [3 sources] provided preliminary simulation results for beam management and analysis on beam prediction for initial access.
    - Besides, [8 sources] citing to NR study for beam management and analysis on beam prediction for initial access.
- [1 source] provided preliminary simulation results and analysis on DL Tx beam prediction for spatial and/or temporal domain with additional local UE information.
- [1 source] provided preliminary simulation results and analysis on reinforcement learning-based approach beam selection
- [2 sources] provided preliminary simulation results and analysis on weak DL Tx beam prediction for MU-MIMO operation
- Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table E.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table E AI/ML for beam management and extension

Sub-use case	Sub-case A: Inter-Cell/M-TRP DL Tx beam prediction and management	Sub-Case B: Cross frequency DL Tx beam /beam pair prediction	Sub-Case C: Tx-Rx beam pair prediction	Sub-Case D: Beam prediction for initial access	Sub-Case E: DL Tx beam prediction for spatial and/or temporal domain with additional local UE information	Sub-Case F: reinforcement learning-based approach beam selection	Sub-Case G: Weak DL Tx beam prediction for MU-MIMO operation
Reported companies	(7) Nokia, ZTE, xiaomi, CEWiT, DoCoMo, Lenovo, BJTU (6) Qualcomm, Samsung, LGE, NVIDIA, CEWiT, {Spreadtrum, UNISOC} (citing to NR study)	(6) Futurewei[1], xiaomi[2], Apple[3], Google, LGE, Qualcomm	(2) Ericsson, Nokia	(3) Huawei, vivo, ZTE (8) Qualcomm, Samsung, LGE, Apple Google {Spreadtrum, UNISOC}, Nokia, Lenovo (citing to NR study),	(1) Huawei	(1) Nokia	(2) Google, Panasonic
Model input	Measurements from Set B of one or more TRPs/Cells	Measurements in frequency A	1.Measurements from Set B DL Tx-Rx beam pairs.	Measurements from Set B of SSB	Measurements from Set B And additional local UE	Measurements from a set of DL Tx beam scheduling stats (at the	L1-RSRP for Set B beams

					information (moving direction and speed) as UE side model input	NW), Cross correlation among DL Tx beams	
Model output	Predicted best beam information and/or predicted measurements from Set A of target cell/TRP(s) of current or future time instance	Predicted cell/beam/beam pair[1,3] related information of frequency B of current or future time instance	1.Predicted best DL Tx-Rx beam pairs information from Set A DL Tx-Rx pairs.	Predicted best DL Tx beam information (and/or predicted measurements from Set A of current or future time instance	Predicted Best beam indexes (probability of each Tx beam in Set A to be the Top-1 Tx beam) and/or Predicted measurements from Set A of current or future time instance	Selected beam index for scheduling UE(s)	Predicted DL Tx beam index(es) from Set A beams with lowest RSRPs
Label	Measurements or Top beams of Set A of target cell/TRP(s)	Measurements on cell(s)/beam(s)beam pair(s)[1,3] of frequency B	Measurements or Top beams pairs of Set A Tx-Rx pair	Measurements or Top beams of Set A	Measurements or Top beams of Set A	label-free (online learning)	Measurements from Set A beams
Training types assumption	offline training	offline training	offline training	offline training	offline training; online finetuning (for UE side model)	Online learning	Offline training
KPI	Prediction cell/beam/measurement accuracy, Throughput, RS overhead reduction	Prediction beam /beam pair[1,3]/measurement accuracy, RS overhead reduction	Prediction beam/measurement accuracy, RS overhead reduction, throughput, L1-RSRP difference	Prediction beam accuracy, 95th-percentile Top-1 RSRP error	Prediction beam/measurement accuracy	Throughput, End to end packet latency	Beam prediction accuracy, NES gain
Benchmark	Based on Set A Based on Set B	Measurements on cell(s)/beam(s)/pair (pairs)[1,3] of frequency B	Based on Set A Based on Set B Rel-19 DL Tx beam prediction	Based on Set A Based on Set B	NR beam prediction with AI/ML	Beam with largest RSRP (from the set) consider as the scheduling beam	based on Set A/B beams
Model location for inference	UE-sided model or NW-sided model	UE-sided model or NW-sided model	UE-sided model	UE-sided model or NW-sided model	NW-sided model + UE-sided model without training collaboratio	NW-sided model	UE-side model or NW-sided model

					n		
Collaboration/ interaction between UE and NW	As UE-sided or NW-sided model in NR	As UE-sided or NW-sided model in NR	As UE-sided model in NR	Similar to UE-sided or NW-sided model in NR	As UE-sided or NW-sided model in NR	No collaboration	Similar to UE-sided or NW-sided model in NR
Potential spec impact	1. Inter-Cell/M-TRP beam prediction related singling/procedure 2. Signalling/ procedure related to LCM for NW-sided model or UE-sided model	1. Cross frequency DL Tx beam/Tx-Rx beam pair[1,3] prediction related signalling / procedure 2. Signalling/ procedure related to LCM for NW-sided model or UE-sided model	1.Signalling/ procedure related to LCM for UE-sided model	1. Initial access/Random access related to beam prediction 2. Signalling/ procedure related to LCM for NW-sided model or UE-sided model	1. As NR AI for BM; 2. Signalling/ procedure related to NW-sided model + UE-sided model. 3. Signalling/ procedure related to online finetuning, if any	1. Signalling/ procedure related to exploration phase (to mitigate the impact of exploration	1. UE reports the recommended beams 2. Signalling/ procedure related to LCM for NW-sided model or UE-sided model

6.5 AI/ML for (de)modulation

Observation 2.5
For 6GR AI/ML use cases identification/categorization, [6 sources] provided preliminary simulation results and analysis on AI/ML for (de)modulation.

- [6 sources] provided preliminary simulation results and analysis on modulation constellation design with the help of AI, and with non-AI or AI receiver.
- [3 sources] provided preliminary simulation results and analysis on AI-based modulation and precoding with two-sided model.
- Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in Table F.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table F AI/ML for (de)modulation

Sub-use case	Sub-use case A: AI-based (de)modulation	Sub-use case B: AI-based modulation and precoding
Reported companies	(6)ZTE[1], OPPO[2], vivo[3], Lenovo[4], Xiaomi[5], Futurewei[6]	(3) ZTE, OPPO, Lenovo
Model input	For constellation design 1. Coded bits [1,2,3,4,5,6] 2. Channel characterization and modulation order[4] For AI receiver 1.Received signal[2,3,4,6]	Encoder: Coded bits Decoder: Estimated symbols
Model output	For constellation design 1. Learned constellation [1,2,3,4,5,6] 2. Probability of constellation points [4] For receiver 1. (Soft) LLR[2,3,4,6]	Encoder: modulated symbols after layer mapping Decoder: Soft LLR
Label	Known coded bits	Known coded bits
Training types	Offline training	Offline training

KPI	BLER	BLER
Benchmark	Uniform QAM with legacy receiver	Uniform QAM with legacy receiver and NR layer mapping
Model location for inference	1.NA (AI for constellation design with legacy receiver) [1,2,3,4,5] 2. NA (constellation learned offline jointly with AI receiver and applied during inference)[6] 3.Receiver-sided model[2,3,4,6]	Two-sided model
Collaboration/interaction between UE and NW	NA or Similar to NW-sided model or UE-sided model in NR	Similar to two-sided model in NR
Potential specification impact	1. Constellation design and related signaling/procedure 2. Signaling/ procedure related to LCM for NW-sided model or UE-sided model 3. RAN4 requirements, e.g., EVM	1. Modulation design and layer mapping design 2. Signaling/ procedure related to LCM for two-sided model including inter-vendor collaboration 3. RAN4 requirements, e.g., EVM

6.6 AI-based none-linearity handling at transmitter or receiver

Observation 2.6
For 6GR AI/ML use cases identification/categorization, [5 sources] provided preliminary simulation results and analysis on AI-based none-linearity handling at transmitter or receiver.

- [5 sources] provided preliminary simulation results and analysis on AI-based DPoD/None-linearity compensation at receiver.
- [3 sources] provided preliminary simulation results and analysis on AI-based DPD at transmitter.
- Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in Table G.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table G AI-based none-linearity handling at transmitter or receiver

Sub-use case	Sub-use case A: AI-based DPoD/None-linearity compensation	Sub-use case B: AI-based DPD
Reported companies	(5) Samsung[1], Ericsson[2], OPPO[3], vivo[4], Huawei[5]	(3) vivo[2], Huawei[1], Nokia[3]
Model input	1. Received signal[1,3,4,5]	1. Time domain samples before pre-distortion[1,2,3] 2. Time domain samples for calibration signals.[2]
Model output	1. Compensated signal in time domain[1,2,4,5] 2. Soft bits[2,3]	1.Time domain samples after pre-distortion[1,2,3] 2. Fitted PA model[2]
Label	1. DMRS[1] 2. Known bit sequence[2,3,4] 3. time domain samples from known sequence[5]	1. Time domain samples[1,2,3] 2. Real PA distortion[2]
Training types	Online training/finetune[1] Offline training	Offline training Online training/finetune[2]
KPI	BLER, MPR, EVM, throughput	BLER, EVM, MPR
Benchmark	Without compensation GMP model based NC[1]	No DPD
Model location for inference	NW-sided model	UE-sided model
Collaboration/interaction between UE and NW	Similar to NW-sided model as NR	Similar to UE-sided model as NR
Potential specification impact	1. RAN 4 requirements, e.g. EVM 2. DMRS/Sequence design/selection, Tx power determination	1. RAN4 requirements, e.g. EVM 2. Tx power determination, SRS design and/or PUSCH scheduling design for

	3. Signaling/ procedure related to LCM for NW-sided model	data collection 3. Signaling/ procedure related to LCM for UE-sided model

6.7 AI/ML for SRS

Observation 2.7

For 6GR AI/ML use cases identification/categorization, [5 sources] provided preliminary simulation results and analysis on AI/ML for SRS

- [5 sources] provided preliminary simulation results and analysis on low overhead SRS with AI/ML
- [1 source] provided preliminary simulation results and initial analysis on low PAPR SRS sequence design with help of AI/ML
- [1 source] provided preliminary simulation results and initial analysis on AI/ML based SRS power imbalance compensation

Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and analysis in Table H.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table H AI/ML for SRS

Use case	Sub-Case A: Low overhead SRS with AI/ML	Sub-Case B: Low PAPR SRS sequence design	Sub-Case C: AI/ML based SRS power imbalance compensation
Reported companies	(5) {Spreadtrum, UNISOC}, vivo, Huawei[1], Kyocera, LGE	(1) vivo	(1) Huawei
Model input	Measurement of channel with low overhead SRS of frequency/temporal/spatial domain 1a. assistance info of estimated channel of DMRS[1]	Sequence index	UL measured channel matrix from SRS with IL imbalance
Model output	Estimated channel	Learn sequences	DL channel matrix with IL compensated
Label	Ideal channel information	Label free	UL SRS measurement without IL (assuming it is compensated by UE at certain conditions) or DL CSI-RS measurement
Training types	Offline training	Offline training	offline training
KPI	SCGS, throughput	PAPR, SGCS, Cross-correlation between SRS sequences	SGCS
Benchmark	With legacy SRS With ideal channel information	Legacy SRS sequence	1. SRS without IL imbalance; 2. non-AI based SRS IL imbalance compensation
Model location for inference	NW-sided model	NW-sided model or Without model for inference	NW-sided model
Collaboration/interaction between UE and NW	Similar to NW-sided model in NR	No collaboration for no model Similar to NW-sided model in NR	Similar to NW-sided model in NR
Potential specification impact	1.Sparse SRS design 1a. Indication of the association between DMRS port and SRS port[1] 2. Signalling/ procedure related to LCM for NW-sided model	1. SRS design 2. Signaling/procedure related to DLable/ULable SRS sequence, when applicable 3. Signalling/ procedure related to LCM for NW-sided	1. Inference: UE reporting on the IL range for ensuring generalization 2. Signalling/procedure related to LCM for NW-sided model

		model, when applicable	

6.8 AI-enabled UL precoder indication

Observation 2.8

For 6GR AI/ML use cases identification/categorization, [5 sources] provided preliminary simulation results and analysis on AI-enabled UL precoder indication with detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in Table I.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table I AI-enabled UL precoder indication

Use case	AI-enabled UL precoder indication
Reported companies	(5) vivo[1], Fujitsu[2], Samsung[3], LGE[4], ZTE[5]
Model input of decoder or model output of encoder	UL precoder indicator/compressed UL precoder
Model output of decoder or model input of encoder	(Reconstructed) eigenvectors of UL channel
Label	Estimated eigenvectors of UL channel based on SRS measurement
Training types	offline training online finetune[1]
KPI	SCGS, BLER
Benchmark	NR TPMI codebook
Model location for inference	No model for inference [1,2,3] Two-sided model[1,2,4, 5]
Collaboration/interaction between UE and NW	Similar to one-sided model or no collaboration in NR [1,2,3] Similar as two-sided model in NR[1,2,4, 5]
Potential specification impact	1.The signaling/procedure related to the download/upload of UL codebooks/compressed UL precoder 2. LCM procedure to facilitate the training of the downloadable/uploadable UL codebooks when no model for inference, [1,2,3] 3. Signalling/ procedure related to LCM for two-sided model including inter-vendor collaboration, when applicable[1,2]

6.9 AI/ML based waveform for PAPR reduction

Observation 2.9

For 6GR AI/ML use cases identification/categorization, [3 sources] provided preliminary simulation results and analysis on AI/ML based waveform for PAPR reduction with detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis in Table J.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table J AI/ML based waveform for PAPR reduction

Use case	AI/ML based waveform for PAPR reduction
Reported companies	(3) vivo[1], Samsung[2], Huawei[3]
Model input	Symbols in frequency domain
Model output	For model output of encoder for UE-sided/NW-part of two-sided model: transformed/precoded symbols in frequency domain For output of decoder for NW-part of two-sided model: 1. LLR[1,3] 2. Symbols in frequency domain[2]
Label	Label free[2,3] Known bit sequences or its LLR[1,3]
Training types	offline training
KPI	BLER, CCDF of PAPR(UL), throughput (DL)
Benchmark	DFT-s-OFDM
Model location for inference	Two-sided model UE-sided model (for frequency domain shaping)[1,3]

Collaboration/interaction between UE and NW	Similar to two-sided model in NR No collaboration for UE-sided model[1]
Potential specification impact	1. Signaling/ procedure related to LCM for two-sided model including inter-vendor collaboration, when applicable 2. Signaling/ procedure related to LCM for UE-sided model [1,3] 3. Related RAN4 requirements

6.10 AI/ML based HARQ-ACK feedback

Observation 2.10
For 6GR AI/ML use cases identification/categorization, [2 sources] provided preliminary simulation results and analysis on AI/ML based HARQ-ACK feedback with detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis in Table K.
Note: whether/how to capture the observation in the TR is a separate discussion.
Table K. AI/ML based HARQ-ACK feedback

Use case	AI/ML based HARQ-ACK feedback
Reported companies	(2) Qualcomm, vivo
Model input	HARQ ACK/NACK bit sequence
Model output	Learned sequences/modulated symbols
Label	HARQ-ACK/NACK bit sequence
Training types	Offline training
KPI	BLER
Benchmark	NR RM code for up to 11bits with Maximum Likelihood (ML) receiver
Model location for inference	No model for inference
Collaboration/interaction between UE and NW	No collaboration
Potential specification impact	1.Learned sequence/modulated symbols design 2.Downloadable sequence/modulated symbols related signalling/ procedure for HARQ-ACK 3. Related RAN4 requirements

6.11 AI/ML for PDCCH

Observation 2.11
For 6GR AI/ML use cases identification/categorization, [2 sources] provided preliminary simulation results and analysis on AI/ML for PDCCH

- [2 sources] provided preliminary simulation results and analysis on prior-information-aided DCI decoding,
- [one source] provided preliminary simulation results and analysis on lossless DCI compression,

Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table L
Note: whether/how to capture the observation in the TR is a separate discussion.

Table L AI/ML for PDCCH

Use Case	Sub-Case A: Prior-Information-Aided DCI Decoding	Sub-Case B: Lossless DCI Compression
Reported companies	(2) CMCC, OPPO	(1) CMCC
Model input	Historical DCI payloads (including last successfully decoded DCI payload)	Historical DCI payload Received compressed DCI payload
Model output	Predicted LLR for intended DCI payload	Predicted DCI payload
Label	DCI payload sequences	DCI payload sequences
Training types	Offline training at the UE side	Offline training at the NW side, and model delivery to UE side
KPI	BLER performance	BER and sample-level prediction

		accuracy; DCI overhead reduction
Benchmark	Traditional DCI decoder	Traditional DCI design
Model location for inference	UE-sided model	UE-sided model + NW-sided model
Collaboration/interaction between UE and NW	Similar to UE-sided model in NR	Model transfer from NW to UE
Potential specification impact	1. Signalling/configuration design for prior-information-aided DCI decoder. 2. Signalling/ procedure related to LCM for UE-sided model	1. Signalling/configuration design for Lossless DCI Compression. 2. Signalling/ procedure related to LCM including model transfer

6.12 AI/ML for power control

Observation 2.12
For 6GR AI/ML use cases identification/categorization, [2 sources] provided preliminary simulation results and analysis on AI/ML for power control
[2 sources] provided preliminary simulation results and analysis on pathloss prediction in the spatial, temporal, and/or frequency domain, to use the predicted pathloss in UL (PUSCH/PUCCH/PRACH/SRS) power control.
[one source] provided preliminary simulation results and analysis on UL closed-loop power control with an NW-sided AI/ML model, where the model predicts the optimal power adjustment (or TPC command index) for the UE,
Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table M
Note: whether/how to capture the observation in the TR is a separate discussion.

Table M AI/ML for power control

Use Case	Sub-Case A: UL closed-loop power control with an NW-sided AI/ML model, where the model predicts the optimal power adjustment (or TPC command index) for the UE.	Sub-Case B: Pathloss prediction in the spatial, temporal, and/or frequency domain, to use the predicted pathloss in UL(PUSCH/PUCCH/PRACH/SRS) power control.
Reported companies	(1) Nokia	(2) Nokia, Google (Cite R19 results)
Model input	UL SINR measurement, UE Tx power estimate (derived from Pcmax, P0, PL alpha, pathloss measurement), and PUSCH allocation size	L1-RSRP measurements from a sub-set/set of RSs/beams (Set B). input can consider history of measurements
Model output	Predicted TPC command index	Predicted pathloss value(s) (or predicted L1-RSRP(s)) for a set of RSs/beams (Set A). output can consider future instances
Label	Optimal TPC command index (offline learning) label-free (online learning)	Pathloss value(s) (or L1-RSRP(s)) for a set of RSs/beams (Set A)
Training types	Offline and Online learning	Offline training
KPI	UL throughput.	Pathloss prediction accuracy, throughput, RS overhead reduction, Complexity.
Benchmark	1. UL Power control with optimized OLPC parameters 2. UL Power control with optimized OLPC parameters and possibly legacy CLPC algorithms (with 5G TPC tables).	Pathloss estimation based on Set A Pathloss estimation based on Set B
Model location for inference	NW-sided model	UE-sided model NW-sided model
Collaboration/	None	As UE-sided or NW-sided mode in nRl

interaction between UE and NW		
Potential specification impact	Configurability of the values in TPC command tables or an extended TPC command table (compared to NR).	1. Pathloss prediction related signalling/procedure 2. Signalling/ procedure related to LCM for UE-sided or NW-sided model 3. RAN4 performance requirements and test cases, including defining new requirements related to pathloss reference signal (PL-RS) measurement and activation delays of TCI state(s).

6.13 AI/ML for RACH related design

Observation 2.13
For 6GR AI/ML use cases identification/categorization, [3 source] provided preliminary simulation results and analysis on AI/ML for RACH related design

- [2 sources] provided preliminary simulation results and analysis on early contention resolution in RACH,
- [1 source] provided preliminary simulation results and analysis on Low PAPR sequence design for PRACH

Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table N
Note: whether/how to capture the observation in the TR is a separate discussion.

Table N AI/ML for RACH related design

Sub-use case	Sub-Case A: Early contention resolution in RACH	Sub-Case B: Low PAPR sequence design for PRACH
Reported Companies	(2) Ofinno[1], Samsung[2]	(1) vivo
Model input	1.Received PRACH signal (e.g., preamble waveform)[1] 2. Detected PRACH signal with PDP or truncated PDP in time domain for each preamble detection window[2]	The number of PRACH sequences
Model output	Predicted number of UEs that transmitted the same preamble for given PRACH resources	A set of learned sequences
Label	Ground-truth number of UEs that transmitted the same preamble	Label free
Training types assumption	Offline training	Offline training
KPI	Prediction accuracy of UE multiplicity, RACH access delay, first-attempt success probability, and false alarm detection rate	PAPR/DCM, Cross-correlation between sequences
Benchmark	First-attempt success rate based on legacy PRACH receiver	Legacy PRACH sequence
Model location for inference	NW-side model	Without model for inference
Collaboration/interaction between UE and NW	Similar to NW-sided model in NR	No collaboration
Potential spec impact	1. Signaling/procedure related to random access, including Mgs.3 grant for more than one UEs selected the same PRACH sequence 2. Signaling/procedure related to LCM for NW-sided model	Standardized learned sequences for PRACH

6.14 Site Specific Learning for AI/ML using RAN Digital Twin

Observation 2.14
For 6GR AI/ML use cases identification/categorization, [3 sources] provided preliminary simulation results and analysis on Site Specific Learning for AI/ML using RAN Digital Twin,
Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table O
Note: whether/how to capture the observation in the TR is a separate discussion.

Table O Site Specific Learning for AI/ML using RAN Digital Twin

Sub-use case	Site Specific Learning for AI/ML using RAN Digital Twin
Reported Companies	(3) DeepSig[1], NVIDIA[2], Huawei[3]
Model input	1.Received signal/estimated channel at DMRS and received signal on data, and the channel information generated by digital twin[1,3] 2.Noisy least square channel estimate on DMRS REs[2] (The model is trained/finetuned with digital twin generated data)
Model output	1.Decoded bit[1,3] 2.Denoised channel estimate on DMRS REs[2] (The model is trained/finetuned with digital twin generated data)
Label	Ground truth of target bit[1,3] Ground truth of channel[2]
Training types assumption	Offline training
KPI	BLER, throughput
Benchmark	Legacy receiver without the help of digital twin
Model location for inference	NW-sided model
Collaboration/interaction between UE and NW	Similar to NW-sided model in NR
Potential spec impact	Signalling/procedure related to LCM for NW-sided model (including the generation of RAN digital twin[3])

6.15 Digital twin construction related use cases

Observation 2.15
For 6GR AI/ML use cases identification/categorization, [1 source] provided preliminary simulation results and analysis on digital twin construction related use cases,

- [1 source] provided preliminary simulation results and analysis on sensing based RAN digital twin construction with NW-side AI/ML model,
- [1 source] provided preliminary simulation results and analysis on AI/ML-enabled RAN digital twin with distributed model,

Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table P
Note: whether/how to capture the observation in the TR is a separate discussion.

Table P Digital twin construction related use cases

Sub-use case	Sub-Case A: Sensing based RAN digital twin construction with NW-side AI/ML model	Sub-Case B: AI/ML-enabled RAN digital twin with distributed model
Reported Companies	(1) Huawei	(1) Huawei
Model input	Point cloud sensed by the BS with mono-static sensing and sensed/reported by UEs with bi-static sensing	UE-part models: local sparse point cloud NW-part model: latent space information from multiple UEs
Model output	3D point cloud representing the static environment	UE-part models: compressed latent space information NW-part model: global point cloud
Label	Ground truth point cloud	Ground truth point cloud
Training types assumption	Offline training	Offline training (adopted in simulation)

		Online finetuning (can be optionally considered)
KPI	Sensing accuracy metric: root mean square error (RMSE) of point cloud. RMSE= $\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i-\hat{x}_i\right)^2}$ is the square root of the average of the squared errors between each sensed point (x_i in forms of coordinates) and ground truth point ($\hat{x}_i$ in forms of coordinates) in the point cloud including n points with {x, y, z} dimensions.	1. Overhead metric: Feedback bits per point 2. Sensing accuracy metric: intersection-over-union (IoU), edge detection probability
Benchmark	BS side mono-static sensing only to construction RAN digital twin	1. Single UE sensing (to justify sensing accuracy metric of using distributed model). 2. Raw data transmission (to justify overhead metric of using distributed model).
Model location for inference	NW-side model	Distributed model: a NW-side model paired with multiple UE-side models.
Collaboration/ interaction between UE and NW	Similar to NW-sided model in NR	Similar to two-sided model: UE reporting of compressed sensing results for inference. Inter-vendor training collaboration between NW side and UE side.
Potential spec impact	1. Signaling/procedure related to bi-static sensing results reported from UE 2. Signalling/procedure related to LCM for NW-sided model	1. Sensing results reported from UE in forms of compressed latent message 2. Signalling/procedure related to LCM for two-sided model including inter-vendor collaboration

6.16 AI for positioning

Observation 2.16

For 6GR AI/ML use cases identification/categorization, [1 source] provided preliminary simulation results and analysis on AI for positioning, and detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table Q

Note: whether/how to capture the observation in the TR is a separate discussion.

Table Q AI for positioning

Sub-use case	Sub-Case A: Positioning Case 1	Sub-Case B: Positioning Case 3a	Sub-Case C: Positioning Case 3b
Reported Companies	(1) Ericsson	(1) Ericsson	(1) Ericsson
Model input	Time domain samples (PDP, DP), or timing/power measurements of multiple paths	Time domain samples (PDP, DP), or timing/power measurements of multiple paths	Time domain samples (PDP, DP), or timing/power measurements of multiple paths
Model output	UE location coordinates	Intermediate positioning measurements (e.g., ToA, LOS/NLOS indicator)	UE location coordinates
Label	UE location coordinates provided by PRU or LMF	ToA; binary value of LOS/NLOS indicator; Label provided by LMF based on UE location coordinates	UE location coordinates provided by PRU or LMF
Training types assumption	Offline training	Offline training	Offline training
KPI	CDF percentiles of horizonal accuracy	CDF percentiles of horizonal accuracy	CDF percentiles of horizonal accuracy
Benchmark	DL-TDOA	UL-TDOA	UL-TDOA
Model location for inference	UE-side model	gNB-side model	LMF-side model

Collaboration/ interaction between UE and NW	LMF provides to UE all assistance information from legacy UE-based DL-TDOA, other than info #7 (Geographical coordinates of the TRPs)	gNB and LMF coordinate to configure SRS transmission by UE	gNB and LMF coordinate to configure SRS transmission by UE
Potential spec impact	Signaling related to the AI/ML functionality; Training data generation; Data collection; Model monitoring	Training data generation; Data collection; Model monitoring	Training data generation; Data collection; Model monitoring

6.17 For AI/ML related service

Observation 2.17

For 6GR AI/ML related service, for 6GR AI/ML use cases identification/categorization,

- [8 sources] provided preliminary simulation results and analysis on improved scheduling/HARQ for token traffic,
- [1 source] provided preliminary simulation results and analysis on transmission for audio token traffic

Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis in Table R.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table R for AI/ML related service

Use case	Sub-Case A: Improved scheduling/HARQ for token traffic	Sub-Case B: Transmission for audio token traffic
Reported companies	(8) Huawei[1], OPPO[2], Google[3], CMCC[4], Lenovo[5], CAICT[6], Pengcheng Lab[7], BUPT[8]	(1)vivo
Model input	Tokenizer model: • Input: Raw data (e.g., image/video/audio/text, etc.) De-tokenizer model: • Input: Tokens	Tokenizer model: • Input: Raw data (e.g., audio, etc.) De-tokenizer model: • Input: Received tokens
Model output	Tokenizer model: • Output: Tokens (e.g., tokenized image/video/audio/text) De-tokenizer model: • Output: Inference results for downstream tasks/Raw data (e.g., image/video/audio, etc.)	Tokenizer model: • Output: Tokens (e.g., tokenized audio), in the form of bits or modulated symbols De-tokenizer model: • Output: reconstructive raw data (e.g., audio, etc.)
Label	Training at OTT, transparent to RAN	Original audio data
KPI	Supported number of UEs, achievable throughput	Coverage gain for acceptable PSEQ/MOS level, supported number of UEs
Benchmark	NR scheduling/HARQ mechanism without knowledge of Token traffic	Legacy design without aware of audio token traffic Note: AI/ML based source coding for both benchmark and proposed use cases
Model location for inference	Transparent to RAN	The tokenizer model is at higher layer of the UE The de-tokenizer model is at higher layer of the NW/OTT server Note: the model may require cross layers operation
Collaboration/interaction between UE and NW	Transparent to RAN	Collaboration/interaction for two-sided model Note: the model may require cross layers operation
Potential specification	• Service awareness in RAN	• Service awareness in RAN

impact	• Token error identification, new scheduling and HARQ	• New scheduling and HARQ • New protocol design cross (e.g., physical, MAC and higher) layers for audio

7 Appendix

7.1 RAN 1 #122

Agreement

For 6GR AI/ML use cases identification/categorization, for each (sub-)use case proposed, proponent companies are encouraged to study and report the following:

- Definition of each (sub-)use case, including at least AI/ML model input/output
- The evaluation assumption, methodology, KPIs, benchmark, and preliminary simulation results
- Assumption on training types, e.g.,
  - offline training, online training/finetuning
  - Label construction (if applicable), including whether/how to obtain label data for model training
- Assumption on model location for inference, e.g., UE-sided model, NW-sided model, and two-sided model
- Collaboration/interaction between UE and NW, e.g.,
  - no collaboration/interaction
  - UE/Network collaboration targeting at separate or joint ML operation
- High level potential specification impact

7.2 RAN1 #122bis

7.2.1 *Low overhead CSI-RS or CSI prediction with AI/ML*

Observation

For 6GR AI/ML use cases identification/categorization, [24 sources] provided preliminary simulation results and analysis on low overhead CSI-RS or CSI prediction with AI/ML.

- [23 sources] provided preliminary simulation results and analysis on frequency and/or spatial domain CSI prediction with sparse/low overhead CSI-RS with AI/ML. Detailed evaluation assumptions (model input/output/label/benchmark/KPI/ training type) and initial analysis can be found in Table A.
- [6 sources] provided preliminary simulation results (or by citing to NR study for CSI time domain prediction) and analysis on CSI time domain prediction with AI/ML wherein [3 sources] assumed Rel-19 CSI prediction while [3 sources] assumed differently. Detailed evaluation assumptions (model input/output/label/benchmark/KPI training type) and initial analysis can be found in Table B.
- [4 sources] provided preliminary simulation results and analysis on CSI prediction cross carrier/band/frequency block with AI/ML. Detailed evaluation assumptions (model input/output/label/benchmark/KPI/training type) and initial analysis can be found in Table B.
- [2 sources] provided preliminary simulation results and analysis on CSI prediction across analog beams with AI/ML. Detailed evaluation assumptions (model input/output/label/benchmark/KPI training type) and initial analysis can be found in Table B.
- [1source] provided preliminary simulation results and analysis on, CSI prediction with linear projection as pre-processing. Detailed evaluation assumptions (model input/output/label/benchmark/KPI training type) and initial analysis can be found in Table B.
- ~~and time domain CSI prediction combining CSI-RS and DMRS measurements (MediaTek).~~

Note: whether/how to capture the observation in the TR is a separate discussion.

Table A

Sub-use case	Sub-Case A: Frequency and/or spatial domain CSI prediction with sparse/low overhead CSI-RS with AI/ML
Reported	(23) Ericsson[1], ZTE[2], vivo[3], OPPO, Xiaomi, CMCC, Huawei[4], Samsung, Fujitsu, Apple,

companies	Qualcomm[5], Kyocera[6], Nokia[7], {Spreadtrum, UNISOC}[8], Interdigital[9], Lenovo, LGE[10], DoCoMo[11], CEWiT, IITM, IIT Kanpur, Tejas, {CATT, CICTCI}[12]
Model input (for decoder of 2-sided model, when applicable)	1. Measurement of channel with sparse/low overhead CSI-RS (majority) 1a. Additional long-term multi-path power/angle/delay info information as assistance information[4] 2. Reported CSI for NW-sided model[3,4,5]
Model output (for decoder of 2-sided model, when applicable)	1. Full channel matrix (majority) 2. Eigenvector [3] for NW-sided model 3. Channel matrix/eigenvector with different/targeted antenna on/off patterns[3, 12]
Label	1. Estimated/ideal channel matrix based on full CSI-RS density(majority) 2. Ideal precoding matrix with full dimension[3] 3. Estimated/ideal channel matrix/eigenvector with different/targeted antenna on/off patterns[3, 12]
Training types	Offline training(majority) Online finetuning for UE-sided model (for NW-sided model + UE sided model without training collaboration)[4]
KPI	NMSE, SGCS, throughput, ratio of CSI-RS overhead
Benchmark	1. non-AI based on full CSI-RS 2. non-AI based on sparse CSI-RS
Model location for inference	UE-sided model NW-sided model[2,3, 4,5,6] Two-sided model[3] NW-sided model + UE-sided model without training collaboration[4]
Collaboration/ interaction between UE and NW	As UE-sided model in NR As NW-sided model in NR As two-sided model for CSI compression[4] in NR
Potential spec impact	1.Sparse CSI-RS design and corresponding feedback (especially for NW-sided model) 2. Signalling/ procedure related to LCM 3. Inter-vendor collaboration for two-sided model, when applicable

Table B

Sub-use case	Sub-Case B: CSI time domain prediction (as Rel-19 CSI prediction or extension)	Sub-case C: CSI prediction cross carrier/band/frequency band	Sub-Case D: CSI prediction across analog beams	Sub-Case E: prediction with linear projection as pre-processing
Reported Companies	(6) Ericsson[2], BJTU, Samsung, MediaTek[3], LGE, vivo[1]	(4) Samsung, Apple, LGE, DoCoMo[1]	(2) Samsung, vivo[1]	(1) Huawei
Model input	1. Channel matrix over K CSI-RS occasions 2. Measurements of interference over K CSI-RS occasions[1] 3. Channel matrix over K CSI-RS occasions with >20ms periodicity[3] 4 Channel matrix with one P CSI-RS with 20ms periodicity and K-1 AP CSI-RS[2]	Channel matrix of carrier/band/frequency block A	Channel matrix of Set B of beams	K past CSI information after linear projecting
Model output	1. Channel matrix of future instances 2. Interference in future instances[1]	Channel matrix of carrier/band/frequency block B	Channel matrix of Set A of beams	Predicted CSI information after linear projecting at a future time instance
Label	Measurement in future time occasions.	Channel matrix of carrier/band/frequency	Channel matrix of Set A of beams	Ground-truth CSI information after

		block B		linear projecting, based on the measurement at the future time instance
Training types assumption	offline training	offline training	offline training	Online finetuning
KPI	NMSE, SGCS, throughput, [ratio of CSI-RS overhead]	SGCS, NMSE, throughput, ratio of CSI-RS overhead	SGCS, NMSE, throughput, ratio of CSI-RS overhead	SGCS
Benchmark		1.Ground truth of target frequency block 2. Sample and hold	Ground truth of Set A of beams	1.Non-AI based CSI prediction 2.AI-based CSI prediction based on CSI information without linear projection
Model location for inference	UE-sided model NW-sided model[1]	UE-sided model NW-sided model[1]	UE-sided model NW-sided model[1] Two-sided model[1]	UE-sided model
Collaboration/ interaction between UE and NW	As UE-sided model in NR As NW-sided model in NR[1]	As UE-sided model in NR	As UE-sided model in NR	Similar to UE-sided model in NR
Potential spec impact	1. As AI based CSI prediction in NR 2. Reporting content, signalling and procedure for LCM for extension cases[1]	1. Cross carrier/frequency switching procedure enhancement based on predicted CSI 2. signalling/ procedure related to LCM	1.CSI-RS configuration for predicted beams 2. signalling/ procedure related to LCM	Signaling/ procedure related to LCM considering online finetuning

7.2.2 *Low overhead DMRS with AI/ML receiver*

Observation
For 6GR AI/ML use cases identification/categorization, [23 sources] provided preliminary simulation results and analysis on low overhead DMRS with AI/ML receiver.

- [22 sources] provided preliminary simulation results and analysis on sparse orthogonal DMRS in frequency and/or time domain with AI/ML receiver.
- [11 sources] provided preliminary simulation results and analysis on superimposed pilot with AI/ML receiver.
- [5 sources] provided preliminary simulation results and analysis on DMRS free with AI/ML receiver.
- Detailed evaluation assumptions (model input/output/label/benchmark/KPI/ training type) and initial analysis can be found in Table C.

Note: whether/how to capture the observation in the TR is a separate discussion.
Table for detailed assumptions of each category will be added.

Table C, For low overhead DMRS with AI/ML receiver

Sub-use case	Sub-case A: Sparse orthogonal DMRS in frequency and/or time domain	Sub-case B: Superimposed pilot	Sub-case C: DMRS free
Reported companies	(23) Nokia[1], Futurewei[2], Ericsson[3], ZTE[4], {Spreadtrum, UNISOC}[5], Interdigial[6], vivo[7], xiaomi[8], CMCC[9], {CATT, CICTCI}[10], Fujitsu[11], Apple[12], Samsung[13], Kyocera[14], Lenovo[15], Huawei[16],	(12) vivo [1], CMCC[2], ZTE[3], Lenovo[4], Huawei[5], OPPO[6], NVIDIA[7], LGE[8] Xiaomi[9] , InterDigital[10] , DocoMo[11] Kyocera[12]	(5) InterDigital[1], Huawei[2], NVIDIA[3], MediaTek[4], Lenovo[5]

	Qualcomm [17], Ofinno[18], NVIDIA[19], MediaTek[20], Lekha[21], LGE[22], DocoMo[23]		
Model input	1. Received signal/estimated channel at DMRS and received signal on data [1,13, 22,15,3,17,10,4, 20,6,18,23] 1a. additionally noise variance [1,13] 2. Received signal/estimated channel at DMRS[2,7, 8,11,12,13,16,5,23]	1. Received signal and DMRS sequence (superimposed signal) (Majority) 2. Estimated channel (in delay doppler domain) from the received signal ~~of target REs~~ (superimposed signal) [1] For Tx side of two-sided model: modulated symbols and DMRS symbol[5]	Received signal For Tx side of two-sided model: coded bit[2,5]
Model output	1. Estimated channel at target data and/or DMRS REs[2,4,5,7,8,9,11, 12, 13,16,17,18,19,21,22,23] 1a. Estimated noise variance [12] 2. LLRs[1,2,3, 4, 6,10, 13,15,19,20, 22] 3. Filtering coefficients for channel estimation [7]	1. Estimated channel at target data REs[1,3,4,5,6,8] 2.LLR[2,3, 5,6,7,8,11,12] 3.Estimated modulation symbols[9] For Tx side of two-sided model: superimposed signal[5]	1.LLR (majority) 2.Estimated channel[2] For Tx side of two-sided model: modulated data symbols [5,2]
Label	1. Ideal channel information [2,5,7,8,9,11,12,13,15,16,17,18,22,23] 2. Known sequence/data[1,2,3, 4,10, 13,15,16,20,22] 3. Label free (unsupervised)[6, 21] 4. Estimated channel using legacy DMRS pattern with legacy receiver[8] 5. Estimated channel of adjacent RE (self-supervised)[13]	1. Known sequence/data [2,3,11,12] 2. Ideal channel information[1,8] 3.Transmitted modulation symbols[9]	1. Known sequence/data 2 ideal channel information[2] 3. Label free[1]
Training types assumption	offline training	offline training	offline training
KPI	MSE, BLER, throughput	MSE, BLER, throughput	MSE, BLER, throughput
Benchmark	With ideal channel information With conventional receiver with sparse or legacy DMRS	With ideal channel informal With conventional receiver with legacy DMRS overhead	With ideal channel information With conventional receiver with legacy DMRS overhead
Model location for inference	UE-sided model for DL or NW-sided model for UL	UE-sided model for DL NW-sided model for UL Two-sided model[5]	UE-sided model for DL[1] NW-sided model for UL[3,4] Two-sided model[2,5]
Collaboration/ interaction between UE and NW	Similar to UE-sided or NW-sided model as NR	Similar to UE-sided or NW-sided model as NR Similar to two-sided model as NR	Similar to UE-sided model as NR Similar to NW-sided model as NR Similar to two-sided model as NR
Potential spec impact	1. DMRS design 2. RAN 4: Demod requirement 3. Signalling/ procedure related to LCM for UE and/or NW sided model Etc.	1. DMRS design 2. RAN 4: Demod requirement 3. Signalling/ procedure related to LCM for UE and/or NW sided model or two-sided model (including inter-vendor calibration), when applicable Etc.	1. RAN 4: Demod requirement 2. Signalling/ procedure related to LCM for UE and/or NW sided model or two-sided model (including inter-vendor calibration), when applicable 3. Constellation design and related

			signalling/procedure Etc.

7.2.3 ***CSI compression and feedback***

Observation

For 6GR AI/ML use cases identification/categorization, [13 sources] provided preliminary simulation results and analysis on CSI compression and feedback.

- [10 sources] provided preliminary simulation results and analysis on CSI compression with joint source and channel coding (JSCC)
- [11 sources] provided preliminary simulation results and analysis on CSI compression with joint source, channel coding and modulation (JSCM)
- [2 sources] provided preliminary simulation results and analysis on CSI feedback with downloadable basis/codebook.
- [3 sources] provided preliminary simulation results (or cite to NR AI/ML for CSI compression simulation results) and analysis on CSI reconstruction with CSI feedback with SRS (assuming separate source and channel coding).
- Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table D.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table D

Sub-use case	Sub-case A: CSI compression with JSCC	Sub-case B: CSI compression with JSCM	Sub-case C: DLable basis/codebook	Sub-case D: CSI reconstruction with CSI feedback with SRS (assuming SSCC)
Reported companies	(10) ZTE[1], Samsung[2], vivo[3], {Pengcheng, ZGC}, Lenovo, OPPO, MediaTek[4], Fujitsu, BJTU[5], {BUPT, ZGC}[6]	(11) BJTU[1], Samsung[2], OPPO[3], {Pengcheng, ZGC}[4],vivo, CMCC, ZTE, {BUPT, ZGC}[7], Fujitsu[8], Apple, Lenovo	(2) ZTE[1], Samsung	(3) Qualcomm, vivo, Samsung
Model input of decoder or model output of encoder, when applicable	1. Compressed CSI bits 1a. additionally estimated channel based on SRS[2,3] 1b. (for training), assuming the model input via error bits caused by in UL transmission after legacy channel decoding[4]	1. Compressed CSI complex values via UE-sided model 2. Compressed CSI complex values via a projection matrix[1,2,3] 3. Received signal at sparse CSI-RS and CSI-RS sequence [1,4]	1.Amplitudes and phases obtained by a look up table based on feedback CSI bits 2. Selected basis[1]	1. Compressed CSI bits 2. Estimated channel based on SRS
Model output of decoder or model input of encoder, when applicable	1. (Reconstructed) Eigenvectors 2. (Reconstructed) Explicit H[1,2,3,4]	1. (Reconstructed) Eigenvectors 2. (Reconstructed) Explicit H[2]	Reconstructed Eigenvectors	(Reconstructed) Eigenvectors
Label	1.Eigenvectors 2.Explicit H[1,2,3,4]	1.Eigenvectors 2.Explicit H[2]	Eigenvectors	Eigenvectors
Training types	Offline training	Offline training	Offline training	Offline training
KPI	SGCS, NMSE, SE, UE complexity	SGCS, NMSE, SE, UE complexity	UPT vs overhead	SGCS, UPT
Benchmark	eType II NR separate source and channel coding	eType II NR separate source and channel coding JSCM with two-sided	eType II	eType II NR AI/ML CSI compression without SRS

		model[1,2,3]		SRS without CSI feedback
Model location for inference	Two-sided model	Two-sided model NW-sided model[1,2,3]	NW-sided model	Two-sided model
Collaboration/ interaction between UE and NW	Similar to two-sided model in NR	Similar to two-sided model in NR For NW-sided model: no collaboration or Similar to NW-sided model in NR	No collaboration or Similar to NW-sided model in NR	Similar to two-sided model in NR
Potential specification impact	1. Necessary signalling/ procedure to support JSCC 2. Signalling/ procedure related to LCM for two-sided model including inter-vendor collaboration	1. Necessary signalling/ procedure to support JSCM 2. Projection matrix design for NW-sided model, when applicable 3. Signalling/ procedure related to LCM with NW-sided model or two-sided model including inter-vendor collaboration, when applicable 4. RAN4 requirements, e.g., EVM	1. Downloadable basis/codebook related signalling/ procedure 2. Signalling/ procedure related to LCM with NW-sided model	1. Necessary signaling/procedure to support lower overhead and/or simpler CSI feedback 2. Signalling/ procedure related to LCM for two-sided model including inter-vendor collaboration

7.2.4 ***AI/ML for beam management and extension***

Observation

For 6GR AI/ML use cases identification/categorization, [13 sources] provided preliminary simulation results and analysis on AI/ML for beam management and extension.

- [7 sources] provided preliminary simulation for DL Tx beam management and analysis on inter-cell/inter-TRP/M-TRP DL Tx beam prediction and management.
    - Besides, [5 sources] citing to NR study for DL Tx beam management and analysis on inter-cell/inter-TRP/M-TRP DL Tx beam prediction and management.
- [3 sources] provided preliminary simulation results and analysis on cross frequency DL Tx beam prediction.
- [2 sources] provided preliminary simulation results and analysis on Tx-Rx beam pair prediction.
- [2 sources] provided preliminary simulation results for beam management and analysis on beam prediction for initial access.
    - Besides ,[5 sources] citing to NR study for beam management and analysis on beam prediction for initial access.
- [1 source] provided preliminary simulation results and analysis on DL Tx beam prediction for spatial and/or temporal domain with additional local UE information.
- [1 source] provided preliminary simulation results and analysis on reinforcement learning-based approach beam selection
- Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table E.

Note: whether/how to capture the observation in the TR is a separate discussion.

able E-1 AI/ML for beam management and extension

Sub-use case	Sub-case A: Inter-Cell/M-TRP DL Tx beam prediction and	Sub-Case B: Cross frequency DL Tx beam prediction	Sub-Case C: Tx-Rx beam pair prediction	Sub-Case D: Beam prediction for initial access	Sub-Case E: DL Tx beam prediction for spatial and/or temporal	Sub-Case F: reinforcement learning-based approach

	management				domain with additional local UE information	beam selection
Reported companies	(7) Nokia, ZTE, xiaomi, CEWiT, DoCoMo, , Lenovo, BJTU (5) Qualcomm, Samsung, LGE, NVIDA, CEWiT (citing to NR study)	(3) Futurewei[1], xiaomi[2], Apple[3]	(2) Ericsson, Nokia	(2) Huawei, vivo (5) Qualcomm, Samsung, LGE, ZTE, Apple (citing to NR study)	(1) Huawei	(1) Nokia
Model input	Measurements from Set B of one or more TRPs/Cells	Measurements in frequency A	Measurements from Set B DL Tx-Rx beam pairs.	Measurements from Set B of SSB	Measurements from Set B And additional local UE information (moving direction and speed) as UE side model input	Measurements from a set of DL Tx beam scheduling stats (at the NW), Cross corelation among DL Tx beams
Model output	Predicted best beam information and/or predicted measurements from Set A of target cell/TRP(s) [of current or future time instance]	Predicted cell/beam related information of frequency B [of current or future time instance]	Predicted best DL Tx-Rx beam pairs information from Set A DL Tx-Rx pairs.	Predicted best DL Tx beam information (and/or predicted measurements from Set A [of current or future time instance]	Predicted Best beam indexes (probability of each Tx beam in Set A to be the Top-1 Tx beam) and/or Predicted measurements from Set A [of current or future time instance]	Selected beam index for scheduling UE(s)
Label	Measurements [or Top beams] of Set A of target cell/TRP(s)	Measurements on cell(s)/beam(s) of frequency B	Measurements [or Top beams pairs] of Set A Tx-Rx pair	Measurements [or Top beams] of Set A	Measurements [or Top beams] of Set A	label-free (online learning)
Training types assumption	offline training	offline training	offline training	offline training	offline training; online finetuning (for UE side model)	Online learning
KPI	Prediction cell/beam/measurement accuracy, Throughput,	Prediction beam/measurement accuracy, RS overhead	Prediction beam/measurement accuracy, RS overhead	Prediction beam accuracy	Prediction beam/measurement accuracy	Throughput, End to end packet latency

	RS overhead reduction	reduction	reduction			
Benchmark	Based on Set A Based on Set B	Measurements on cell(s)/beam(s) of frequency B	Based on Set A Based on Set B	Based on Set A Based on Set B	NR beam prediction with AI/ML	Beam with largest RSRP (from the set) consider as the scheduling beam
Model location for inference	UE-sided model or NW-sided model	UE-sided model or NW-sided model	UE-sided model	UE-sided model or NW-sided model	NW-sided model + UE-sided model without training collaboration	NW-sided model
Collaboration/ interaction between UE and NW	As UE-sided or NW-sided model in NR	As UE-sided or NW-sided model in NR	As UE-sided model in NR	Similar to UE-sided or NW-sided model in NR	As UE-sided or NW-sided model in NR	No collaboration
Potential spec impact	1. Inter-Cell/M-TRP beam prediction related singling/procedure 2. Signalling/ procedure related to LCM for NW-sided model or UE-sided model	1. Cross frequency DL Tx beam prediction related signalling /procedure 2. Signalling/ procedure related to LCM for NW-sided model or UE-sided model	1.Signalling/ procedure related to LCM for UE-sided model	1. Initial access related to beam prediction 2. Signalling/ procedure related to LCM for NW-sided model or UE-sided model	1. As NR AI for BM; 2. Signalling/ procedure related to NW-sided model + UE-sided model. 3. Signalling/ procedure related to online finetuning, if any	1. Signalling/ procedure related to exploration phase (to mitigate the impact of exploration).

7.2.5 ***AI/ML for (de)modulation***

Observation

For 6GR AI/ML use cases identification/categorization, [5 sources] provided preliminary simulation results and analysis on (de)modulation.

- [5 sources] provided preliminary simulation results and analysis on modulation constellation design with the help of AI, and with non-AI or AI receiver.
- [3 sources] provided preliminary simulation results and analysis on AI-based modulation and precoding with two-sided model.
- Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in Table F.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table F For (de)modulation

Sub-use case	Sub-use case A: AI-based (de)modulation	Sub-use case B: AI-based modulation and precoding
Reported companies	(5)ZTE[1], OPPO[2], vivo[3], Lenovo[4], Xiaomi[5]	(3) ZTE, OPPO, Lenovo
Model input	For constellation design 1. Coded bits [1,2,3,4,5] 2. Channel characterization and modulation order[4] For AI receiver 1.Received signal[2,3,4]	Encoder: Coded bits Decoder: Estimated symbols
Model output	For constellation design 1. Learned constellation [1,2,3,45]	Encoder: modulated symbols after layer mapping

	2. Probability of constellation points [4] For receiver 1. LLR[2,3,4]	Decoder: Soft LLR
Label	Known coded bits	Known coded bits
Training types	Offline training	Offline training
KPI	BLER	BLER
Benchmark	Uniform QAM with legacy receiver	Uniform QAM with legacy receiver and NR layer mapping
Model location for inference	1.NA (AI for constellation design with legacy receiver) [1,2,3,4,5] 2.Receiver-sided model[2,3,4]	Two-sided model
Collaboration/interaction between UE and NW	NA or Similar to NW-sided model or UE-sided model in NR	Similar to two-sided model in NR
Potential specification impact	1. Constellation design and related signaling/procedure 2. Signaling/ procedure related to LCM for NW-sided model or UE-sided model 3. RAN4 requirements, e.g., EVM	1. Modulation design and layer mapping design 2. Signaling/ procedure related to LCM for two-sided model including inter-vendor collaboration 3. RAN4 requirements, e.g., EVM

7.2.6 ***AI-based none-linearity handling at transmitter or receiver***

Observation

For 6GR AI/ML use cases identification/categorization, [5 sources] provided preliminary simulation results and analysis on AI-based none-linearity handling at transmitter or receiver.

- [5 sources] provided preliminary simulation results and analysis on AI-based DPoD/None-linearity compensation at receiver.
- [2 sources] provided preliminary simulation results and analysis on AI-based DPD at transmitter.
- Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in Table G.

Note: whether/how to capture the observation in the TR is a separate discussion.

Sub-use case	Sub-use case A: AI-based DPoD/None-linearity compensation	Sub-use case B: AI-based DPD
Reported companies	(5) Samsung[1], Ericsson[2], OPPO[3], vivo[4], Huawei[5]	(2) vivo[2], Huawei[1]
Model input	1. Received signal[1,3,4,5]	Time domain samples before pre-distortion
Model output	1. Compensated signal in time domain[1,2,4,5] 2. Soft bits[2,3]	Time domain samples after pre-distortion
Label	1. DMRS[1] 2. Known bit sequence[2,3,4] 3. time domain samples from known sequence[5]	Time domain samples
Training types	Online training/finetune[1] Offline training	Offline training Online training/finetune[2]
KPI	BLER, MPR, EVM, throughput	BLER, EVM, MPR
Benchmark	Without compensation	No DPD
Model location for inference	NW-sided model	UE-sided model
Collaboration/interaction between UE and NW	Similar to NW-sided model as NR	Similar to UE-sided model as NR
Potential specification impact	1. RAN 4 requirements, e.g. EVM 2. DMRS/Sequence design/selection, Tx power determination	1. RAN4 requirements, e.g. EVM 2. Tx power determination 3. Signaling/ procedure related to LCM

	3. Signaling/ procedure related to LCM for NW-sided model	for UE-sided model

7.2.7 *Low overhead SRS with AI/ML*

Observation

For 6GR AI/ML use cases identification/categorization, [4 sources] provided preliminary simulation results and analysis on low overhead SRS with AI/ML

[1 source] provided preliminary simulation results and initial analysis on low PAPR SRS sequence design with help of AI/ML

Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and analysis in Table I.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table I SRS with AI/ML

Use case	Low overhead SRS with AI/ML	Low PAPR SRS sequence design
Reported companies	(4) {Spreadtrum, UNISOC}, vivo, Huawei, Kyocera	(1) vivo
Model input	Measurement of channel with low overhead SRS of frequency/temporal domain	Sequence index
Model output	Estimated channel	Learn sequences
Label	Ideal channel information	Label free
Training types	Offline training	Offline training
KPI	SCGS, throughput	PAPR, SGCS, Cross-correlation between SRS sequences
Benchmark	With legacy SRS With ideal channel information	Legacy SRS sequence
Model location for inference	NW-sided model	NW-sided model or Without model for inference
Collaboration/interaction between UE and NW	Similar to NW-sided model in NR	No collaboration for no model Similar to NW-sided model in NR
Potential specification impact	1.Sparse SRS design 2. Signalling/ procedure related to LCM for NW-sided model	1. SRS design 2. Signaling/procedure related to DLable/ULable SRS sequence, when applicable 3. Signalling/ procedure related to LCM for NW-sided model, when applicable

7.2.8 *AI-enabled UL precoder indication*

Observation

For 6GR AI/ML use cases identification/categorization, [3 sources] provided preliminary simulation results and analysis on AI-enabled UL precoder indication with detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in Table H.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table H AI-enabled UL precoder indication

Use case	AI-enabled UL precoder indication
Reported companies	(3) vivo[1], Fujitsu[2], Samsung[3]
Model input of decoder or model output of encoder	UL precoder indicator/compressed UL precoder
Model output of decoder or model input of encoder	(Reconstructed) eigenvectors of UL channel
Label	Estimated eigenvectors of UL channel based on SRS measurement
Training types	offline training

	online finetune[1]
KPI	SCGS, BLER
Benchmark	NR TPMI codebook
Model location for inference	No model for inference [1,3] Two-sided model[1,2]
Collaboration/interaction between UE and NW	Similar to one-sided model in NR [1,3] Similar as two-sided model in NR[1,2]
Potential specification impact	1.The signaling/procedure related to the download/upload of UL codebooks/compressed UL precoder 2. LCM procedure to facilitate the training of the downloadable/uploadable UL codebooks when no model for inference, [1,3] 3. Signalling/ procedure related to LCM for for two-sided model including inter-vendor collaboration, when applicable[1,2]

7.2.9 *AI/ML based waveform for PAPR reduction*

Observation

For 6GR AI/ML use cases identification/categorization, [3 sources] provided preliminary simulation results and analysis on AI/ML based waveform for PAPR reduction with detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis in Table J.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table J AI/ML based waveform for PAPR reduction

Use case	AI/ML based waveform for PAPR reduction
Reported companies	(3) vivo[1], Samsung[2], Huawei[3]
Model input	Symbols in frequency domain
Model output	For model output of encoder for UE-sided/NW-part of two-sided model: transformed/precoded symbols in frequency domain For output of decoder for NW-part of two-sided model: 1. LLR[1,3] 2. Symbols in frequency domain[2]
Label	Label free[2,3] Known bit sequences or its LLR[1,3]
Training types	offline training
KPI	BLER, CCDF of PAPR(UL), throughput (DL)
Benchmark	DFT-s-OFDM
Model location for inference	Two-sided model UE-sided model (for frequency domain shaping)[1]
Collaboration/interaction between UE and NW	Similar to two-sided model in NR No collaboration for UE-sided model[1]
Potential specification impact	1. Signaling/ procedure related to LCM for two-sided model including inter-vendor collaboration, when applicable 2. Signaling/ procedure related to LCM for UE-sided model [1] 3. Related RAN4 requirements

7.2.10 *AI/ML based HARQ-ACK feedback*

Observation

For 6GR AI/ML use cases identification/categorization, [2 sources] provided preliminary simulation results and analysis on AI/ML based HARQ-ACK feedback with detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis in Table K.

Note: whether/how to capture the observation in the TR is a separate discussion.

Table K. AI/ML based HARQ-ACK feedback

Use case	AI/ML based HARQ-ACK feedback
Reported companies	(3) Qualcomm, vivo
Model input	HARQ ACK/NACK bit sequence

Model output	Learned sequences/modulated symbols
Label	HARQ-ACK/NACK bit sequence
Training types	Offline training
KPI	BLER
Benchmark	NR RM code for up to 11bits with Maximum Likelihood (ML) receiver
Model location for inference	No model for inference
Collaboration/interaction between UE and NW	No collaboration
Potential specification impact	1.Learned sequence/modulated symbols design 2.Downloadable sequence/modulated symbols related signalling/ procedure for HARQ-ACK 3. Related RAN4 requirements

7.2.11 *improved scheduling/HARQ for token traffic*

Observation
For 6GR AI/ML related service, for 6GR AI/ML use cases identification/categorization, [2 sources] provided preliminary simulation results and analysis on improved scheduling/HARQ for token traffic
Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis in Table L.
Note: whether/how to capture the observation in the TR is a separate discussion.

Table L for improved scheduling/HARQ for token traffic

Use case	Improved scheduling/HARQ for token traffic
Reported companies	(2) Huawei[1], OPPO[2]
Model input	Tokenizer model: • Input: Raw data (e.g., image/video/audio, etc.) De-tokenizer model: • Input: Tokens
Model output	Tokenizer model: • Output: Tokens (e.g., tokenized image/video/audio) De-tokenizer model: • Output: Inference results for downstream tasks/Raw data (e.g., image/video/audio, etc.)
Label	Training at OTT, transparent to RAN[1,2]
Training types	Offline training at OTT, transparent to RAN
KPI	Supported number of UEs, achievable throughput
Benchmark	NR scheduling/HARQ mechanism without knowledge of Token traffic
Model location for inference	The tokenizer model is at UE or NW/OTT server (e.g., an encoder). The de-tokenizer model is at NW/OTT server or UE (e.g., a decoder).
Collaboration/interaction between UE and NW	NA
Potential specification impact	• Service awareness in RAN • Token error identification, new scheduling and HARQ

7.2.12 *Other use cases*

Observation
For 6GR AI/ML use cases identification/categorization,
[one source] provided preliminary simulation results and analysis on pathloss prediction in the spatial, temporal, and/or frequency domain, to use the predicted pathloss in UL (PUSCH/PUCCH/PRACH/SRS) power control.
[one source] provided preliminary simulation results and analysis on UL closed-loop power control with an NW-sided AI/ML model, where the model predicts the optimal power adjustment (or TPC command index) for the UE,
[one source] provided preliminary simulation results and analysis on prior-information-aided DCI decoding,
[one source] provided preliminary simulation results and analysis on lossless DCI compression,
[one source] provided preliminary simulation results and analysis on early contention resolution in RACH,
[one source] provided preliminary simulation results and analysis on sensing based RAN digital twin construction with NW-side AI/ML model,

[one source] provided preliminary simulation results and analysis on AI/ML-enabled RAN digital twin with distributed model,
[one source] provided preliminary simulation results and analysis on AI/ML based SRS power imbalance compensation,
[one source] provided preliminary simulation results and analysis on Site Specific Learning for AI/ML and RAN Digital Twin,
Detailed evaluation assumptions (model input/output/label/KPI/benchmark) and initial analysis can be found in in Table M
Note: whether/how to capture the observation in the TR is a separate discussion.

Table M -1

Use Case	Prior-Information-Aided DCI Decoding	Lossless DCI Compression	UL closed-loop power control with an NW-sided AI/ML model, where the model predicts the optimal power adjustment (or TPC command index) for the UE.	Pathloss prediction in the spatial, temporal, and/or frequency domain, to use the predicted pathloss in UL(PUSCH/PUCCH /PRACH/SRS) power control.
Reported companies	(1)CMCC	(1)CMCC	(1) Nokia	(1) Nokia
Model input	LLR after demodulation at current transmission, and historical AI/ML based predicted LLR	Historical DCI payload	UL SINR measurement, UE Tx power estimate (derived from Pcmax, P0, PL alpha, pathloss measurement), and PUSCH allocation size	L1-RSRP measurements from a sub-set/set of RSs/beams (Set B). input can consider history of measurements
Model output	Decoded DCI payloads, and predicted LLR for next transmission	Predicted DCI payload	Predicted TPC command index	Predicted pathloss value(s) (or predicted L1-RSRP(s)) for a set of RSs/beams (Set A). output can consider future instances
Label	DCI payload sequences	DCI payload sequences	Optimal TPC command index (offline learning) label-free (online learning)	Pathloss value(s) (or L1-RSRP(s)) for a set of RSs/beams (Set A)
Training types	Offline training at the UE side	Offline training at the NW side, and model delivery to UE side	Offline and Online learning	Offline training
KPI	BLER performance	BER and sample-level prediction accuracy; DCI overhead reduction	UL throughput.	Pathloss prediction accuracy, throughput, RS overhead reduction, Complexity.
Benchmark	Traditional DCI decoder	Traditional DCI design	1. UL Power control with optimized OLPC parameters 2. UL Power control with optimized OLPC parameters and possibly legacy CLPC algorithms (with 5G TPC tables).	Pathloss estimation based on Set A Pathloss estimation based on Set B
Model location for	UE-sided model	UE-sided model +	NW-sided model	UE-sided model

inference		NW-sided model		NW-sided model
Collaboration/ interaction between UE and NW	Similar to UE-sided model in NR	Model transfer from NW to UE	None	As UE-sided or NW-sided mode in nRl
Potential specification impact	1. Signalling/configuration design for prior-information-aided DCI decoder. 2. Signalling/ procedure related to LCM for UE-sided model	1. Signalling/configuration design for Lossless DCI Compression. 2. Signalling/ procedure related to LCM including model transfer	Configurability of the values in TPC command tables or an extended TPC command table (compared to NR).	1. Pathloss prediction related signalling/procedure 2. Signalling/ procedure related to LCM for UE-sided or NW-sided model 3. RAN4 performance requirements and test cases, including defining new requirements related to pathloss reference signal (PL-RS) measurement and activation delays of TCI state(s).

Table M-2

Sub-use case	Early contention resolution in RACH	Sensing based RAN digital twin construction with NW-side AI/ML model	AI/ML-enabled RAN digital twin with distributed model	AI/ML based SRS power imbalance compensation	Site Specific Learning for AI/ML and RAN Digital Twin
Reported Companies	(1) Ofinno	(1) Huawei	(1) Huawei	(1) Huawei	(1) DeepSig
Model input	Received PRACH signal (e.g., preamble waveform)	Point cloud sensed by the BS with mono-static sensing and sensed/reported by UEs with bi-static sensing	UE-part models: local sparse point cloud NW-part model: latent space information from multiple UEs	UL measured channel matrix from SRS with IL imbalance	Received signal/estimated channel at DMRS and received signal on data, and the channel information generated by digital twin
Model output	Predicted number of UEs that transmitted the same preamble for given PRACH resources	3D point cloud representing the static environment	UE-part models: compressed latent space information NW-part model: global point cloud	DL channel matrix with IL compensated	Decoded bit
Label	Ground-truth number of UEs that transmitted the same preamble	Ground truth point cloud	Ground truth point cloud	UL SRS measurement without IL (assuming it is compensated by UE at certain conditions) or DL CSI-RS measurement	Ground truth of target bit

Training types assumption	Offline training	Offline training	Offline training (adopted in simulation) Online finetuning (can be optionally considered)	offline training	Offline training
KPI	Prediction accuracy of UE multiplicity, RACH access delay, first-attempt success probability	Sensing accuracy metric: root mean square error (RMSE) of point cloud. RMSE= $\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \hat{x}_i\right)^2}$ is the square root of the average of the squared errors between each sensed point (x_i in forms of coordinates) and ground truth point ($\hat{x}_i$ in forms of coordinates) in the point cloud including n points with {x, y, z} dimensions.	1. Overhead metric: Feedback bits per point 2. Sensing accuracy metric: intersection-over-union (IoU), edge detection probability	SGCS	BLER, throughput
Benchmark	First-attempt success rate based on legacy PRACH receiver	BS side mono-static sensing only to construction RAN digital twin	1. Single UE sensing (to justify sensing accuracy metric of using distributed model). 2. Raw data transmission (to justify overhead metric of using distributed model).	1. SRS without IL imbalance; 2. non-AI based SRS IL imbalance compensation	Legacy receiver without the help of digital twin
Model location for inference	NW-side model	NW-side model	Distributed model: a NW-side model paired with multiple UE-side models.	NW-sided model	NW-sided model
Collaboration/ interaction between UE and NW	Similar to NW-sided model in NR	Similar to NW-sided model in NR	Similar to two-sided model: UE reporting of compressed sensing results for inference. Inter-vendor training collaboration between NW side and UE side.	Similar to NW-sided model in NR	Similar to NW-sided model in NR

Potential spec impact	1. Signaling/procedure related to Mgs.3 grant for more than one UEs selected the same PRACH sequence 2. Signalling/procedure related to LCM for NW-sided model	1. Signaling/procedure related to bi-static sensing results reported from UE 2. Signalling/procedure related to LCM for NW-sided model	1. Sensing results reported from UE in forms of compressed latent message 2. Signalling/procedure related to LCM for two-sided model including inter-vendor collaboration	1. Inference: UE reporting on the IL range for ensuring generalization 2. Signalling/procedure related to LCM for NW-sided model	1. Signalling/procedure related to LCM for NW-sided model

7.3 RAN 1 #123

Agreement

From RAN 1 perspective, the following use cases can be matched to the identified primary agendas of RAN1

Use cases		Primary agendas
Low overhead CSI-RS or CSI prediction with AI/ML		CSI-RS and CSI acquisition
Low overhead DMRS with AI/ML receiver		UL & DL DMRS associated with PUSCH/PDSCH Note: Sub-Case C: DMRS-free may be related to modulation
CSI compression and feedback		CSI acquisition Note: this may be related to uplink control
AI/ML for beam management and extension		Initial access for Sub-case D Beam management for other sub-cases Note: sub-case A/B/D maybe related to mobility
AI/ML for SRS	**Low overhead SRS with AI/ML**	SRS
	Low PAPR SRS sequence design	
	AI/ML based SRS power imbalance compensation	Not RAN1-led
AI-enabled UL precoder indication		UL MIMO
AI-based non-linearity handling at transmitter or receiver		Not RAN1-led Note: this may be related to DMRS, SRS, Power control in RAN 1
AI/ML for (de)modulation		modulation Note: Sub-case B may be related to MIMO Note: assuming no change to DMRS
AI/ML based waveform for PAPR reduction		Waveform
AI/ML based HARQ-ACK feedback		Channel coding Note: this may be related to uplink control
PDCCH related	**Prior-Information-Aided DCI Decoding**	DL control
	Lossless DCI Compression	
Power control related	**UL closed-loop power control**	Power control
	Pathloss prediction	
RACH related design	**Early contention resolution in RACH**	Random access/PRACH

	Low PAPR PRACH sequence design	
Site Specific Learning for AI/ML using RAN Digital Twin		Depending on corresponding use case where site specific learning is applicable, e.g., DMRS
Digital twin construction related use cases	**AI/ML-enabled RAN digital twin with distributed model**	ISAC
	Sensing based RAN digital twin construction with NW-side AI/ML model	ISAC
AI for positioning		Positioning related agenda, if any

To accommodate AI/ML service, e.g., token traffic	if any impact, most likely scheduling/HARQ

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8 Reference

[1] R1-2508324 Discussion on AI/ML in 6GR interface FUTUREWEI
[2] R1-2508341 Views on AI/ML Operation and Use Cases for 6G Radio Air Interface Nokia
[3] R1-2508351 Discussion on AI/ML-driven use cases for 6GR BJTU
[4] R1-2508365 AI/ML Use Cases for 6GR Air Interface Ericsson
[5] R1-2508366 AI/ML in 6GR interface Kyocera
[6] R1-2508392 Discussion on AIML in 6GR interface Spreadtrum, UNISOC
[7] R1-2508437 Discussion on AI/ML in 6GR interface vivo
[8] R1-2508460 Discussion on AI/ML in 6GR interface CMCC
[9] R1-2508515 AI/ML for 6G Air Interface InterDigital, Inc.
[10] R1-2508539 Discussion on AI/ML in 6GR air interface TCL
[11] R1-2508549 Discussion on AIML in 6GR interface NEC
[12] R1-2508582 AI/ML in 6GR interface CATT, CICTCI
[13] R1-2508643 AI/ML use cases and framework for 6GR Air Interface AT&T
[14] R1-2508689 Discussion on AI/ML in 6GR interface Xiaomi
[15] R1-2508697 Discussion on AI-based Smart Radio for 6G Air Interface ZTE Corporation, Sanechips
[16] R1-2508732 AIML use cases for 6GR air interface OPPO
[17] R1-2508740 Views on AI/ML in 6GR air interface Huawei, HiSilicon
[18] R1-2508753 Discussion on AI/ML in 6GR interface LG Electronics
[19] R1-2509516 (R1-2508807) AI/ML Use cases and framework for 6GR Samsung
[20] R1-2508827 AI/ML Use Cases for 6G RAN Tejas Network Limited

[21] R1-2508916 Discussion on AI/ML in 6GR interface Lekha Wireless Solutions
[22] R1-2508933 Discussion on AI/ML in 6GR Fujitsu
[23] R1-2508947 AI/ML in 6GR Air Interface Google
[24] R1-2508978 Discussion on AI/ML in 6GR interface ETRI
[25] R1-2508999 Discussion on AI/ML in 6GR interface Panasonic
[26] R1-2509001 Views on improved scheduling/HARQ for token traffic CAICT
[27] R1-2509004 AI/ML in 6GR Lenovo
[28] R1-2509008 AIML in 6GR air interfaceKAIST
[29] R1-2509021 AI and ML in 6GR air interface NVIDIA
[30] R1-2509022 Discussion on AI/ML in 6GR IMU
[31] R1-2509039 Views on AI/ML in 6GR interface Ofinno
[32] R1-2509046 Discussion on AI ML in 6GR air interface Hanbat National University
[33] R1-2509050 Discussion on AI/ML for 6GR interface Ruijie Networks Co. Ltd
[34] R1-2509078 Use cases for AI/ML in 6GR Interface Sony
[35] R1-2509115 On AI/ML for 6G air interface Apple
[36] R1-2509116 Discussion on AI/ML in 6GR interface AUMOVIO
[37] R1-2509148 AI/ML in 6GR Air Interface MediaTek Inc.
[38] R1-2509184 Discussions on AI/ML in 6GR interface Sharp
[39] R1-2509236 AI/ML in 6GR air interface Qualcomm Incorporated
[40] R1-2509251 Discussion on AI/ML in 6GR interface Transsion Holdings
[41] R1-2509287 Discussion on AI/ML for 6GR air interface NTT DOCOMO, INC.
[42] R1-2509302 Use cases for AI/ML in 6GR interface KT Corp.
[43] R1-2509354 AI/ML in 6GR Interface CEWiT
[44] R1-2509373 Discussion on AI/ML in 6GR Interface Indian Institute of Tech (M)
[45] R1-2509400 On new use cases for AI/ML in 6GR interface Vodafone, Deutsche Telekom, Nokia
[46] R1-2509411 Discussion on AI/ML-enabled use cases for 6GR BUPT, ZGC Institute of Ubiquitous-X Innovation and Application, Pengcheng Laboratory
[47] R1-2509417 New use cases for AI/ML in 6GR interface Pengcheng Laboratory, ZGC Institute of Ubiquitous-X Innovation and Application
[48] R1-2509433 Discussion on AI/ML use cases for 6GR air interface IIT Kanpur
[49] R1-2509521 Discussion on AI/ML in 6GR air interface SK Telecom, Nokia, NTT DOCOMO, NTT, Inc.

Reference for RAN 1#122bis
[50] R1-2506742 Discussion on AI/ML in 6GR interface FUTUREWEI
[51] R1-2506757 Views on AI/ML Operation and Use Cases for 6G Radio Air Interface Nokia
[52] R1-2506762 AI/ML Use Cases for 6GR Air Interface Ericsson
[53] R1-2506778 Discussion on AI-based Smart Radio for 6G Air Interface ZTE Corporation, Sanechips
[54] R1-2506780 Discussion on AI/ML-driven use cases for 6GR BJTU
[55] R1-2506784 Discussion on AI/ML in 6GR air interface TCL
[56] R1-2506820 Discussion on AIML in 6GR interface Spreadtrum, UNISOC
[57] R1-2506821 AI/ML for 6G Air Interface InterDigital, Inc.
[58] R1-2506904 Discussion on AI/ML in 6GR interface vivo
[59] R1-2506951 AI/ML in 6GR interface Kyocera
[60] R1-2506995 Discussion on AI/ML in 6GR interface Xiaomi
[61] R1-2507020 Discussion on AI/ML in 6GR interface CMCC
[62] R1-2507029 Discussion on AI/ML in 6GR interface Lekha Wireless Solutions
[63] R1-2507064 Views on AI/ML in 6GR air interface Huawei, HiSilicon
[64] R1-2507107 Views on AI/ML in 6GR interface CATT, CICTCI
[65] R1-2507182 AIML use cases for 6GR air interface OPPO
[66] R1-2507203 AI/ML in 6GR interface Tejas Network Limited
[67] R1-2507259 AI/ML Use cases and framework for 6GR Samsung
[68] R1-2507288 Discussion on AI/ML in 6GR Fujitsu
[69] R1-2507305 Discussion on AIML in 6GR interface NEC
[70] R1-2507377 Discussion on AI/ML in 6GR interface Panasonic
[71] R1-2507378 AI/ML Use Cases for 6G NTU
[72] R1-2507400 Discussion on AI/ML in 6GR interface LG Electronics
[73] R1-2507409 Views on AI/ML in 6GR air interface SK Telecom
[74] R1-2507433 AI and ML in 6GR air interface NVIDIA
[75] R1-2507471 Views on AI/ML in 6GR interface Ofinno
[76] R1-2507488 Discussion on AI/ML Use-cases in 6GR Lenovo

[77] R1-2507492 Use cases for AI/ML in 6GR interface KT Corp.
[78] R1-2507512 Discussion on AI/ML in 6GR interface ETRI
[79] R1-2507522 AI/ML in 6GR Air Interface Google
[80] R1-2507525 New use cases for AI/ML in 6GR interface Pengcheng Laboratory, ZGC Institute of Ubiquitous-X Innovation and Application
[81] R1-2507547 Discussion on AI/ML in 6GR interface Continental Automotive
[82] R1-2507601 Discussion on the potential AI/ML use cases for 6GR interfaceSony
[83] R1-2507604 Discussion on AI/ML for 6GR interface Ruijie Networks Co. Ltd
[84] R1-2507605 Discussion on AI/ML Use Cases for 6GR DeepSig Inc
[85] R1-2507613 AI/ML Framework and Use Cases for 6GR Air Interface MediaTek Inc.
[86] R1-2507633 Discussion on AI/ML in 6GR interface Transsion Holdings
[87] R1-2507683 Discussion on 6G AI/ML use cases Apple
[88] R1-2507727 AI/ML in 6GR air interface Qualcomm Incorporated
[89] R1-2507752 AI/ML use cases and framework for 6GR Air Interface AT&T
[90] R1-2507776 Discussions on AI/ML in 6GR interface Sharp
[91] R1-2507779 Discussion on AI/ML in 6GR Shanghai Jiao Tong University, Toyota
[92] R1-2507821 Discussion on AI/ML for 6GR air interface NTT DOCOMO, INC.
[93] R1-2507826 AI/ML in 6GR air interface KAIST
[94] R1-2507827 Discussion on AI/ML-enabled use cases for 6GR BUPT, ZGC Institute of Ubiquitous-X Innovation and Application
[95] R1-2507842 Discussion on AI/ML in 6GR air interface Hanbat National University
[96] R1-2507872 Discussion on AI/ML in 6GR Interface Indian Institute of Tech (M)
[97] R1-2507883 On new use cases for AI/ML in 6GR interface Vodafone, Deutsche Telekom
[98] R1-2507907 AI/ML in 6GR Interface CEWiT
[99] R1-2507910 Discussion on AI/ML in 6GR -Physical Layer Rakuten Mobile, Inc
[100] R1-2507940 Discussion on AI/ML in 6GR air interface IIT Kanpur